Virtuoso® Layout Pro: T9 Virtuoso Design Planner

Course Version IC 25.1

Lab Manual **Revision 1.0**

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Virtuoso Layout Pro: T9 Virtuoso Design Planner

Module 1: About This Course

There are no labs for this module.

Module 2: Setting Up the Design Planning Environment

Lab 2-1 Setting Up the Working Directory

Objective: To set up your design environment to run the Design Planning tasks.

The Virtuoso® Design Planner is a connectivity-based tool in the Layout EXL that optimizes layout planning and supports seamless layout planning, placement, and routing. The Design Planner supports built-in automatic placement and congestion analysis capabilities that allow optimizing layout planning at the top level, block level, and cell level. The Design Planner offers a new method for efficient physical layout planning, providing a smooth transition between placement and routing.

The Design Planner supports:

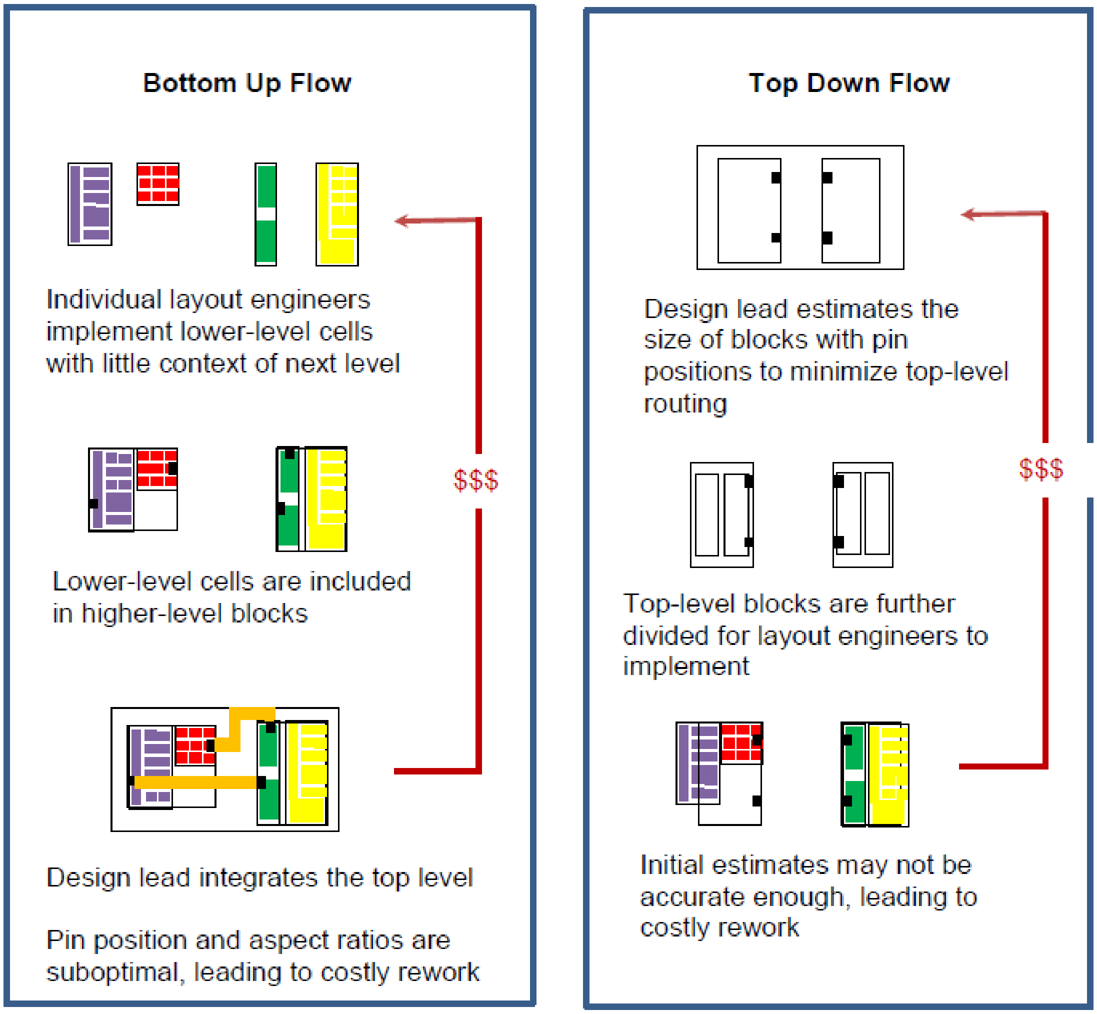
- **Hierarchical schematic-driven layouts** combine the best of bottom-up and top-down design methodologies while avoiding the shortcomings of each.
- **Hierarchical visualization** allows you to easily view or hide details on any level, anywhere in your design, to view only what you need when you need it.
- **Hierarchical and congestion-aware design planning and placement** that provides automated and assisted generation.
- **Hierarchical routing and congestion analysis** that makes real routing and congestion analysis information available upfront.

Important: Virtuoso Design Planner is **NOT** a replacement for Virtuoso Floorplanner. The two approaches are complementary.

Overview

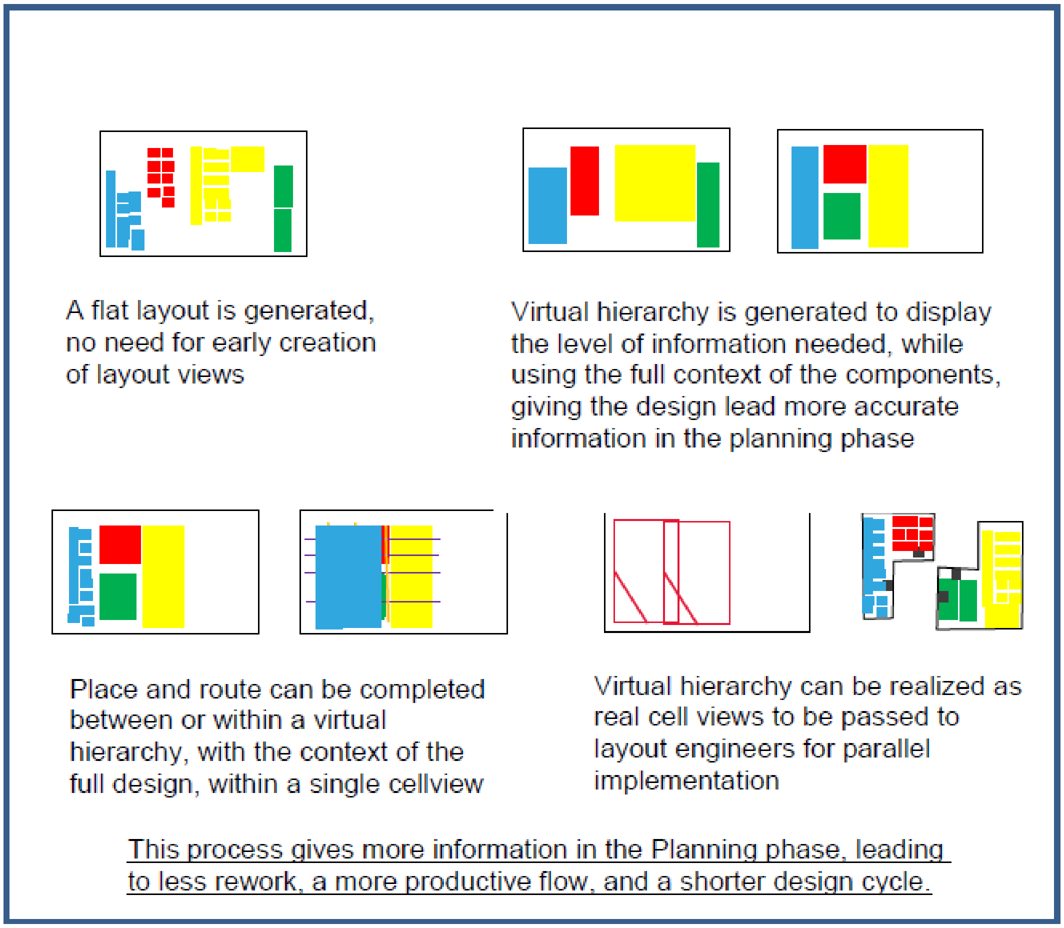
The creation of a top- or block-level layout is a complex process. Physical design planning is mandatory to increase silicon usage and design performance. Although physical planning is a time-consuming task, it is an important stage of the layout design process. Working with a flat hierarchy can be impractical due to the large number of components in a design. However, introducing hierarchy to allow the work to be divided amongst a design team also has its challenges. Irrespective of whether the team works in a bottom-up or a top-down flow, it will face challenges such as creating the layout cells early in the design and going through multiple iterations for Placement and Routing.

Drawbacks of Both the Bottom-Up and Top-Down Flows



Advantages of the Virtuoso Design Planner Flow

The Virtuoso Design Planner offers a new method for efficient layout planning that can be applied at the top, block, or cell level. It introduces the virtual layout hierarchy. This new concept allows the design team to benefit from the advantages of both the flat and hierarchical design flows while providing a smooth transition to layout, place, and route.



Invoking the *virtuoso* Executable

In this lab, you set up your design environment to run the Design Planning tasks in the subsequent labs.

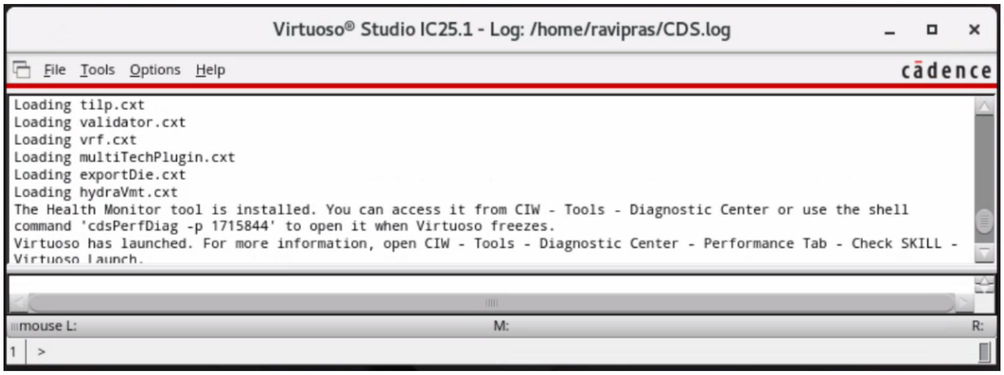
1. In a terminal window, enter the following command from your home (login) directory to get to the lab database directory when you initially logged in:

   ```
   cd ./VLPT9_IC_25_1
   ```

2. To start the Cadence® Virtuoso software, enter this command in the same terminal window:

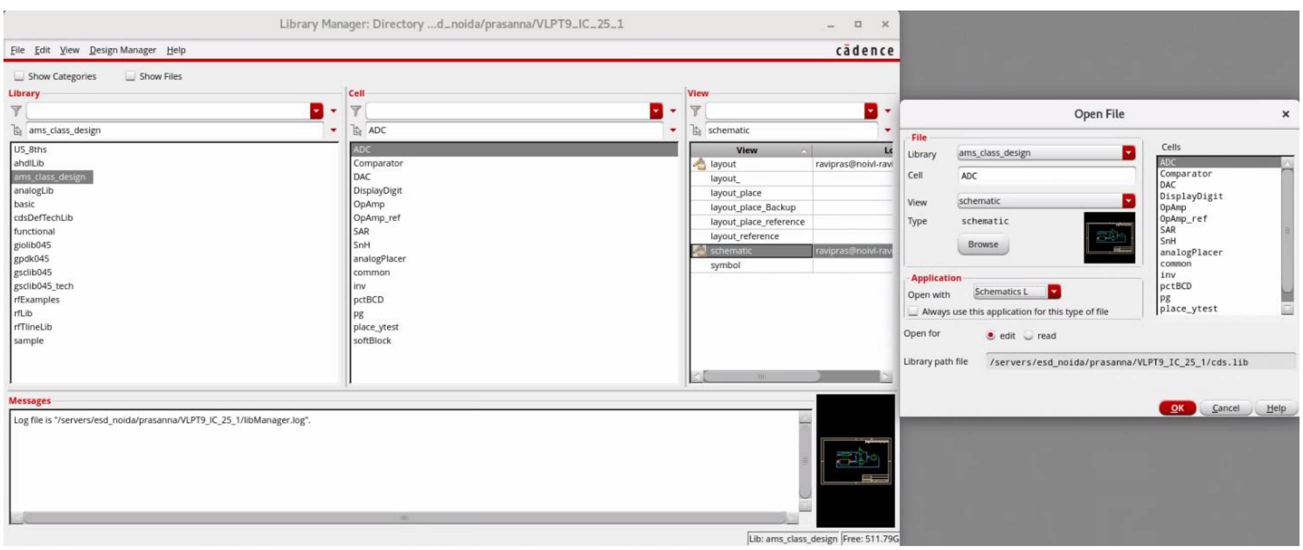
   ```
   virtuoso &
   ```

The Command Interpreter Window (CIW) appears.

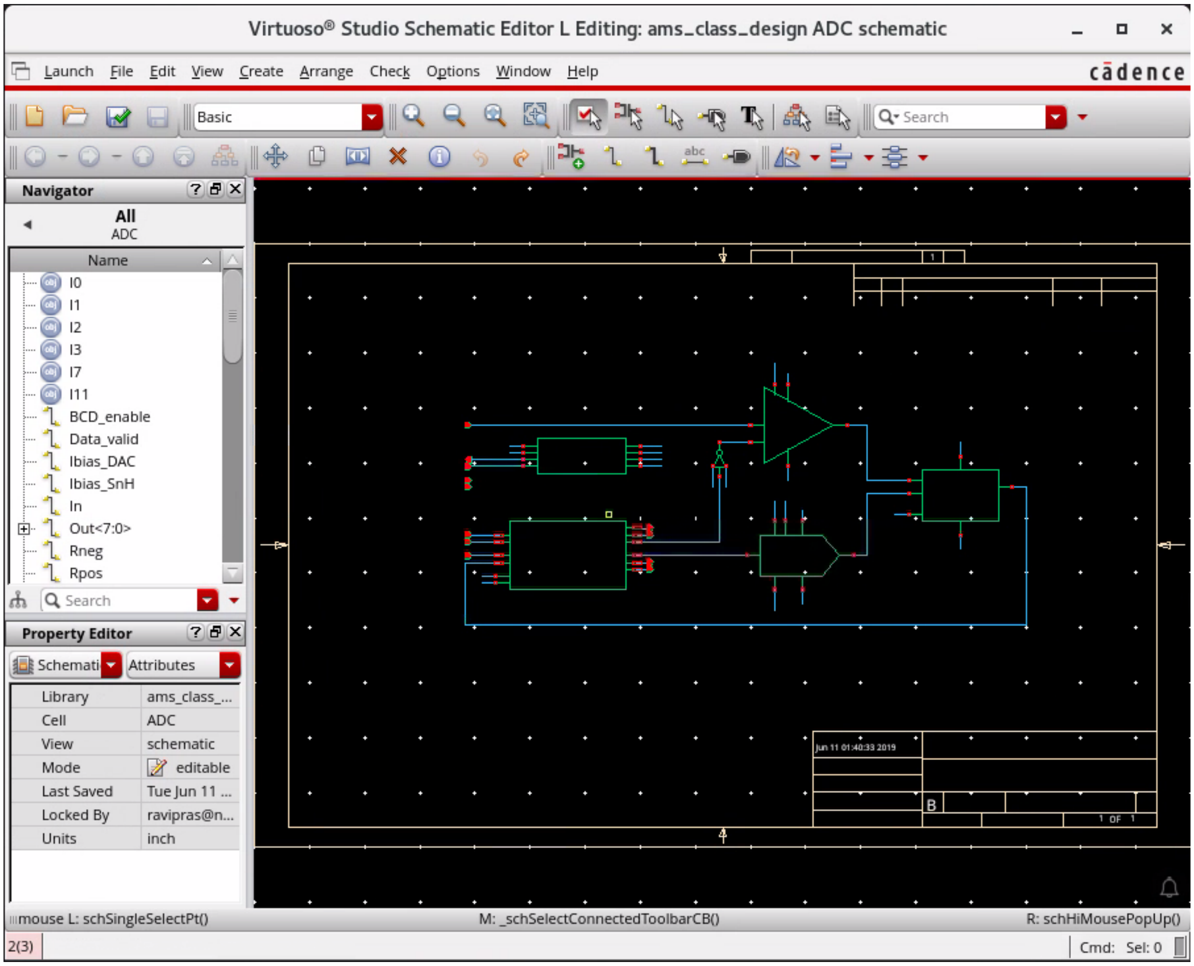


3. Open the following schematic in **L** with the **Library Manager** or choose **File – Open** from the CIW, as shown here.

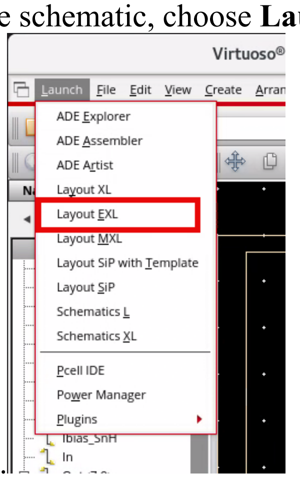
Library	`ams_class_design`
Cell	`ADC`
View	`schematic`



The *ams_class_design/ADC/schematic* window opens in the **L** mode, as shown here.

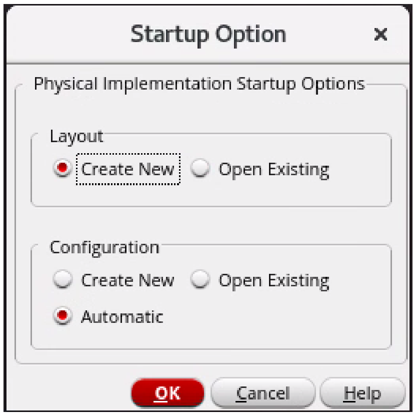


4. In the schematic, choose **Launch – Layout EXL** to bring up the **Startup Option** form.



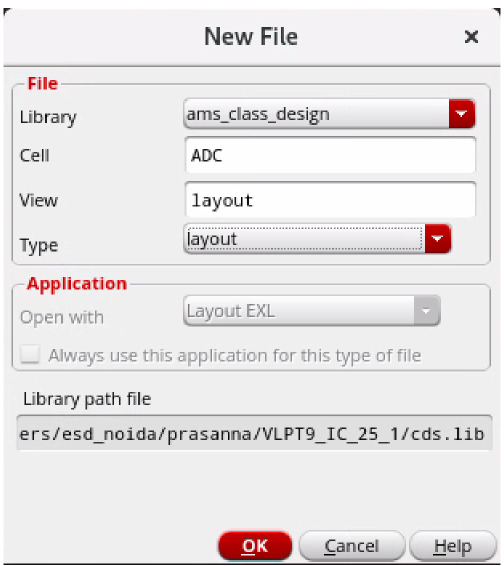
To run the Design Planning tasks in this lab, ensure that you have the Virtuoso Layout Suite EXL or MXL license configured in your IC 25.1 design environment.

5. In the Startup Option form, select the options shown here to create a new layout view and click **OK**.

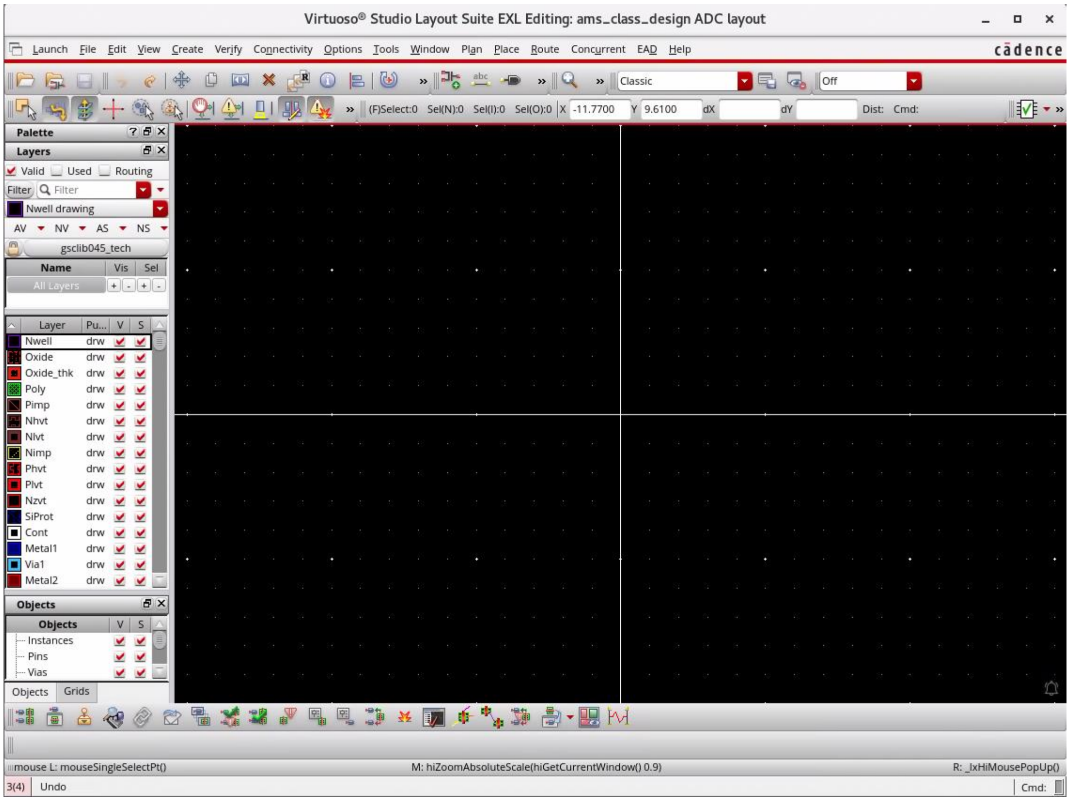


The New File form opens.

6. Fill out the **New File** form as shown here, and click **OK**.



The *ams_class_design/ADC/layout* window opens in the **EXL** mode, as shown here.



Note: In the layout canvas, from the **Window – Assistants** menu, you can enable/disable the assistants as per your requirement.

Note: In the layout canvas, from the **Window – Toolbars** menu, you can make the toolbars appear/disappear as per your requirement.

Note: In the layout canvas, from the **Workspace Configuration** cyclic field, you can choose the required workspace.

7. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned how to:

- Set up your working directory and launch the required schematic and layout designs for design planning.

End of Lab

Lab 2-2 Analyzing the Design Planning Environment

Objective: To analyze the new design planning environment within Virtuoso.

In this lab, you explore the new Design Planning Environment within Virtuoso that supports design planning.

The Design Planning Environment

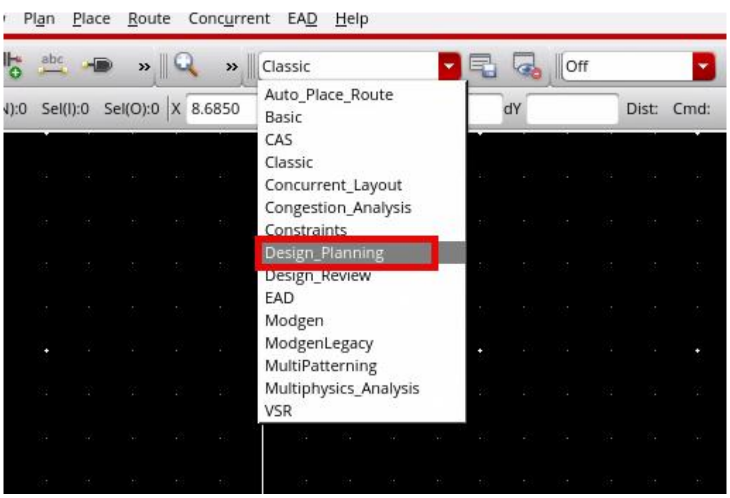
It consists of three main components:

- Design_Planning workspace
- Navigator assistant with Design Planning data sets
- Design Planning toolbar

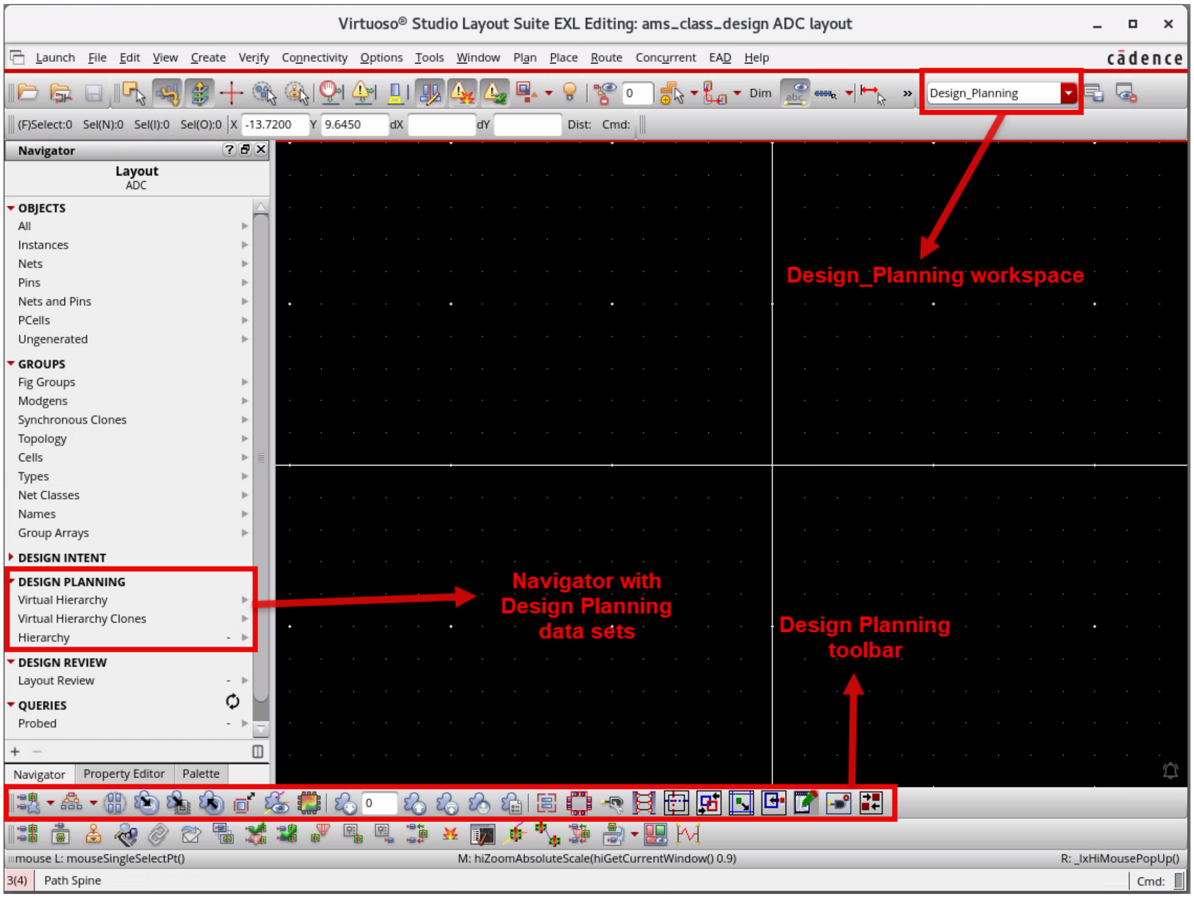
Selecting the *Design_Planning* Workspace in the Layout Canvas

By default, the newly created layout view opens in the *Classic* workspace.

1. From the Workspace Configuration pull-down, select *Design_Planning* to change the workspace.

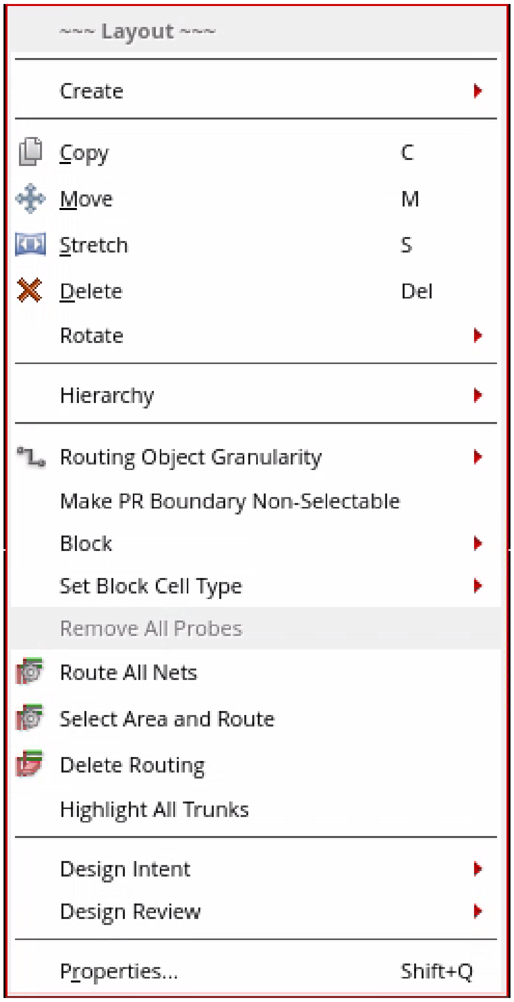


The layout canvas opens with the *Design_Planning* workspace, which enables the Navigator with Design Planning data sets, Palette, & Property Editor assistants, and the Design Planning toolbar, as shown here.



Cadence recommends that all the design planning work be completed using the *Design_Planning* workspace because this is the only workspace that supports all the commands needed for design planning. You are free to customize the workspace to add your favorite toolbars and assistants.

In addition, you can use the Design Planning commands available in the layout context menu. To display the context menu, **right**-click on the layout canvas or use the Navigator assistant. You will see how to use these commands in the subsequent labs.



2. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned:

- About the new Design Planning Environment within Virtuoso that supports design planning.

End of Lab

Module 3: Design Planning

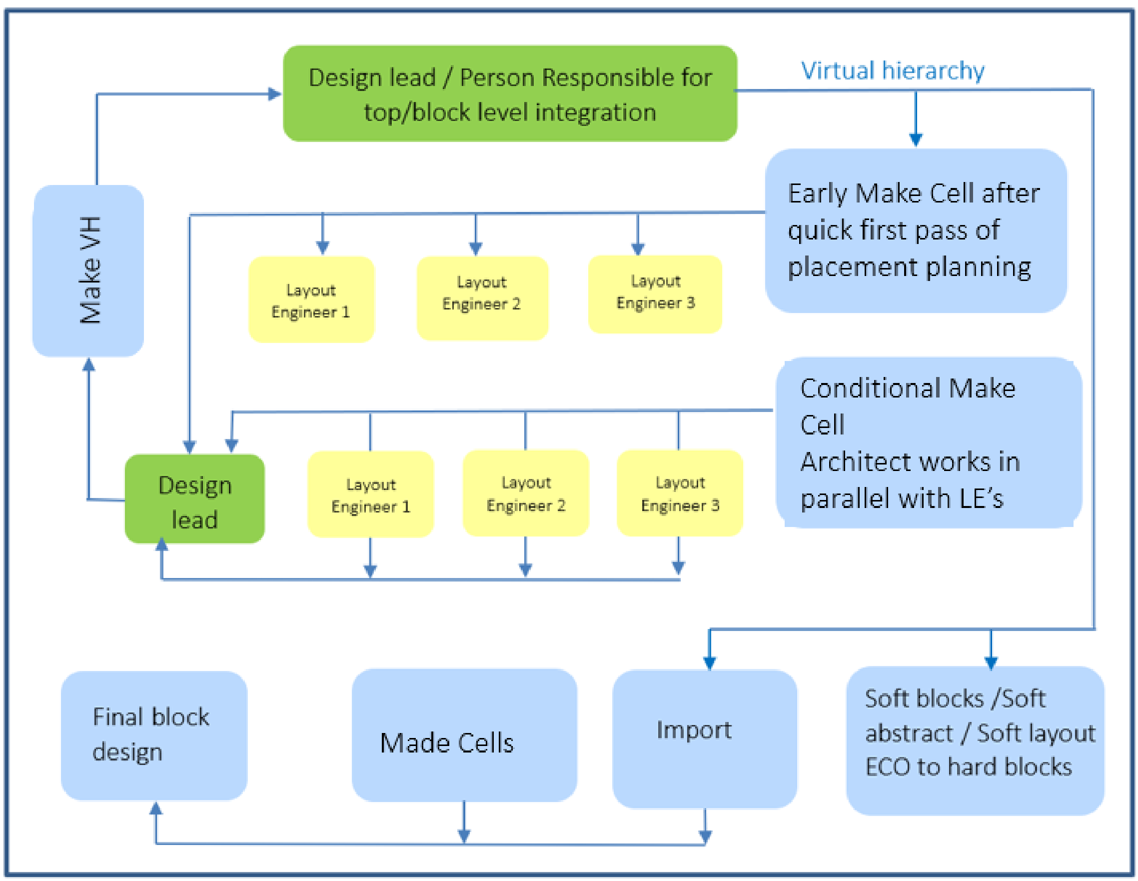
Lab 3-1 Generating the Virtual Hierarchy

Objective: To generate the virtual hierarchy of a design.

The Design Planner introduces a new concept called Virtual Hierarchy. In this lab, you explore how to generate the virtual hierarchy of a design.

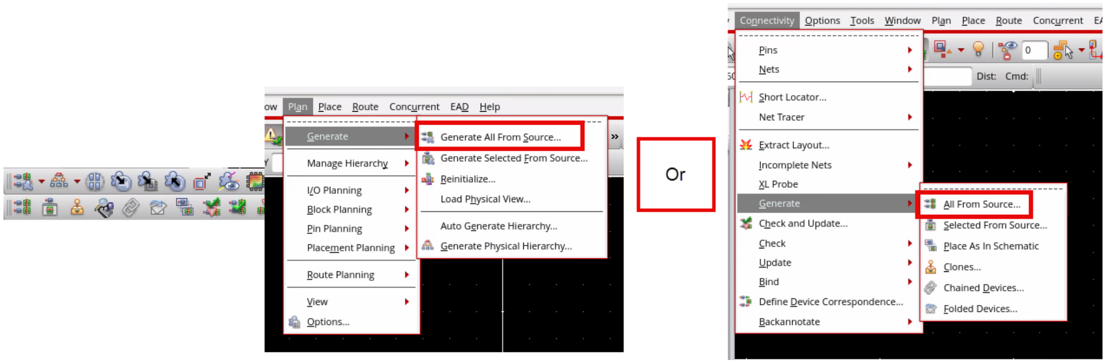
Advantages of the Virtual Hierarchy in the Virtuoso® Design Planner

- Enables better design planning decisions around size and utilization compared to a flat layout.
- Has additional connectivity information that would be hidden in a lower level of hierarchy in a traditional flow.
- Is quickly and easily manipulated, making a design plan available much earlier.



Generating the Layout with Virtual Hierarchy

1. To generate the layout with virtual hierarchy, click the **Generate All From Source** (**GFS**) icon in the Design Planning toolbar or choose the **Plan – Generate – Generate All From Source** menu command or choose the **Connectivity – Generate – All From Source** menu command or click the **Generate All From Source** (GFS) icon in the Layout XL toolbar to launch the Generate Layout form.



The Generate Layout form appears.

2. In the Generate Layout form in the Design Planning group box:

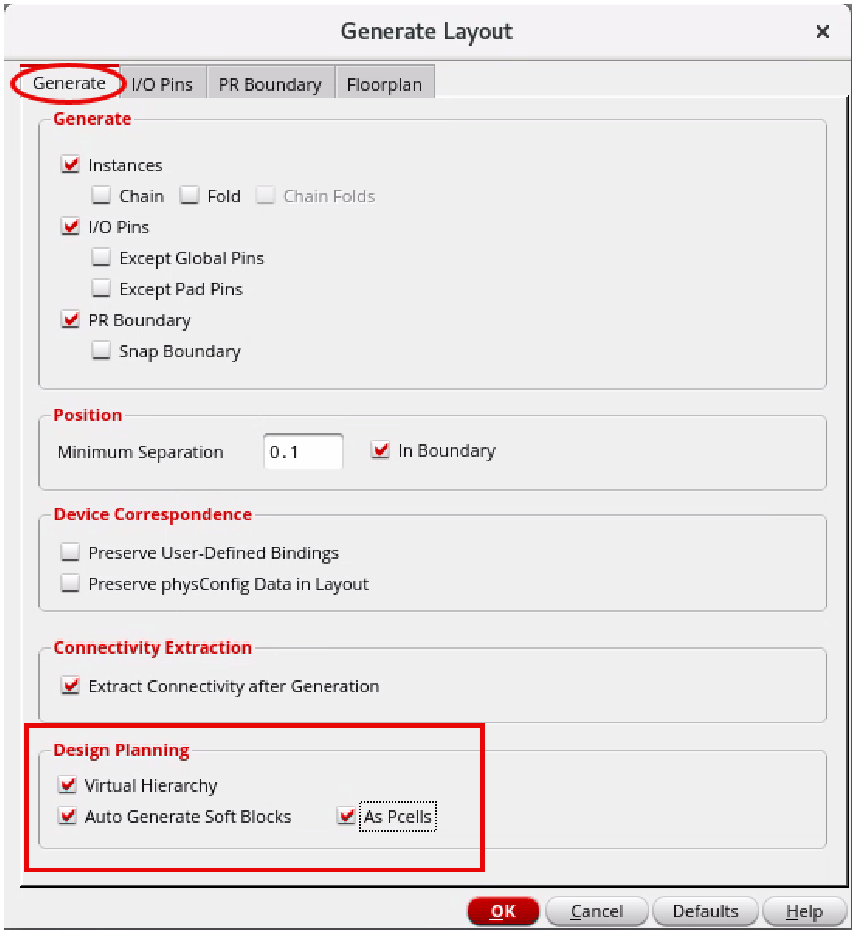
   a. Keep the **Virtual Hierarchy** and **Auto Generate Soft Blocks** options selected.

   b. Keep the other values as default in all the tabs.

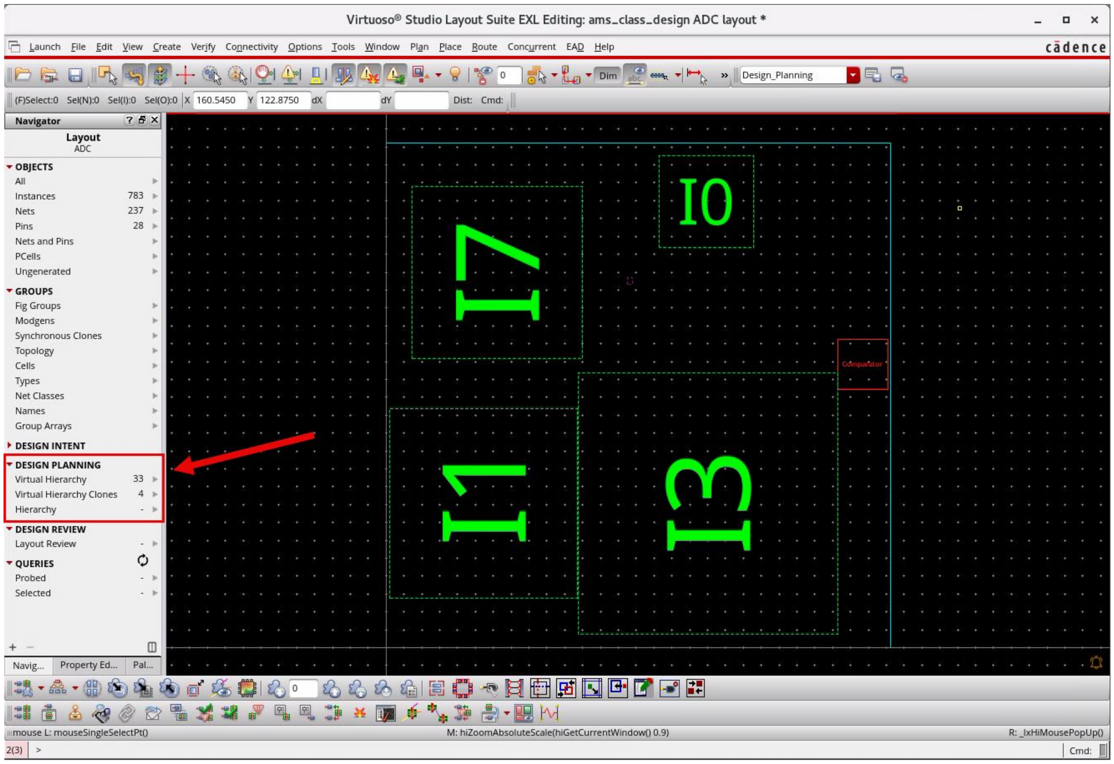
   c. Click **OK** in the form to generate the layout.

   Note: When the **Virtual Hierarchy** option is true, a virtual hierarchy is generated for schematics that have no corresponding layout.

Note: When the **Auto Generate Soft Blocks** option is true, a soft block is created for symbols that do not have corresponding schematics.



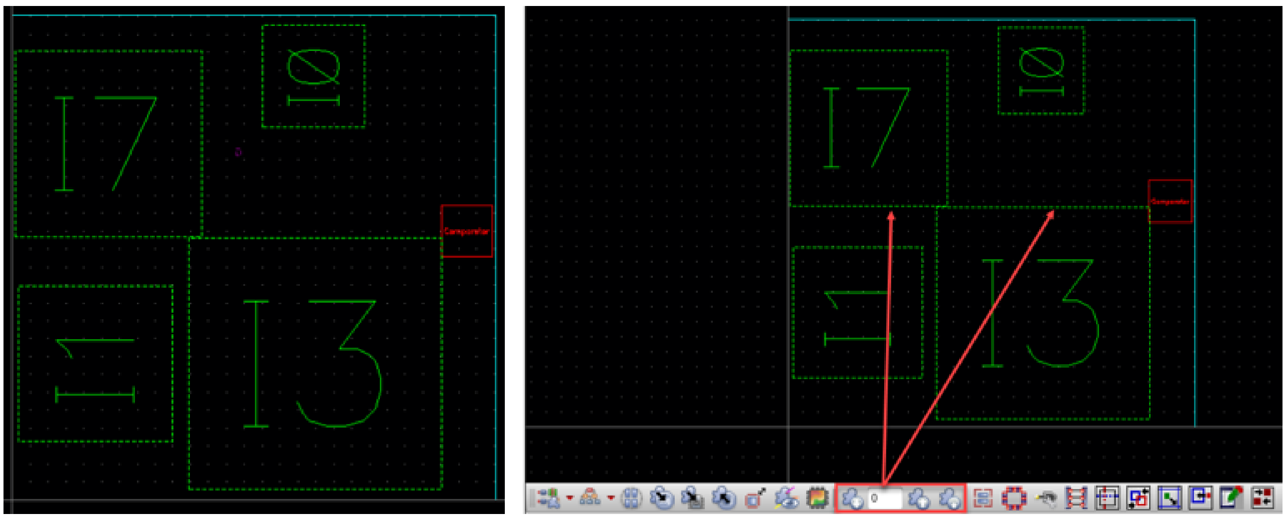
The layout is generated flat (opaque, Display level = 0) containing virtual hierarchies and a soft block, as displayed here.



Note: Here, I0, I1, I3, and I7 are Virtual Hierarchies, I11 is a Virtual Hierarchy Clone, and I2 is a Soft Block.

Identifying the Virtual Hierarchy

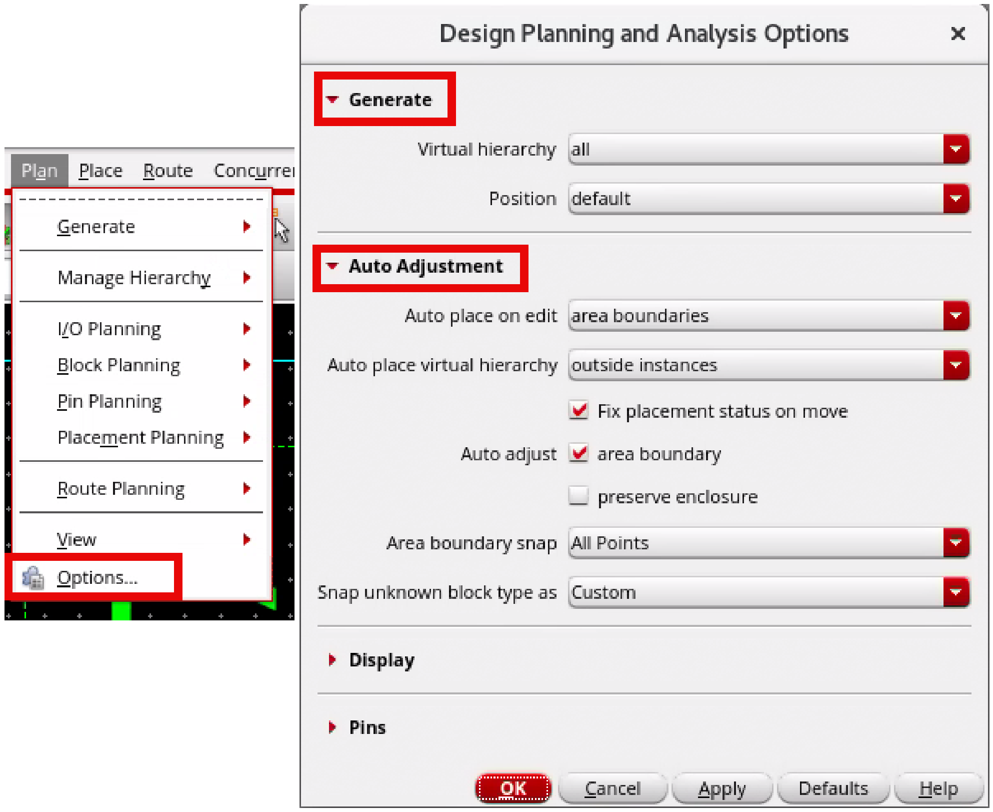
Virtual hierarchies are displayed with a green dashed line to make them stand out from other real hierarchies. By default, the virtual hierarchies are generated opaque with their contents hidden. This can be seen in the Design Planning toolbar, where the display level is set to 0. You have the flexibility to display as much of the hierarchy as is needed. As the virtual hierarchy is flat, you would likely want to hide levels that are not important for the current task. If you can see an object in a virtual hierarchy, you can interact with it, and some commands behave differently depending on the level displayed. This will be explained later in this lab.



Configuring the Design Planning and Analysis Options

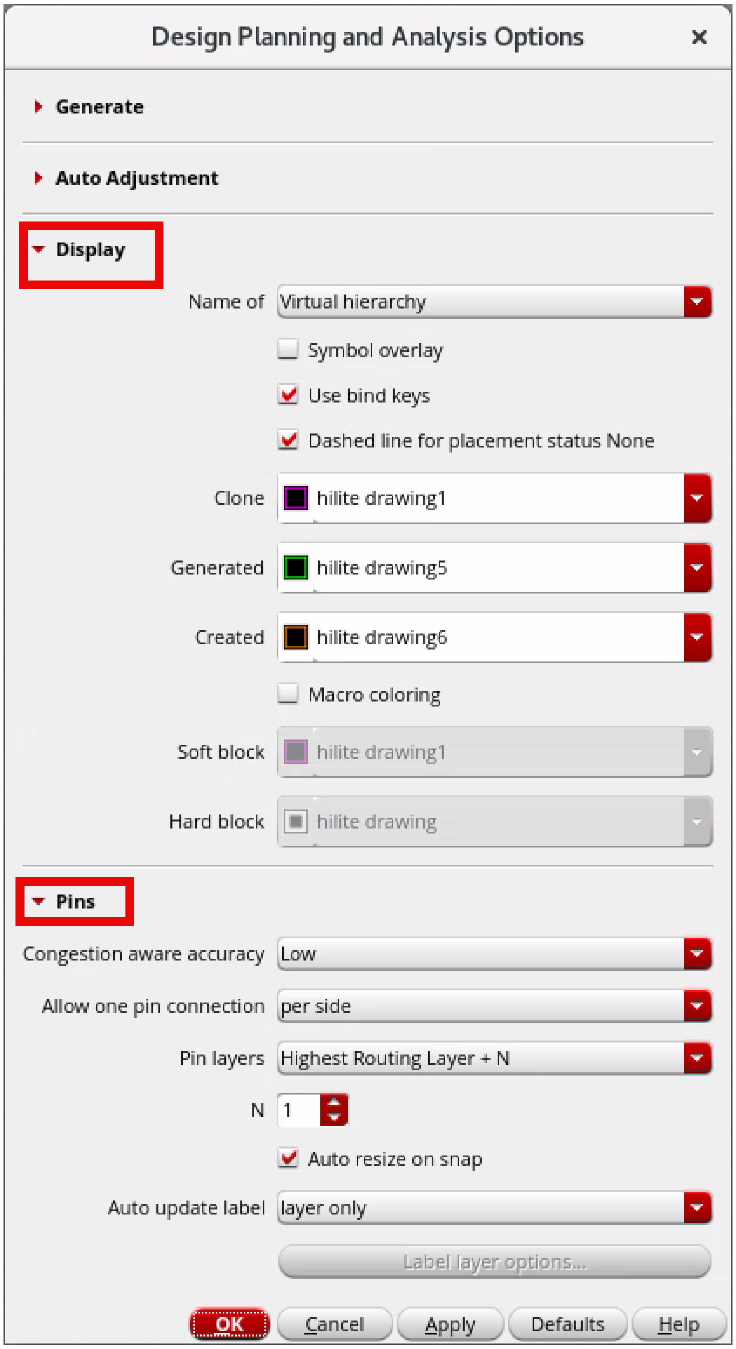
1. From the menu bar, select Plan → Options

2. The **Design Planning and Analysis Options** form appears.



a. **Generate** section - Generates the virtual hierarchy for all levels with default instance positioning.

b. **Auto Adjustment** section - Controls how blocks and boundaries are automatically adjusted, snapped, and repositioned during interactive editing.



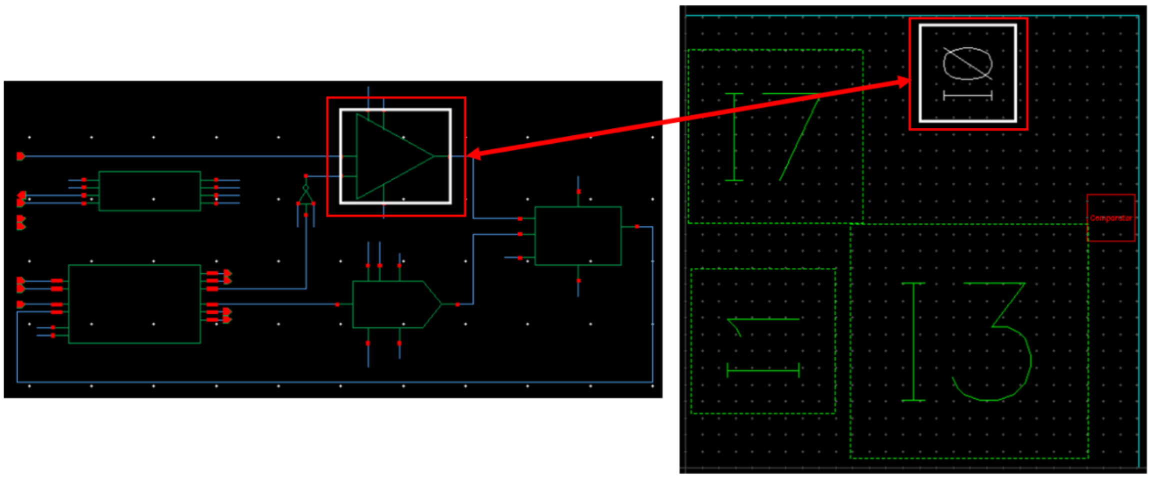
c. **Display** Section - Controls the visual appearance and color-coded highlighting of virtual hierarchy blocks in the layout canvas.

d. **Pins** Section - Controls pin planning behavior, congestion analysis accuracy, and pin layer assignments for routing.

Cross Selection of the Virtual Hierarchies in the Schematic/Layout

1. Select a virtual hierarchy in the layout canvas.

   Selecting a virtual hierarchy in the layout canvas cross selects its counterpart in the schematic showing that the virtual hierarchies are connectivity-compliant and correct by construction, as shown here.



2. Deselect all the selected objects, if any.

3. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned how to:

- Generate the virtual hierarchy of a design.

End of Lab

Lab 3-2 Setting Up the Soft Block Pcells Using CSV in Design Planning

Objective: To set up and generate soft block P-cells using a CSV file within the Design Planning environment in Virtuoso.

In this lab, you explore how to configure soft block P-cells using an Area Estimation CSV file in the Design Planning Environment within Virtuoso. You learn to define block dimensions through the CSV, enable the **As P-cells** option in the Generate Layout form, and regenerate the layout to apply the updated soft block sizes.

Advantages of Soft Block P-cells Using CSV in Design Planning

- Explore the **Design Planning** environment in the Virtuoso Layout Suite used for early-stage chip floor planning.
- Understand how a **Soft Block** is created as a placeholder when a block has a symbol but no schematic or layout.
- Learn how **Soft Block P-cells** act as parameterized versions of Soft Blocks, enabling dynamic resizing and regeneration.
- Use an external **CSV file** to drive block dimensions instead of manually editing each block in the GUI.
- Enable the **As P-cells** option in the Generate Layout form along with Virtual Hierarchy and Auto-Generate Soft Blocks.
- Generate and regenerate the layout to verify that Soft Block dimensions update dynamically based on CSV input.
- Identify that Design Planning functionality is available in the **VLS EXL** and **MXL** tiers of the Virtuoso Layout Suite.
- Apply Soft Block P-cells during early floor planning to estimate block sizes and efficiently plan the overall chip area.

Providing CSV File for Area Estimation

1. Open the **Terminal** from the Linux desktop.
2. Navigate to your working directory:

```
cd ./VLPT9_IC_25_1
```

3. Create a new CSV file named **data.csv** using the nano editor:

```
nano data.csv
```

4. Enter the block dimensions in the following format (header row followed by block entries):

```
lib,cell,width,height
ams_class_design,Comparator,70,70
ams_class_design,SnH,80,80
ams_class_design,common,50,50
```

- Meaning of Each Column:

    a. lib → Library name

    b. cell → Block (cell) name

    c. width → Block width

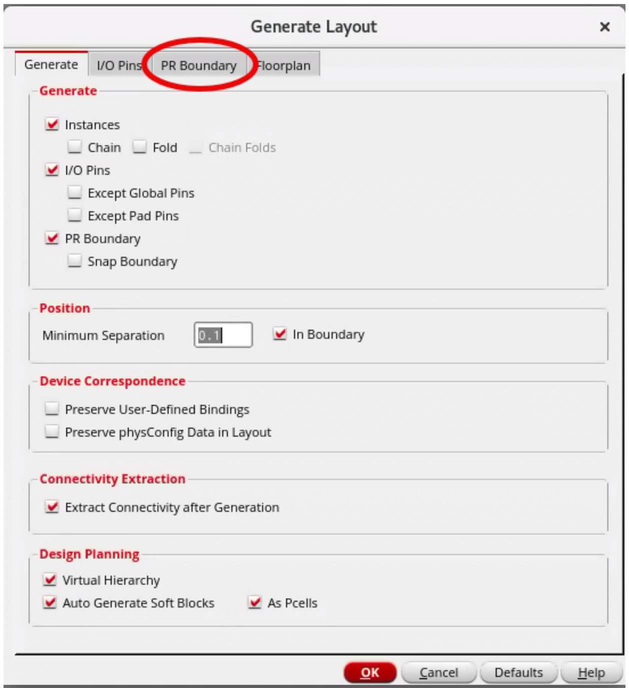
    d. height → Block height

5. Save the file by pressing **Ctrl + O**, press Enter to confirm, and exit using **Ctrl + X.**

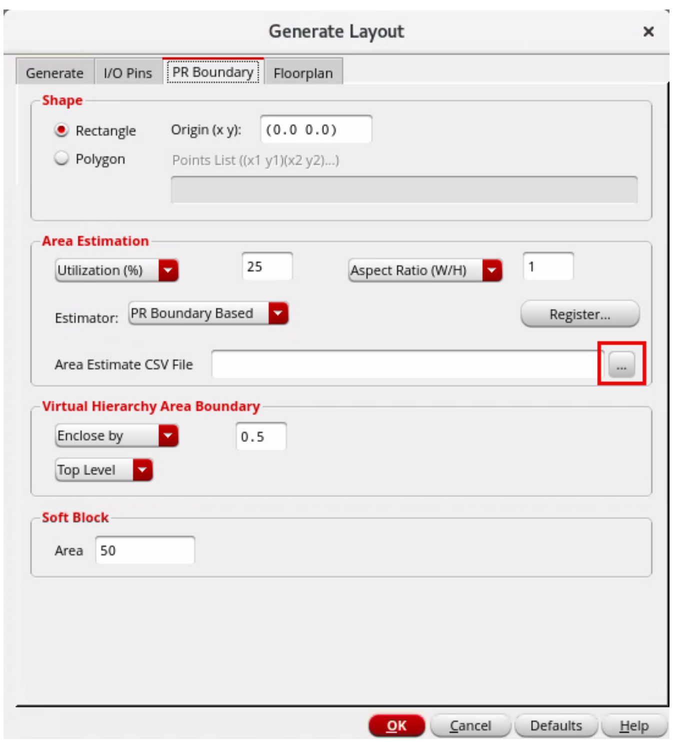
Loading the CSV File in the Generate Layout Form

1. In the Virtuoso Layout EXL window, open the **Generate Layout** form.

2. Go to the **PR Boundary** tab.



3. In the Area Estimate CSV File field, click the three-dot (…) button on the right.

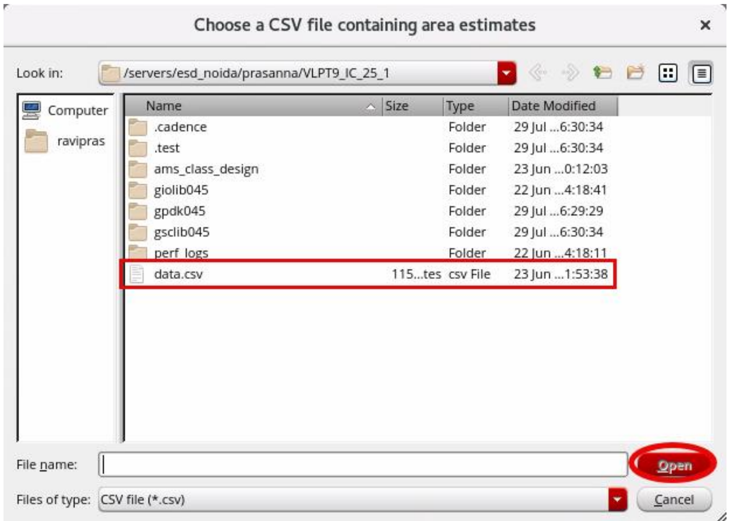


4. In the Choose a CSV file containing area estimates dialog:

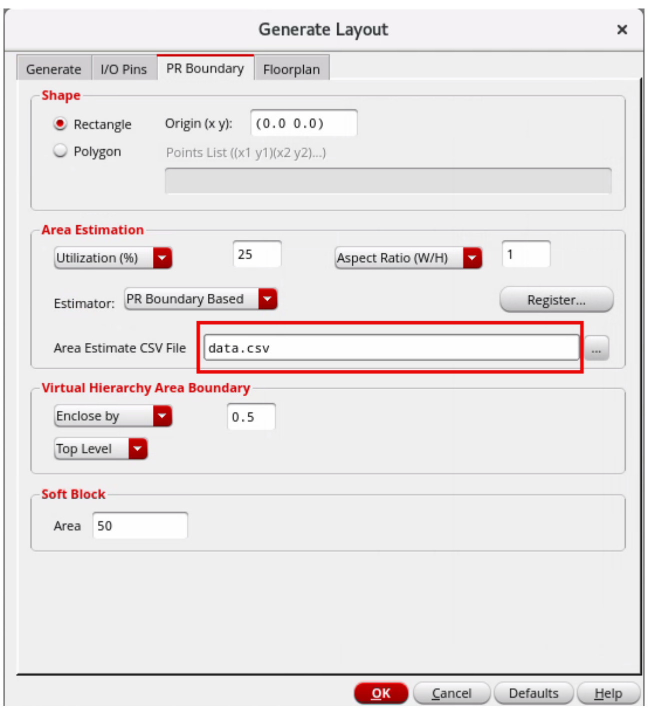
5. Navigate to your working directory.

6. Select the **data.csv** file created earlier.

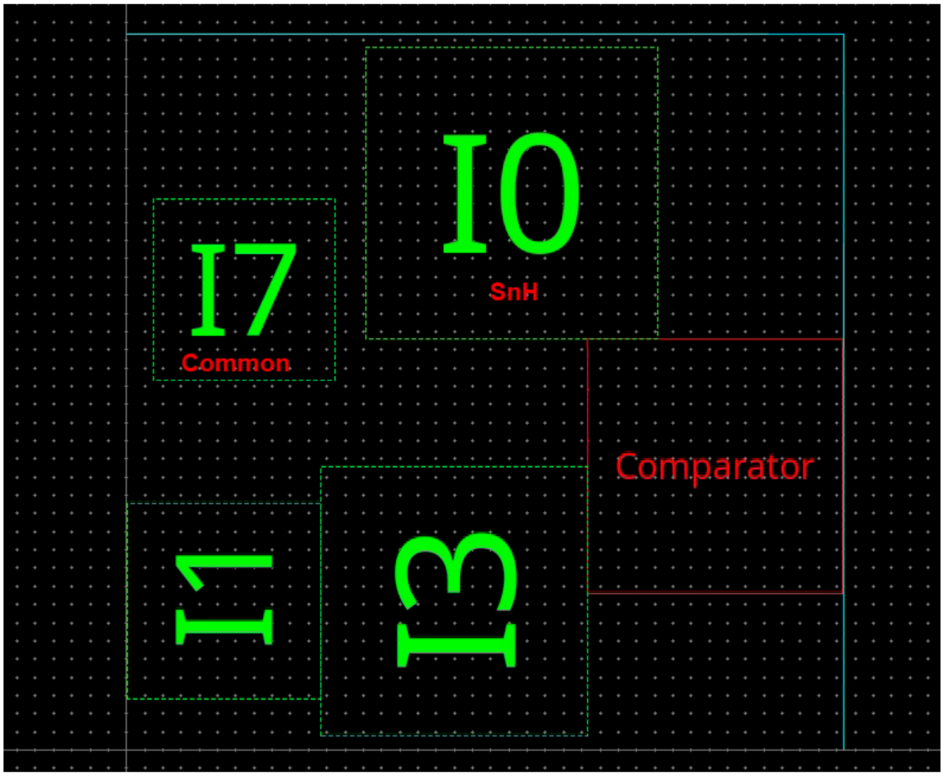
7. Click **Open**.



8. Verify that the file name **data.csv** is displayed in the **Area Estimate CSV File** field.



9. Click **OK** in the Generate Layout form to generate the layout.



Verifying the Generated Soft Block Pcells

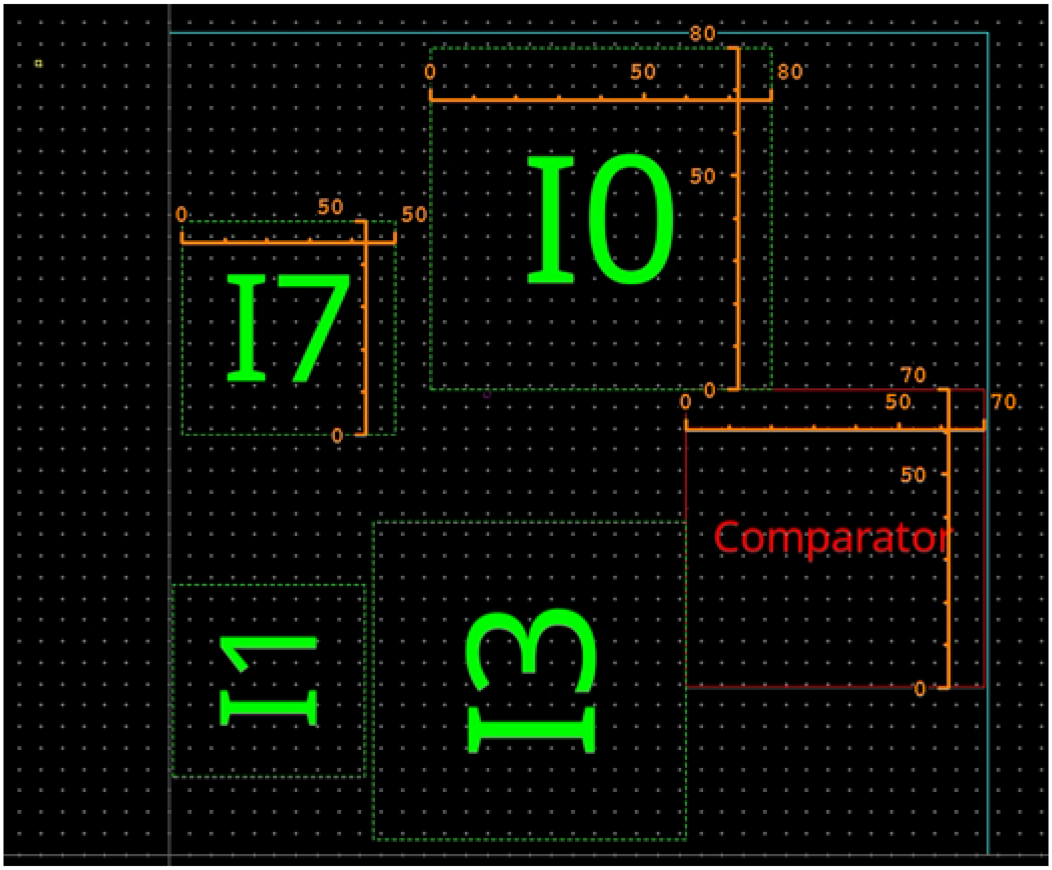
1. Observe the layout canvas — the soft blocks are generated with dimensions defined in the CSV file.

2. Verify the block sizes against the values entered in **data.csv**:

    a. **Comparator** → 70 × 70

    b. **SnH (I0)** → 80 × 80

c. **common blocks (I7)** → 50 × 50



Regenerating the Layout Using a Revised CSV File

When block dimensions change, the layout can be updated by simply loading a revised CSV file — there is no need to delete the existing layout. The tool automatically regenerates all Soft Block P-cells with the new dimensions from the updated file.

1. Open the **Terminal** and navigate to your working directory:

   ```
   cd ./VLPT9_IC_25_1
   ```

2. In the Virtuoso Layout EXL window, open the Generate Layout form and navigate to the PR Boundary tab.

3. Create a new CSV file named **data_1.csv** using the nano editor

   ```
   nano data_1.csv
   ```

4. Enter the updated block dimensions in the same format as before, with the new values:

   ```
   lib,cell,width,height
   ```

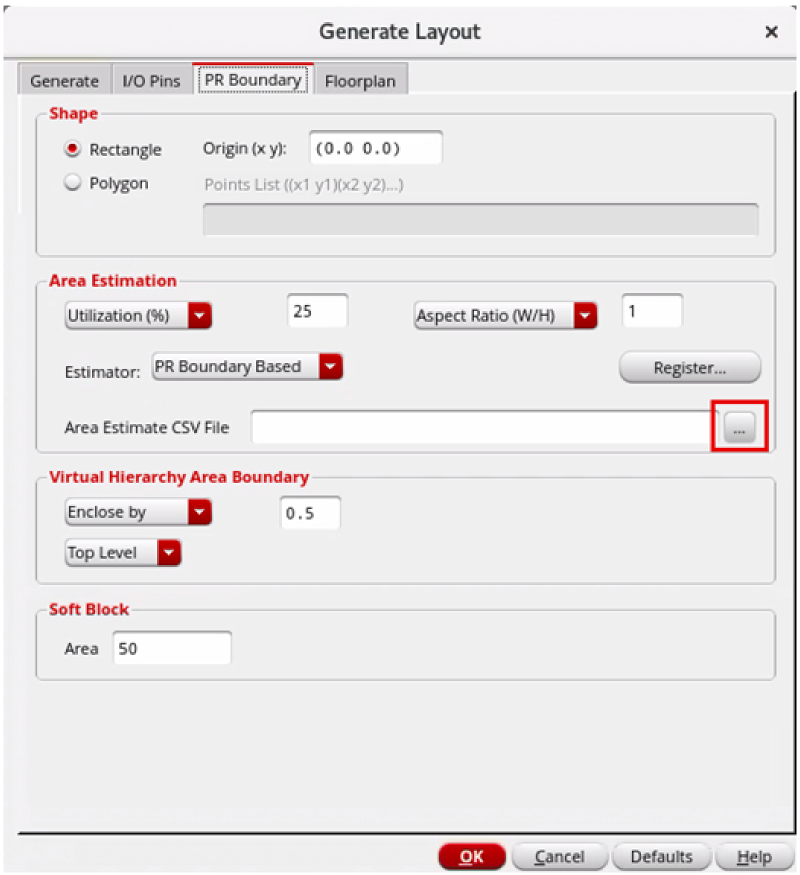


```
ams_class_design,Comparator,100,100
ams_class_design,SnH,120,120
ams_class_design,common,80,80
```

5. Save the file by pressing **Ctrl + O**, press **Enter** to confirm, and exit using **Ctrl + X**.

Loading the Revised CSV File in the Generate Layout Form

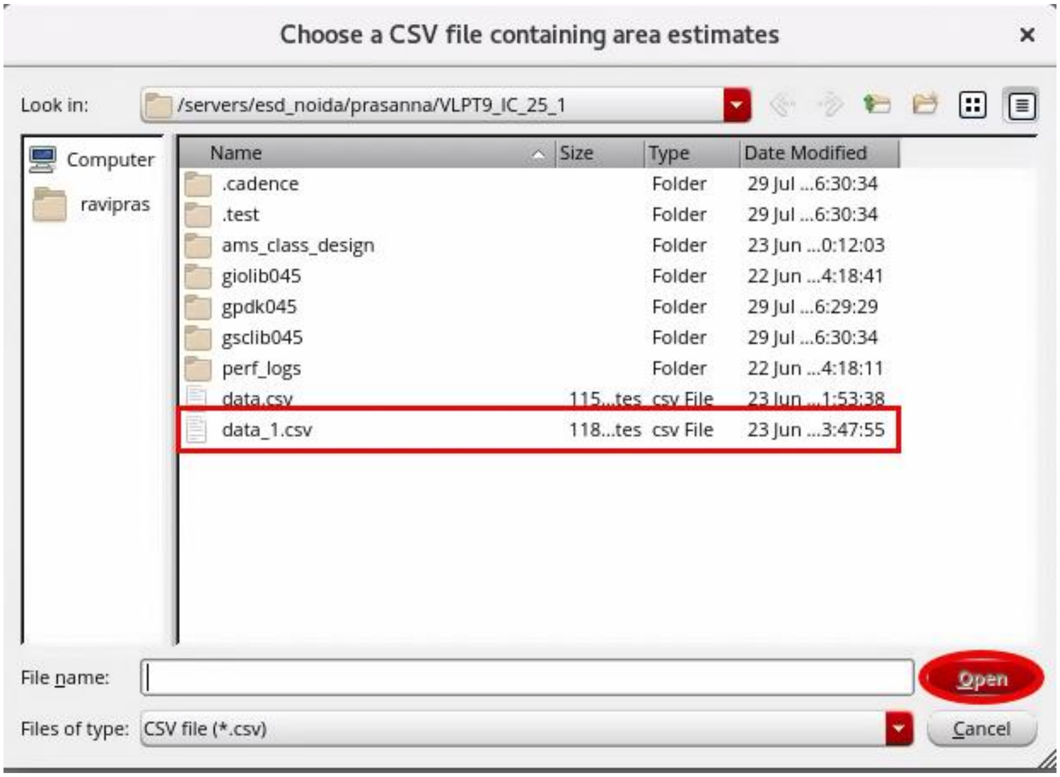
1. In the Area Estimation section, select the browse button (three-dot …) next to the Area Estimate CSV File field.



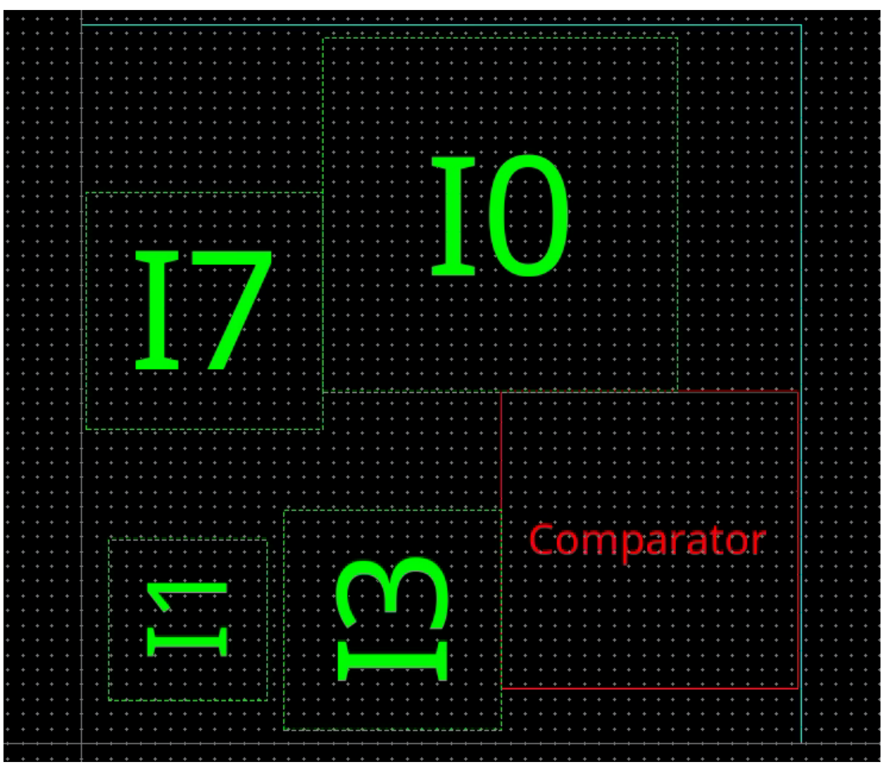
2. In the Choose a CSV file containing area estimates dialog:

   a. Select **data_1.csv** from the file browser.

b. Select **Open**, then click **OK** in the Generate Layout form.



3. Observe the layout canvas — the Soft Block P-cells are automatically regenerated with the updated dimensions from **data_1.csv**, without deleting or recreating any existing block.



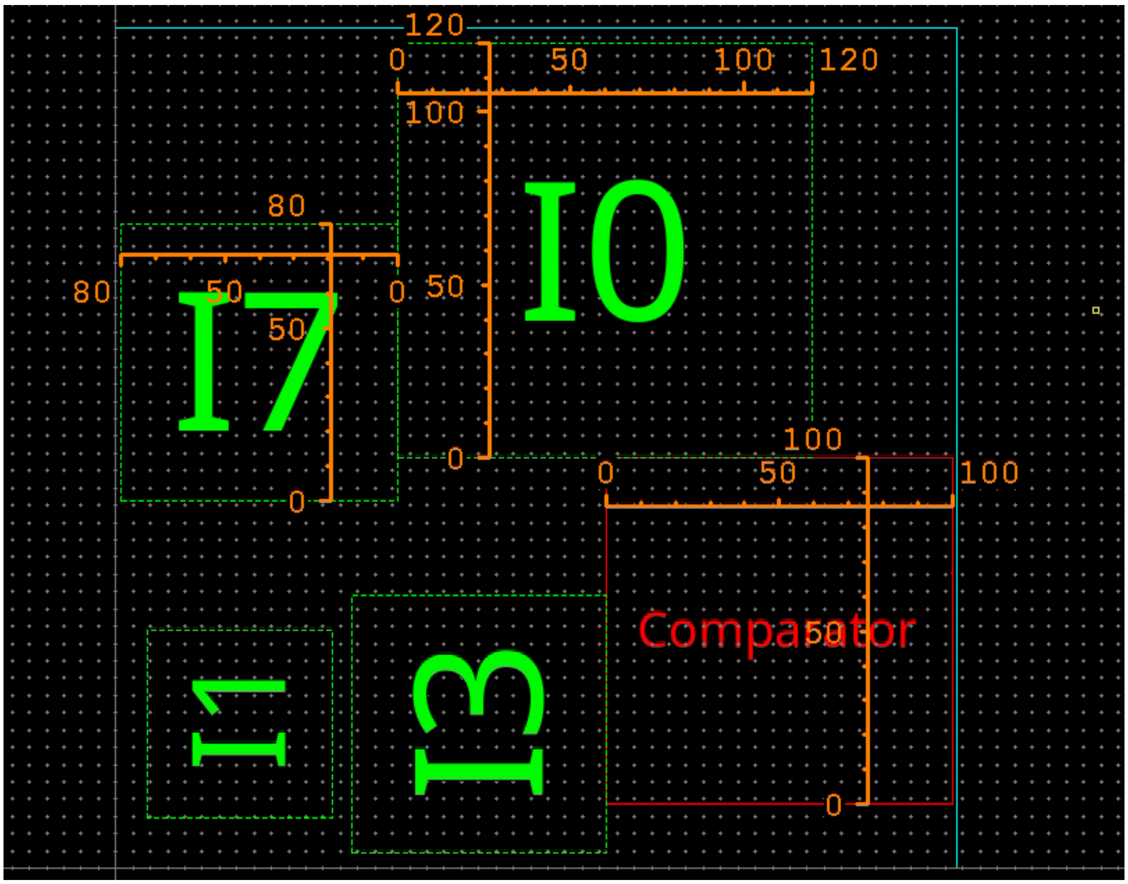
Verifying Hierarchy, Connectivity, and PR Boundary Adjusting Correctly to Reflect the New Block Sizes

1. Verify the block sizes against the values entered in **data_1.csv**:

   a. **Comparator** → 100 × 100

   b. **SnH (I0)** → 120 × 120

   c. **common blocks (I7, I1, I3)** → 80 × 80



End of Lab

Lab 3-3 Accessing the Design Planning Data Sets in the Navigator Assistant

Objective: To access the design planning data sets in the Navigator assistant.

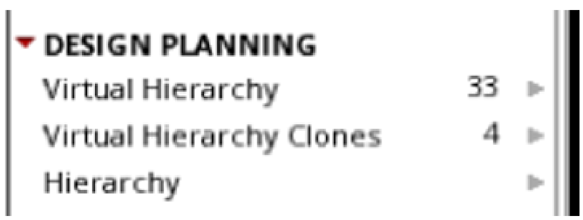
In this lab, you explore how to access the Design Planning data sets in the Navigator assistant.

Identifying the Design Planning Data Sets in the Navigator Assistant

In Layout EXL, the Navigator assistant supports three additional data sets for displaying the generated virtual hierarchies and related components: *Virtual Hierarchy*, *Virtual Hierarchy Clones*, and *Hierarchy*.

In Layout XL, in the Navigator assistant, the counts under the Design Planning data subset show that virtual hierarchies and virtual hierarchy clones have been created. The *Hierarchy* subset does not display a count until you expand it for performance reasons.

Each of these pre-defined data sets, as displayed here, is available under the *DESIGN PLANNING* category in the Navigator assistant.



Accessing the Virtual Hierarchies

The virtual hierarchies generated in the layout are categorized under the *Virtual Hierarchy* data set in the Navigator assistant.

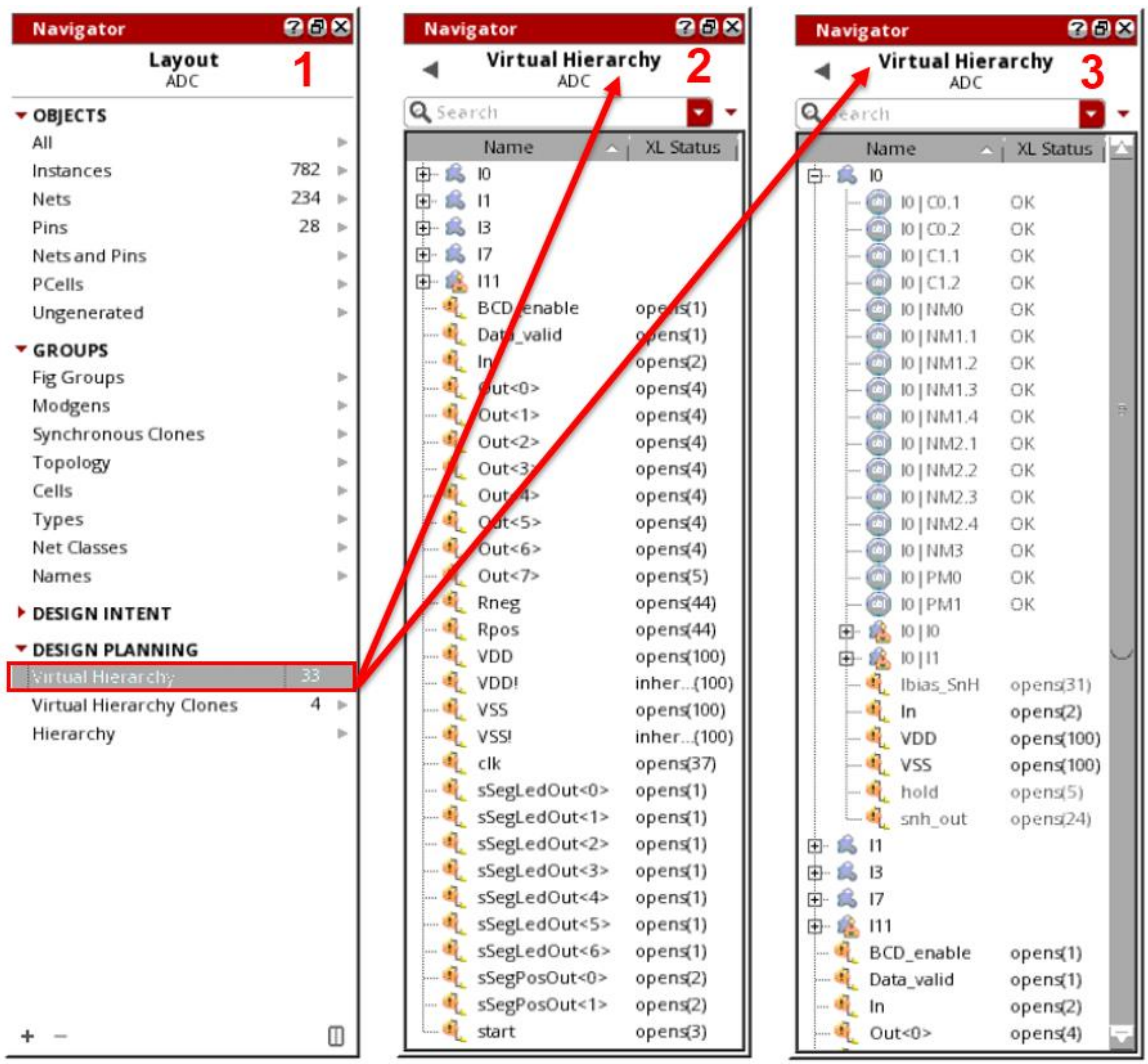
To access the generated virtual hierarchies and interface nets in the layout canvas:

1. In the Navigator assistant, in the Design Planning category, select the **Virtual Hierarchy** data set, as shown in graphic 1.

   The Navigator tree updates to display the available virtual hierarchies and the associated instances and nets.

   The virtual hierarchy is represented using an amoeba-shaped icon. 

Note: If the virtual hierarchy is created as a group, it is represented in the Navigator using the group icon.



2. As shown in the above graphic 2, you can see that there are five top-level virtual hierarchies that have their own icon to help with identification. Top-level nets are also shown in the *Virtual Hierarchy* view.

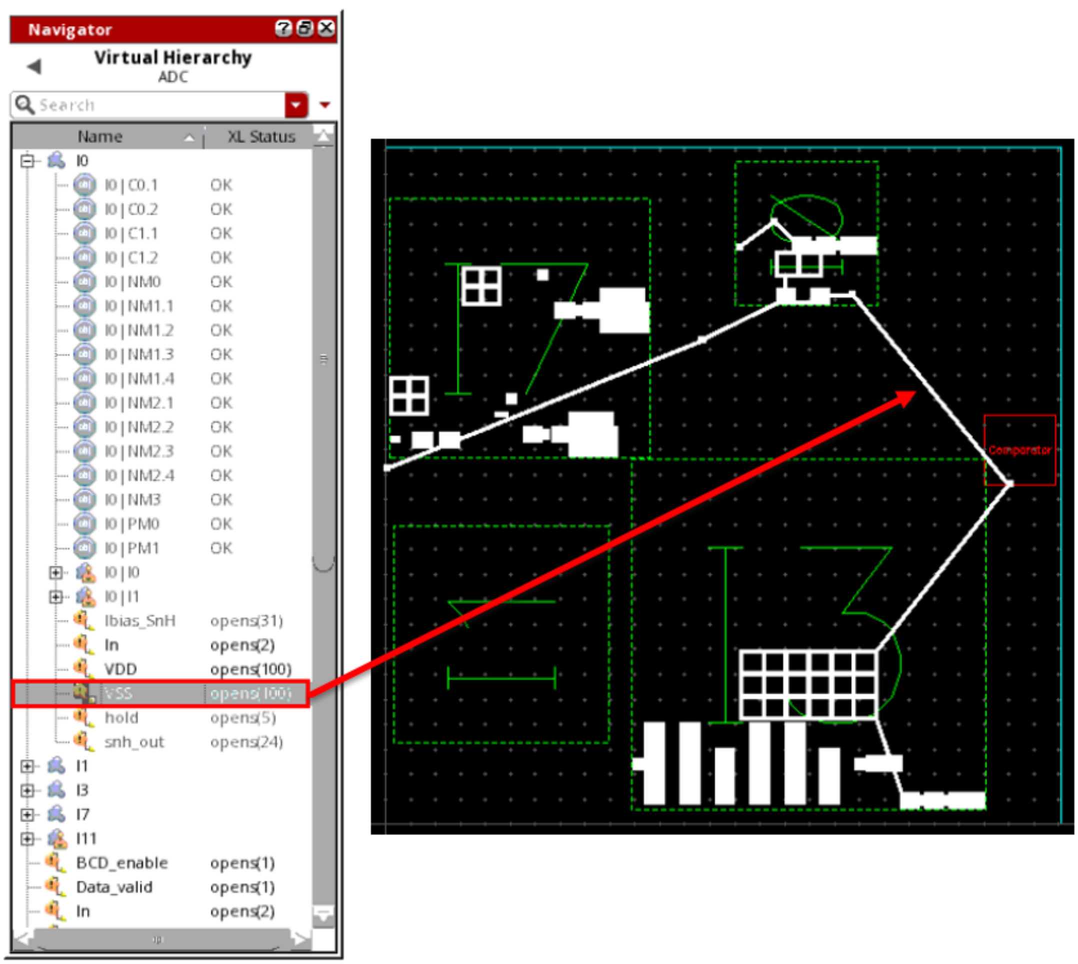
3. As shown in the above graphic 3, virtual hierarchies can be further expanded to show their contents. Here **I0** is expanded. The interface nets are shown at each level. This allows you to easily find the connectivity between the virtual hierarchies. In addition, the Navigator assistant displays the *XL Status* of the virtual hierarchy components.

   Note: A virtual hierarchy can also contain an instance (or more) of another virtual hierarchy as one of the components. Moving such a virtual hierarchy around the layout canvas will display flightlines on the canvas. However, if the virtual hierarchy you move is opaque, which means it has no contents, no flightline will be generated to the virtual hierarchy.

Probing the Interface Nets in the Virtual Hierarchies

1. Select an interface net, for example, **VSS**, associated with a virtual hierarchy in the Navigator tree.

   Layout EXL creates a probe highlighting all the associated instances and nets in the layout canvas, as displayed here.

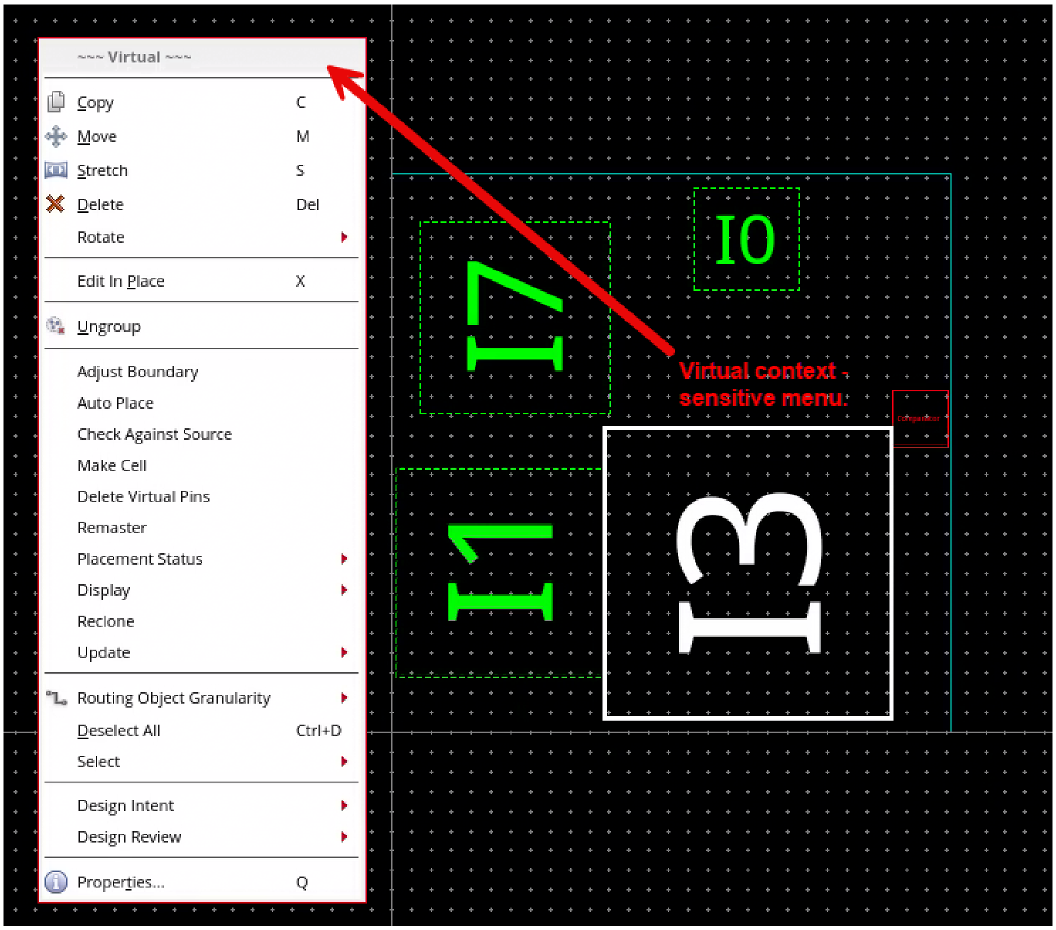


2. Deselect the selected net **VSS**.

Using the Context-Sensitive Menu of the Virtual Hierarchies

1. (Optional Exercise) **Right**-click a virtual hierarchy, for example, **I3**, in the Navigator tree or the layout canvas to perform any of the operations supported through the context menu.

   Note: When invoked from the Navigator assistant, the *Virtual* context-sensitive menu displays an additional command: *Show Content*.



Accessing the Virtual Hierarchy Clones

The virtual hierarchy clones generated in the layout are categorized under the *Virtual Hierarchy Clones* data set in the Navigator assistant.

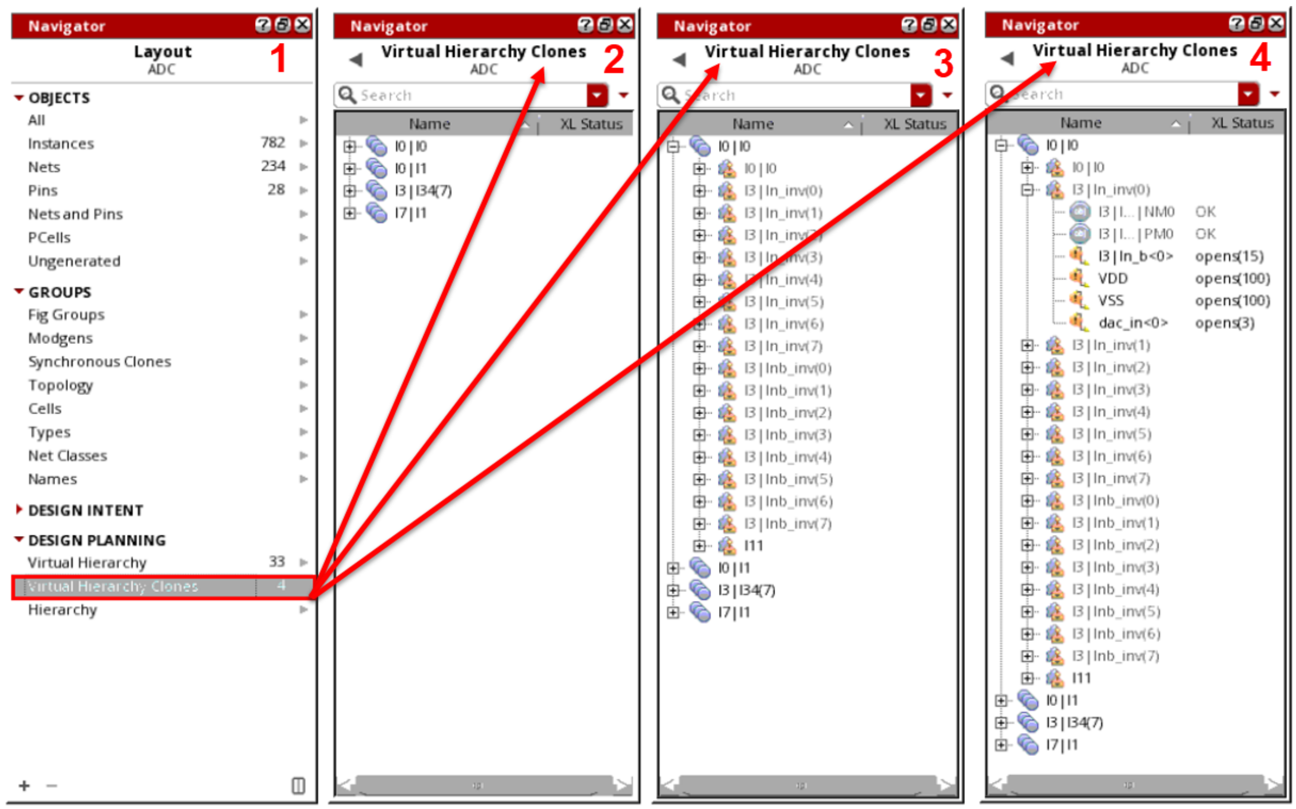
To access the generated virtual hierarchy clones in the layout canvas:

1. In the Navigator assistant, in the Design Planning category, select the **Virtual Hierarchy Clones** data set, as shown in graphic 1.

   The Navigator tree updates to display the available virtual hierarchy clone families.

The virtual hierarchy clone family is represented using the icon shown here in the Navigator assistant.

The corresponding virtual hierarchy clones are represented using the icon shown here in the Navigator assistant.



2. As shown in the above graphic 2, you can see that the view expands, showing four virtual hierarchy clone families.

   These are automatically generated and are given the name of the first clone in the family. Clones are created where virtual hierarchies have the same schematic master. If you are creating layout views, these would be instances of the same layout master. In Design Planner, you use clones to keep the virtual hierarchies identical.

3. As shown in the above graphic 3, click the (**+**) button adjacent to the virtual hierarchy clone family icon (**I0|I0**) in the Navigator tree to view the corresponding clone family members.

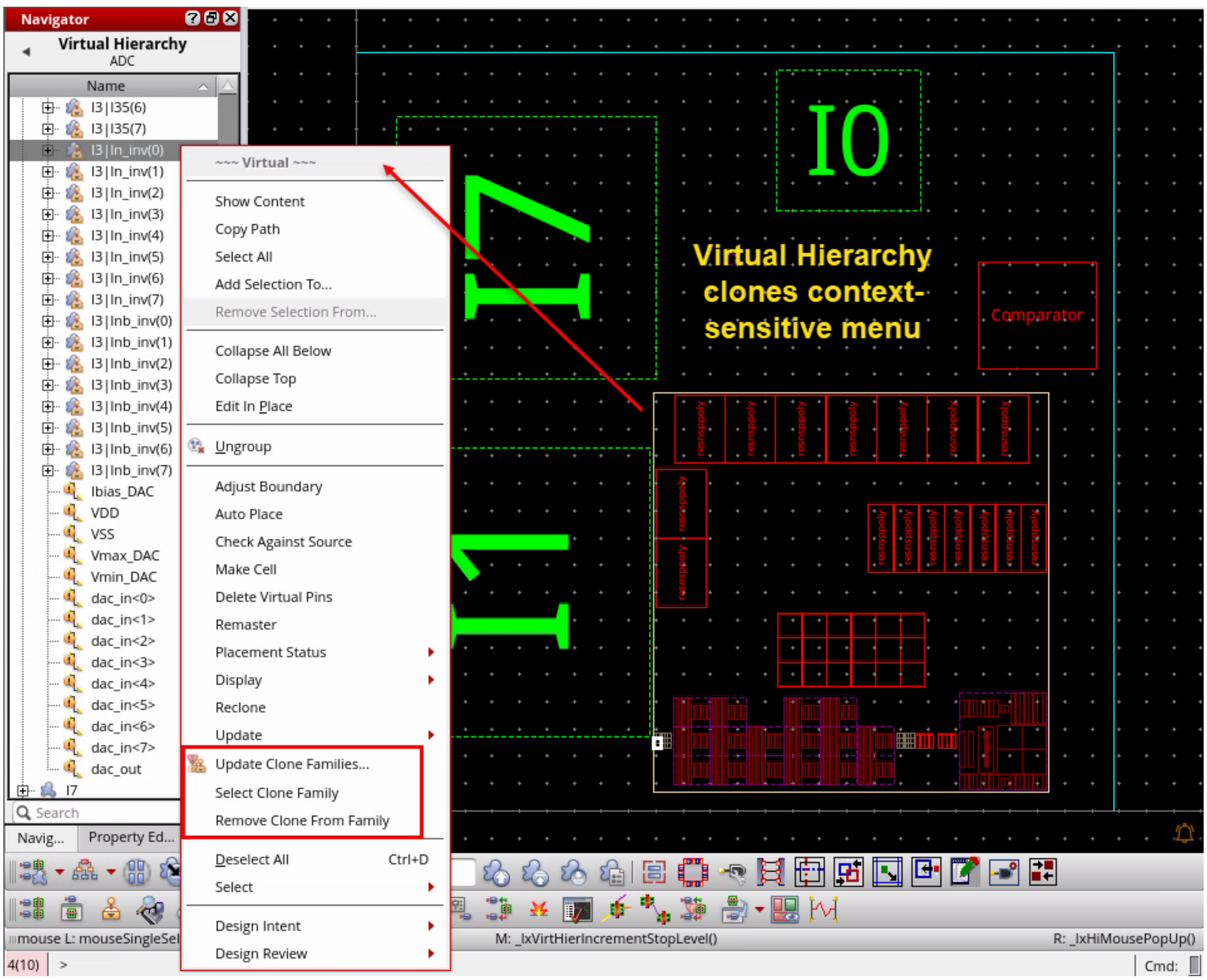
4. As shown in the above graphic 4, click the (**+**) button adjacent to the virtual hierarchy clone icon **I3|In_inv(0)** in the Navigator tree to view the objects inside the virtual hierarchy clone.

Using the Context-Sensitive Menu of the Virtual Hierarchy Clones

1. (Optional Exercise) **Right**-click a virtual hierarchy clone, for example, **I3|In_inv(0)**, in the Navigator tree to perform any operations supported through the context menu.

   Note: You may have to increase the *Display Depth* to **1** in the Design Planning toolbar in the layout canvas for this clone to see the clone commands. Setting Display Depth will be covered in detail in **Lab 3-3**.

   A virtual hierarchy clone supports three additional commands in the context-sensitive menu: *Update Clone Families*, *Select Clone Family*, and *Remove Clone From Family*.



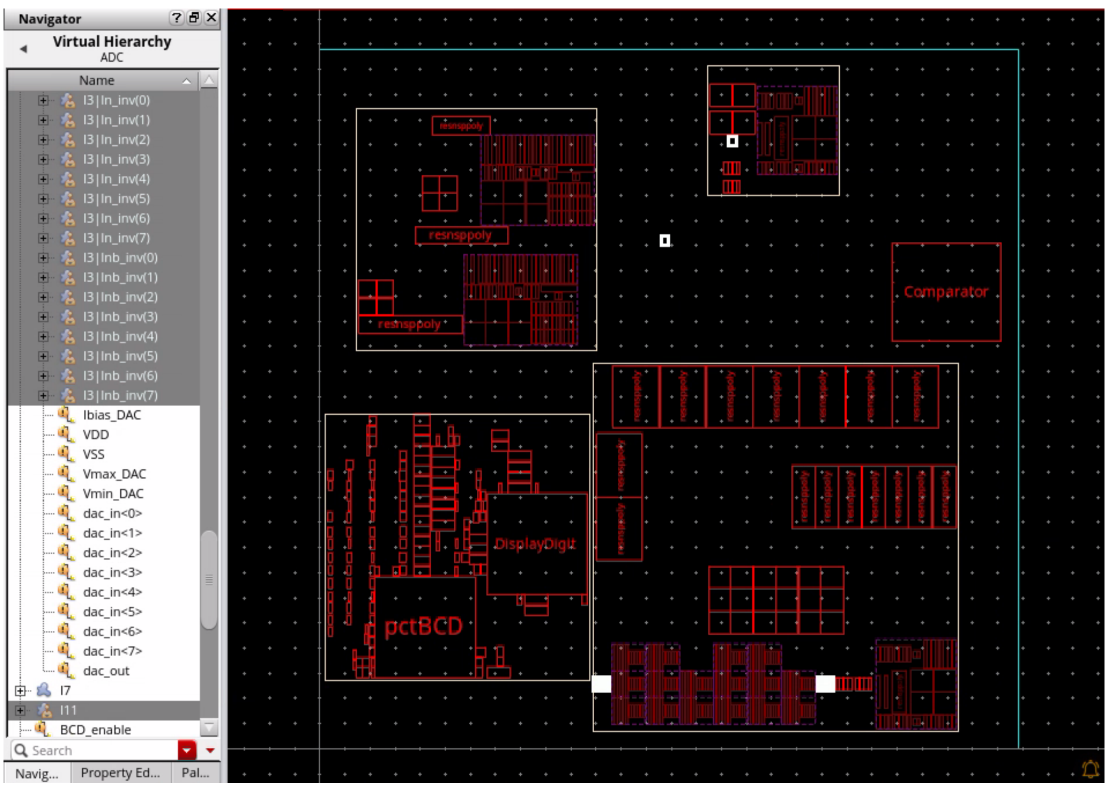
Cross Selection of the Contents in the Virtual Hierarchy Clone Families in Navigator/Layout

1. In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to make all the virtual hierarchies opaque.

2. In the layout canvas, in the Design Planning toolbar, set *Increase Display Depth* to **2** with no virtual hierarchies selected.

   a. Click the (**+**) button next to **I0|I0** in the Navigator assistant to expand the clone family.

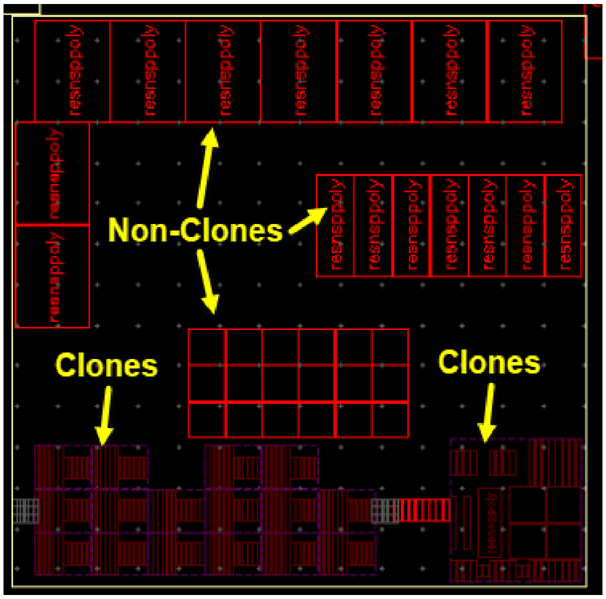
   b. Select the contents of this clone family in the Navigator; they will cross select in the canvas, as displayed here.



As you can see, clones do not need to be in the same virtual hierarchy. The only condition that must be fulfilled for clones to be created is the same schematic master.

Non-Clones and Clones Objects in the Virtuoso Design Planner

Virtual hierarchy clones are controlled by the cloning infrastructure. You **cannot** select their contents without doing an **Edit In Place** first. Initially, the clones appear dimmed on the canvas, which helps with their identification, as shown here.



Accessing the Hierarchical Objects

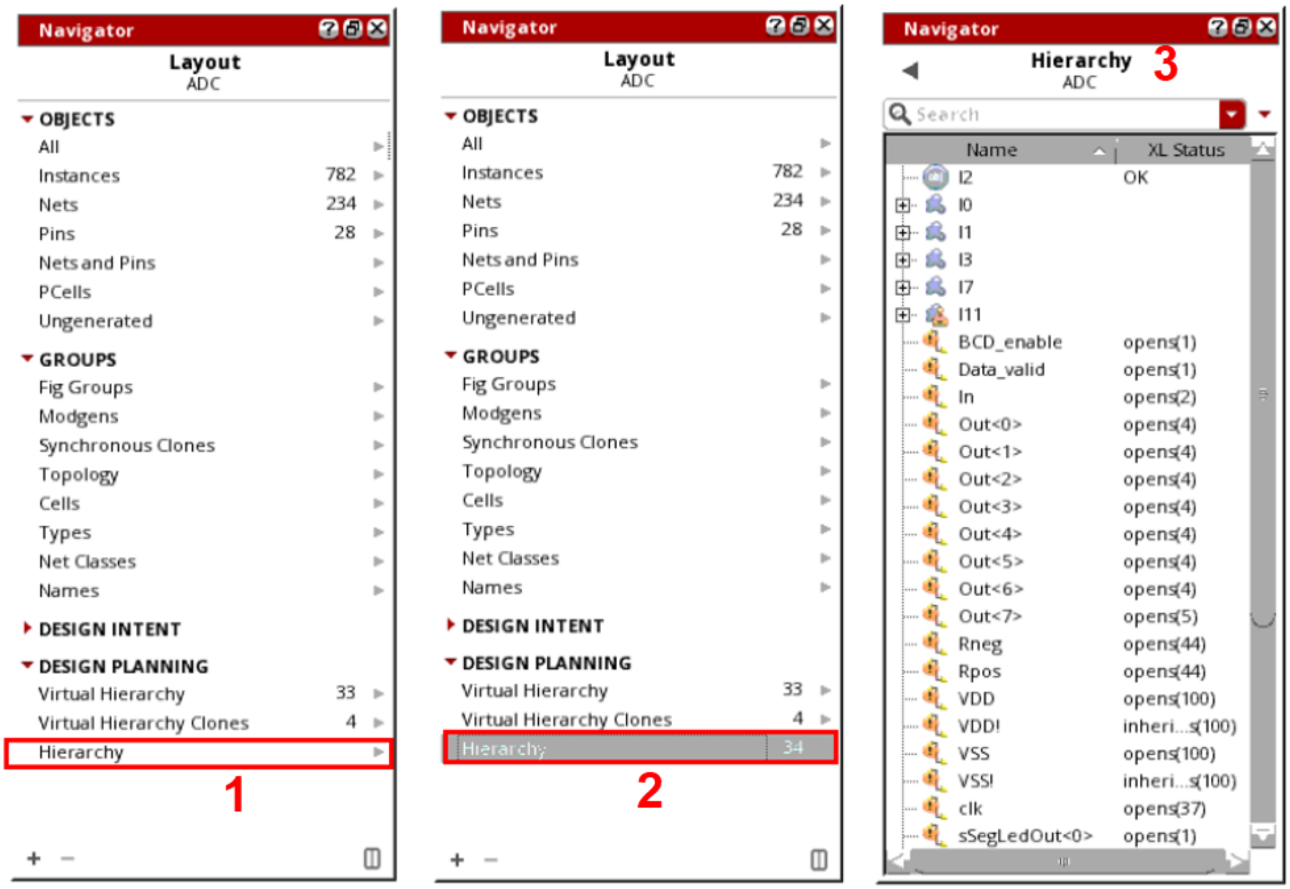
The hierarchical objects associated with the generated virtual hierarchy are categorized under the *Hierarchy* data set in the Navigator assistant.

To access the generated hierarchy in the layout canvas:

1. In the Navigator assistant, in the Design Planning category, click the **Hierarchy** data set, as shown in graphic 1.

   Note: As you have seen earlier, the *Hierarchy* subset does not display a count until you expand it for performance reasons.

2. Click the **back**-arrow button to see that, in the Navigator assistant, the counts under the Design Planning data subset show that hierarchies have been created, as shown in graphic 2.

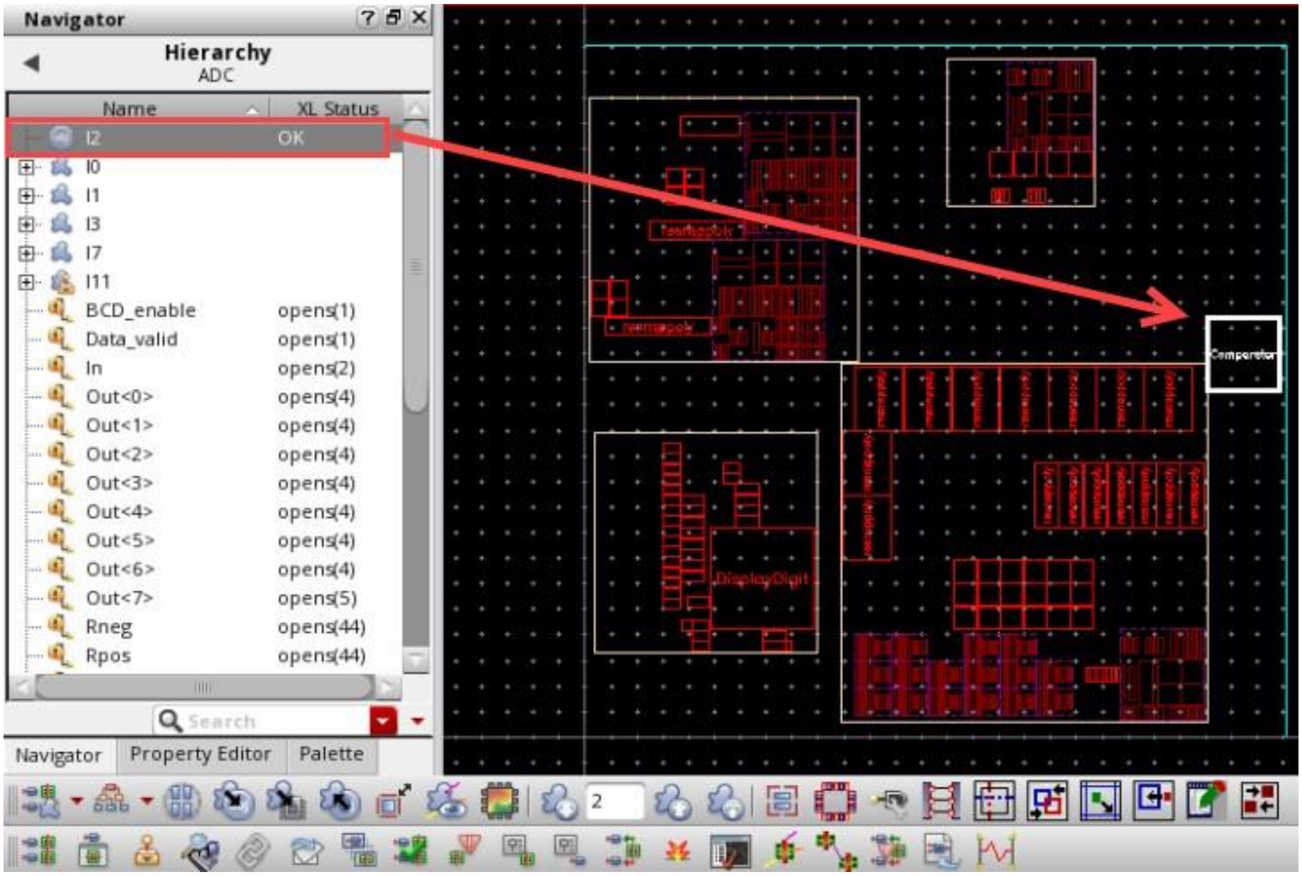


3. Click the **Hierarchy** data set again, as shown in graphic 3.

   The Navigator tree updates to display the available hierarchical instances.

   The *Hierarchy* data set shows both virtual and real hierarchies. You can see that this category has one more entry than the virtual hierarchy – a total of six. This is because **I2** has been created as a soft block, as shown here. The Navigator assistant displays the *XL Status* of the hierarchical objects.

Generated Layout with Depth Level Set as 2



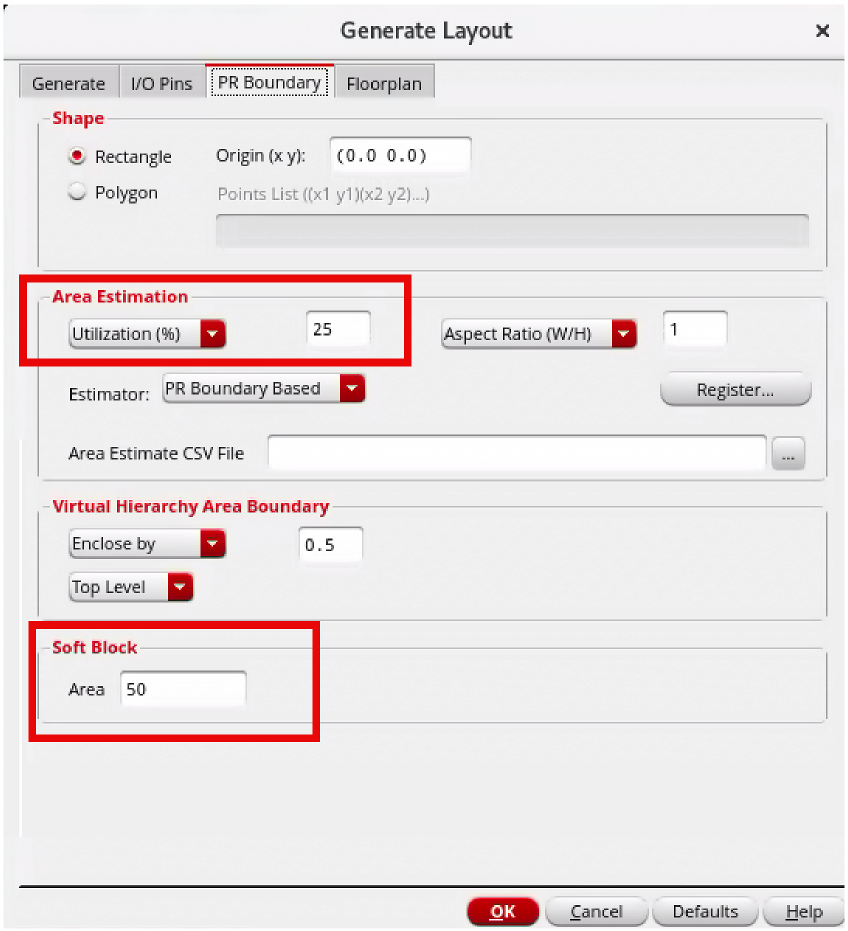
The reason **I2** is created as a soft block is that there is no schematic for the **Comparator** symbol, and the *Auto Generate Soft Blocks* option was set to **true** during generation. This supports early design planning when some symbols may still not have their schematics ready. When generating such symbols in the layout, a cell type *blockBlackBox* is created.

> **Note:** The soft block instance is represented using the icon shown here in the Navigator assistant.

Unselect all the selected blocks, if any.

Calculating the Size of the Generated Soft Block

The size of the generated soft block is determined by the value of the *Area* in the **Soft Block** section and the **Utilization (%)** specified under the **Area Estimation** section on the **PR Boundary** tab of the Generate Layout form, as shown here.



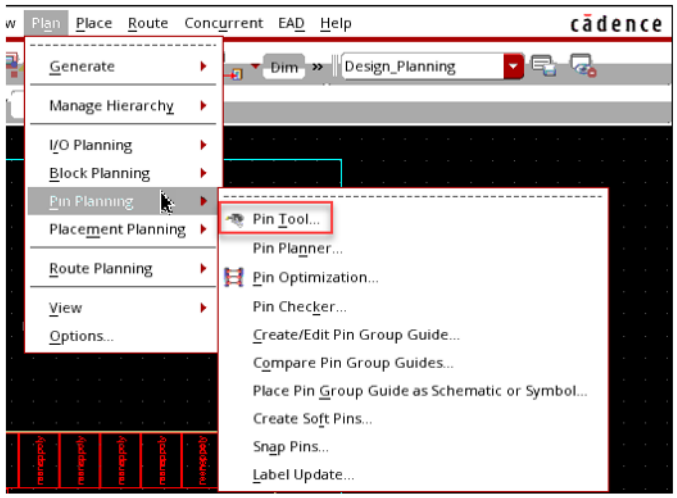
In this example, the area of the generated soft block will be 50 * 4 = 200.

This size will be applied to all the soft blocks created. The designer performing the design planning should change it to a realistic value after generation.

A more accurate size of the soft block could be achieved by registering a generic area estimator. This could work on the symbol master if you have some knowledge of the symbol area. For digital symbols, one could estimate the area based on the number of gates and the gate size or estimate based on the number of pins in a symbol.

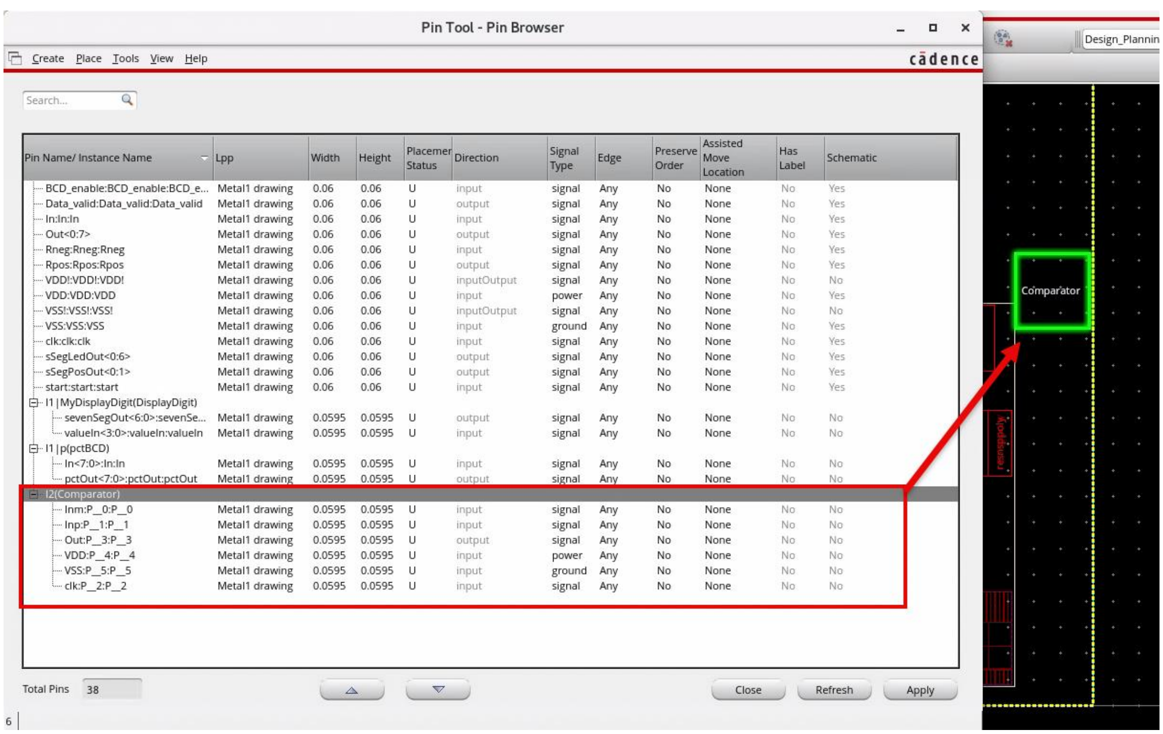
Using the Pin Tool Form for Soft Blocks

1. (Optional Exercise) In the layout EXL canvas, choose the **Plan – Pin Planning – Pin Tool** menu command or click the **Pin Tool** icon in the Design Planning toolbar if you want to review/update the pins in the created soft block.



The Pin Tool form displays.

2. Use the Pin Tool form to select and review/update the created pins in the soft block.



Note: The Virtuoso Floorplanner includes the Pin Tool, which is a unified interface for all pin-related tasks, such as creating pins, resizing pins, planning and optimizing pins, editing pin attributes, and setting their location constraints.

3. **Close** the Pin Tool form after reviewing the created pins.

4. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned how to:

- Access the design planning data sets in the Navigator assistant.

End of Lab

Lab 3-4 Viewing the Virtual Hierarchy

Objective: To view the virtual hierarchy in the design.

In this lab, you explore how to view the virtual hierarchy in the design.

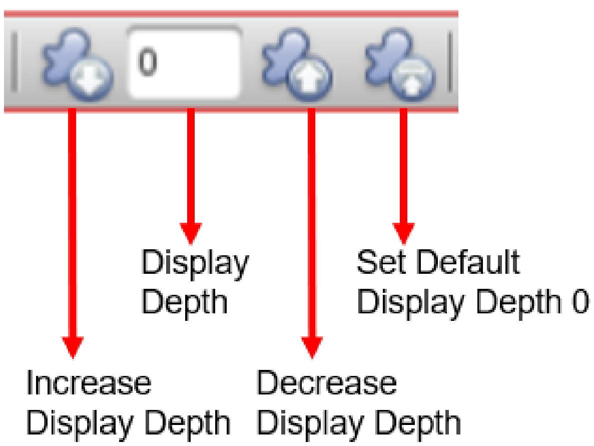
Setting Display Controls

Since the layout is created flat, it is important to control the display of the extra information contained in the layout view.

You can control how much of the virtual hierarchy is visible either globally or individually by setting an override on selected virtual hierarchies. You saw in the generation step that the default display depth is 0. The display depth can be controlled from the toolbar for global and overrides or using the context menu for overrides only. Bindkeys are also supported.

Display Depth Options

The Design Planner toolbar provides the display control options, as shown here.



Tip: Alternatively, you can use bindkeys to control the display depth. Use **Ctrl**, **Shift+** to increment and **Ctrl-** to decrement the display depth.

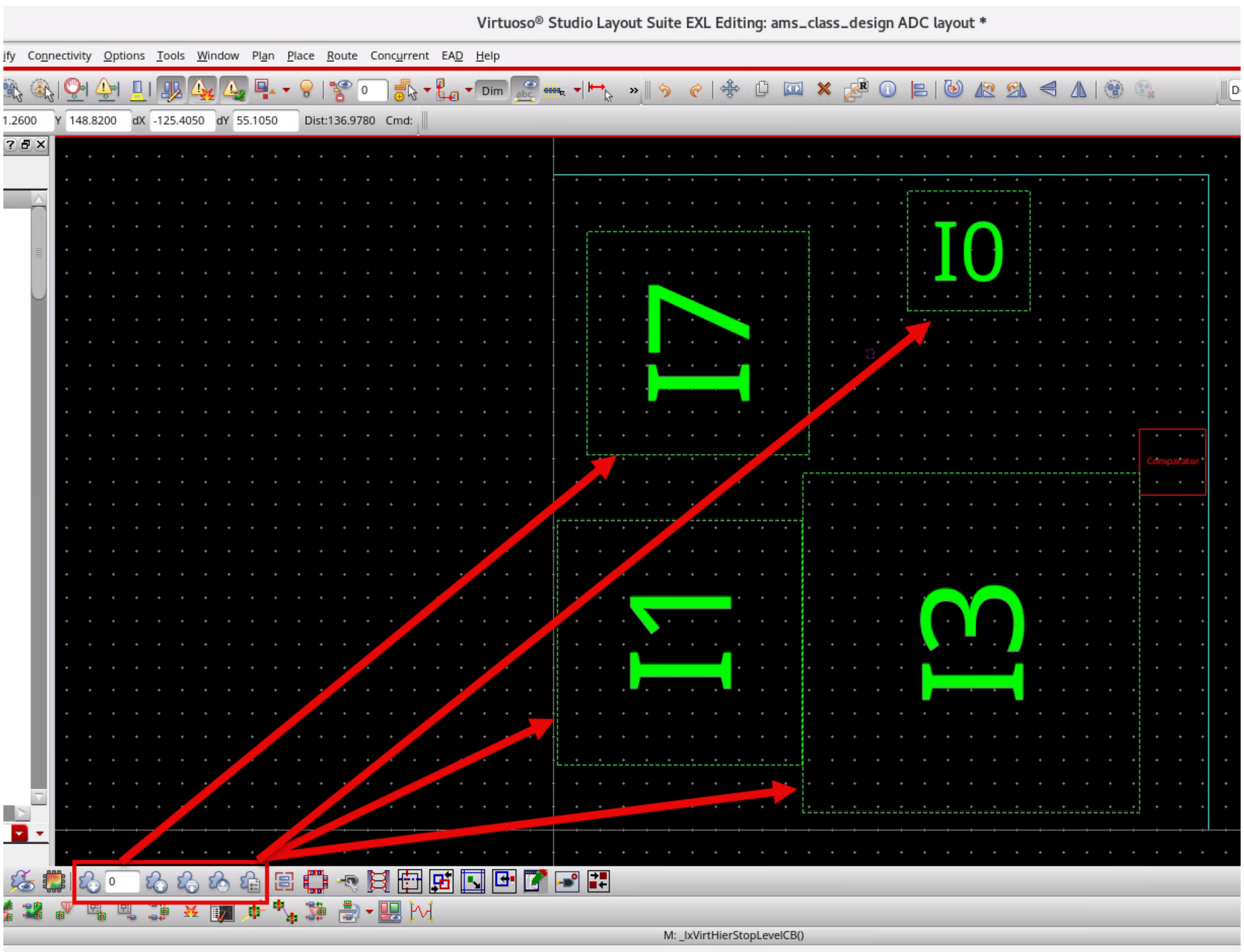
Setting Display Depth

The Design Planner enables you to control the display depth globally for all the components in the design or only for the selected components. You can choose to navigate and display the design.

Making All the Virtual Hierarchies Opaque

You will make all the virtual hierarchies opaque.

1. In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to make all the virtual hierarchies opaque, as shown here.



Controlling the Levels of Virtual Hierarchies Displayed

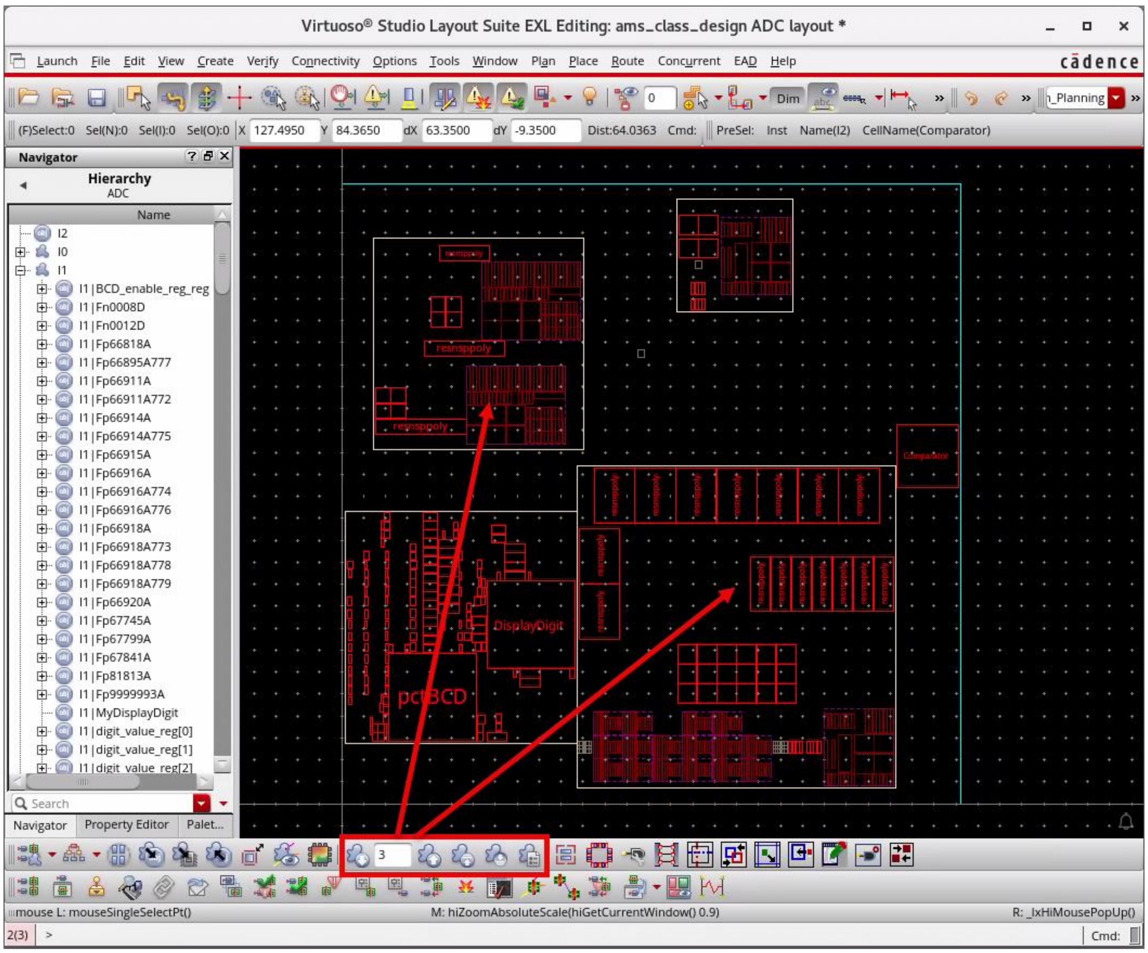
You will control the levels of virtual hierarchies displayed.

1. With no virtual hierarchy selected, click the **Increase Display Depth** button in the Design Planning toolbar, to say, for example, **3**, to control how many levels of virtual hierarchies are displayed.

   The count in the *Global Display Depth* field will change to reflect the current level, as shown here.

   Note: You can also type in the required value if needed.

   Note: You can use the **Decrease Display Depth** toolbar button to return to a lower level of hierarchy.



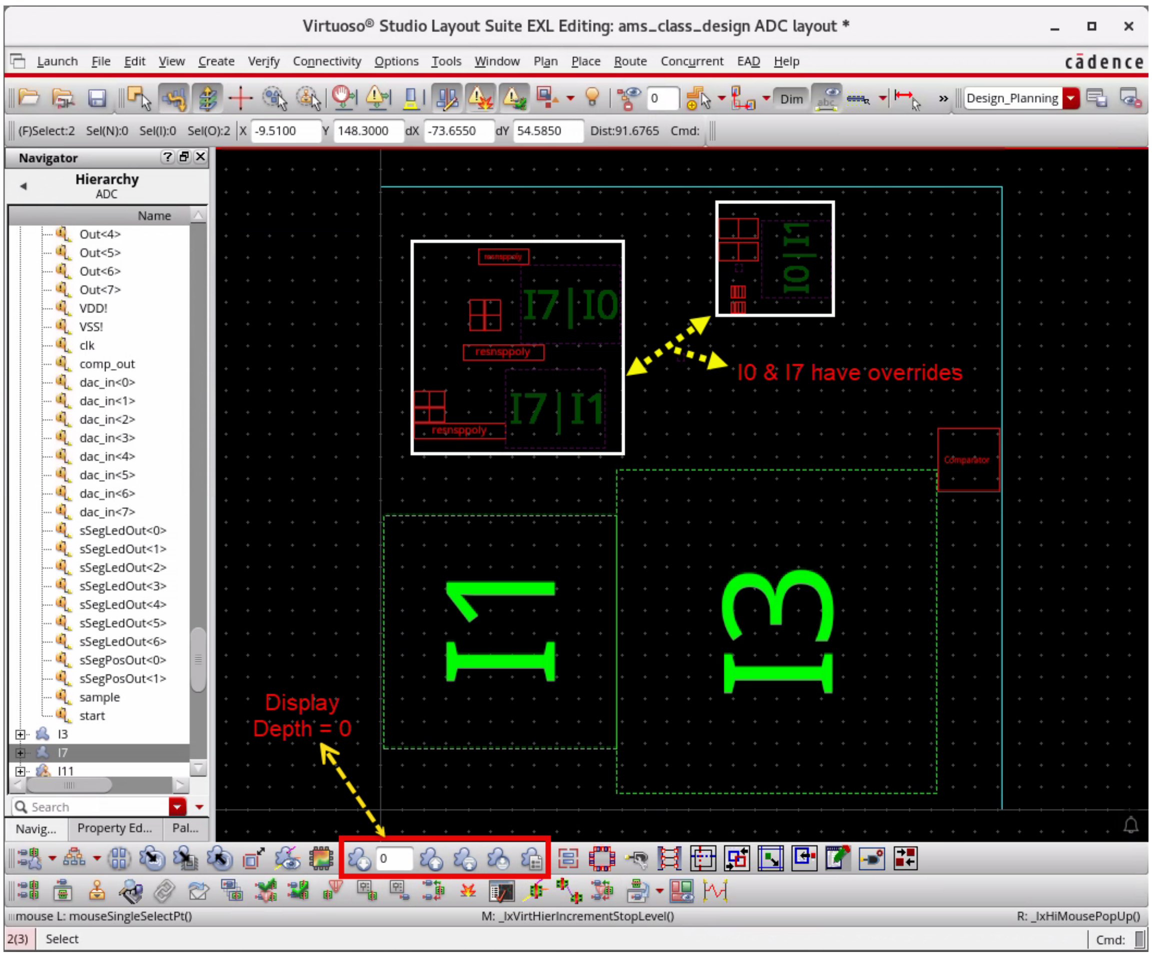
2. In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to return to level **0,** making all the virtual hierarchies opaque again.

Adding Display Overrides for the Virtual Hierarchies

A global display level may not be enough during design planning, and you may wish to see a lower level for some virtual hierarchies that will leave the others opaque. You can use a display override to do the same.

1. With one or more virtual hierarchies selected, for example, after selecting **I0** & **I7**, click the **Increase Display Depth** toolbar button to change the display depth of the selected virtual hierarchies, as needed.

   The count in the *Global Display Depth* field remains **0**, i.e., unchanged, as shown here.

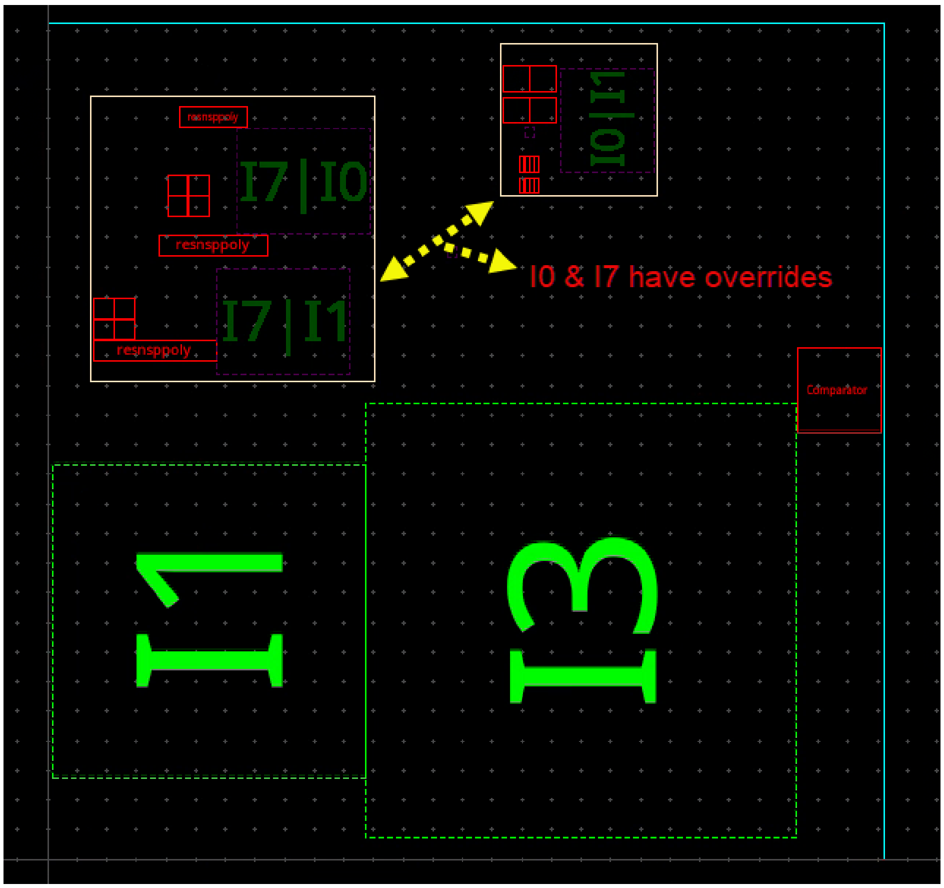
   The display depth is changed for the selected virtual hierarchies (**I0** & **I7**) while the other virtual hierarchies are unchanged. This is due to an override that prevents the global display depth from any changes.

Removing Display Overrides for the Virtual Hierarchies

You will remove the overrides for the virtual hierarchies **I0** & **I7**.

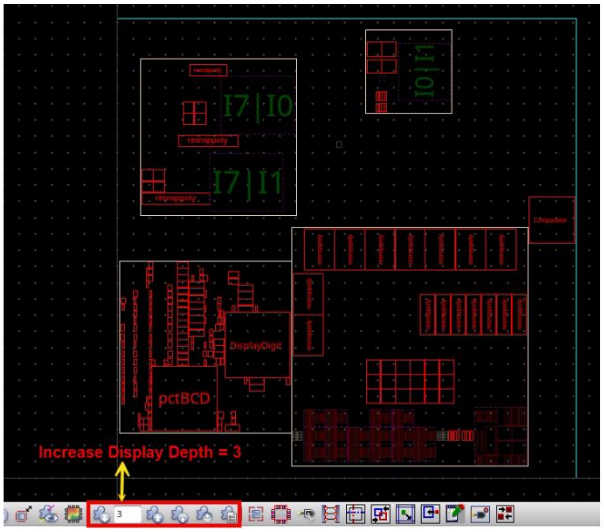
1. In the layout canvas, choose **Edit – Select – Deselect All** or press the bindkey **Ctrl+D** to deselect the objects, if any.

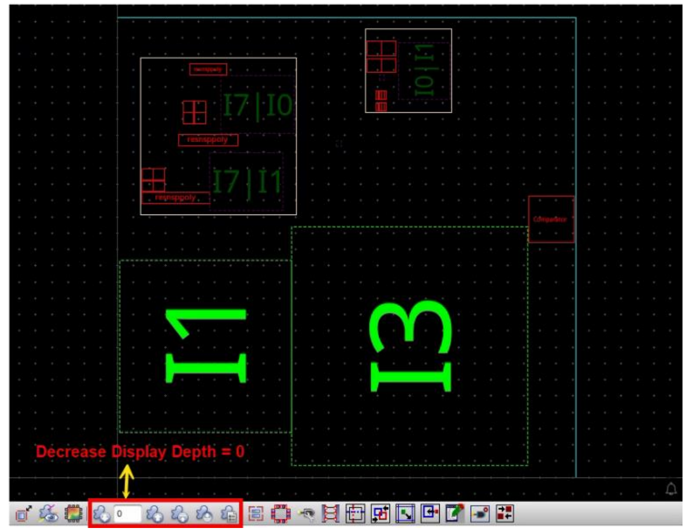
   You can observe that **I0** & **I7** are overridden from the previous step, and they are unselected now, as shown here.

2. With no virtual hierarchy selected, click the **Increase** and/or **Decrease Display Depth** toolbar buttons to change the global display depth, as needed.

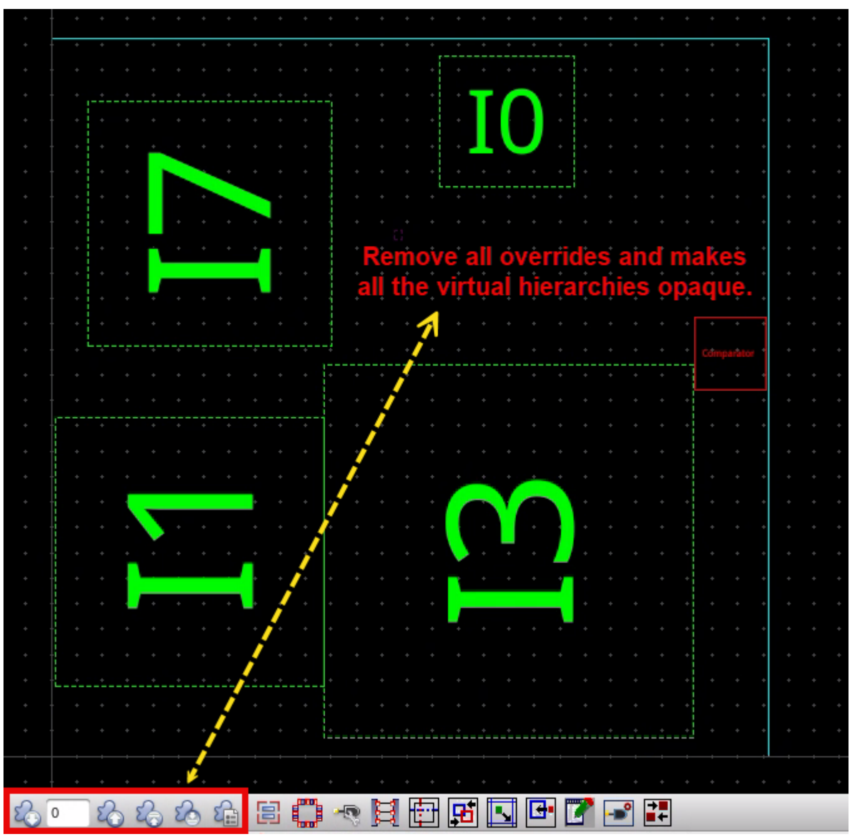
You will notice that the virtual hierarchies with the overrides (**I0** & **I7**) are not changed.





3. In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar.

   The overrides are removed, and the *Display Depth* is restored to **0**, as shown here:



4. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned how to:

- View the virtual hierarchy in the design.

End of Lab

Lab 3-5 Analyzing the Net Connectivity

Objective: To analyze the net connectivity in the design.

In this lab, you analyze the net connectivity in the design.

Using the Net Connectivity

Because the layout is flat, you have access to more net connectivity than would be the case in a bottom-up flow containing a real layout hierarchy or a top-down flow of blackboxes. You can use this enhanced net connectivity to make better design planning decisions.

There are two complementary ways to get net connectivity:

- Selecting the Nets in the Navigator Assistant
- Using the Analyze Connectivity Command

Selecting the Nets in the Navigator Assistant

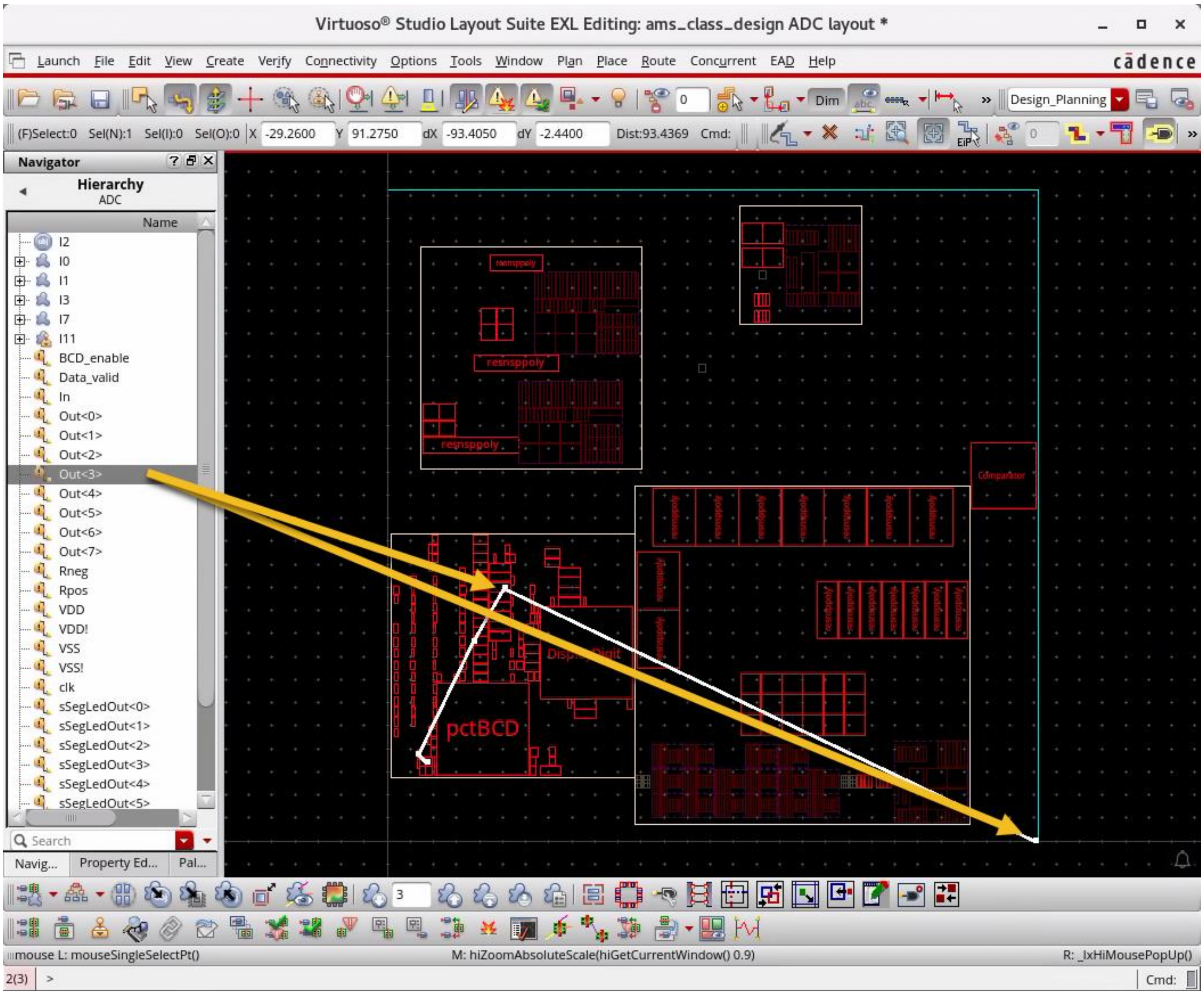
1. With no virtual hierarchy selected, click the **Increase Display Depth** button in the Design Planning toolbar, for example, **3**, to increase the display levels of the virtual hierarchies in the design.

2. In the Navigator assistant, in the Design Planning category, click the **Hierarchy** data set.

   a. Select the net **Out<3>**.

   This view shows the interface nets at each level of the virtual hierarchy.

The net is cross selected in the layout canvas, as displayed here. You can zoom in and observe that the net is connected to a top-level pin **/Out<3>/** and to the virtual hierarchy, **I1**. If **I1** were a real layout, you would only see the connection to its pin, not the instance inside.



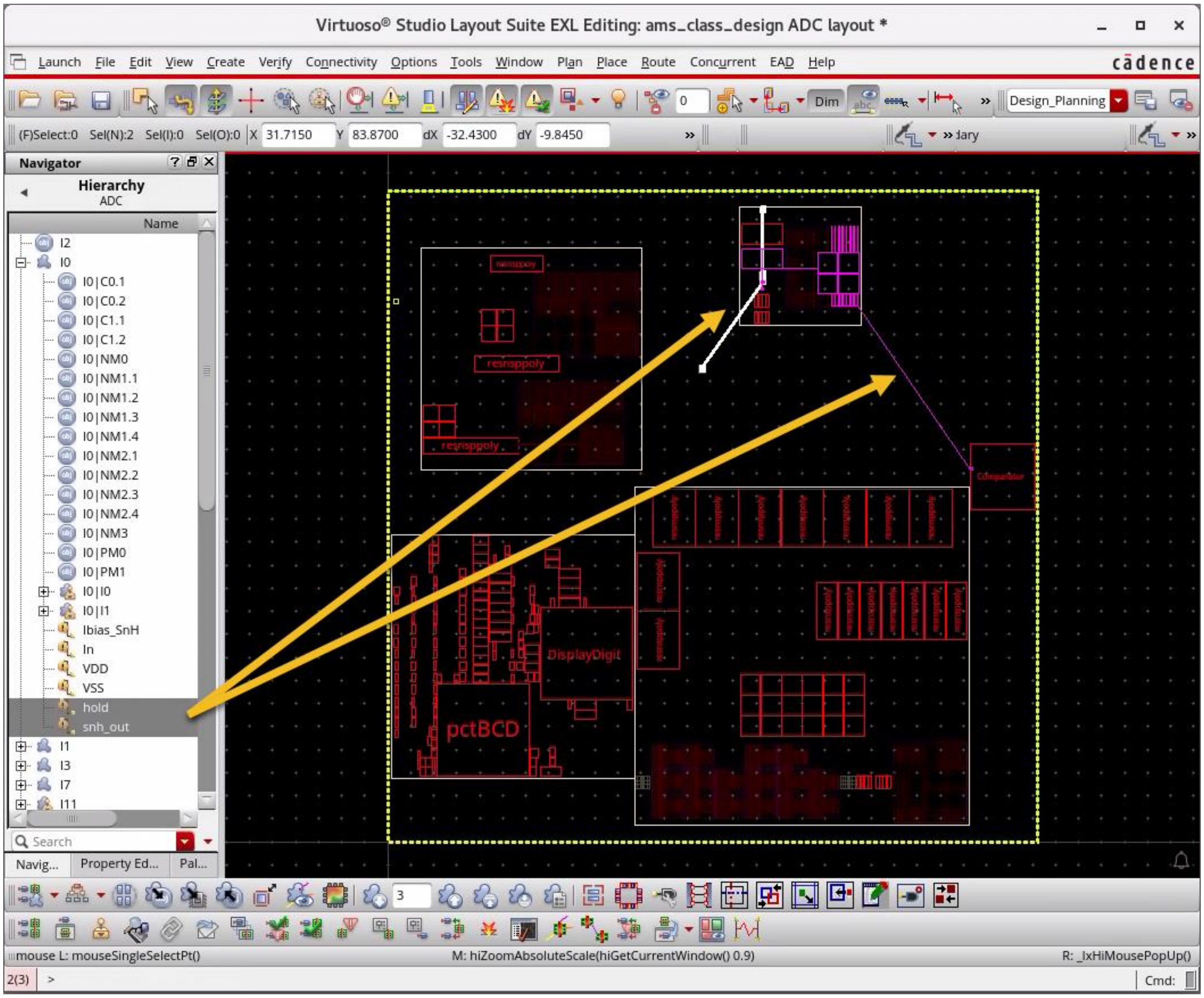
3. Unselect the net **Out<3>**.

4. Click the (**+**) button next to **I0** to expand the virtual hierarchy in the *Hierarchy* data set.

   a. Press the **Ctrl** key and select any interface net(s), for example, **hold**, and **Snh_out** to see the connections.

   The lower-level instances, virtual hierarchy, and interface nets are listed, as shown here.

You can see the connectivity throughout the virtual hierarchy. This extra information will help you make better design planning decisions.



5. Unselect all the selected objects, if any.

Using the Analyze Connectivity Command

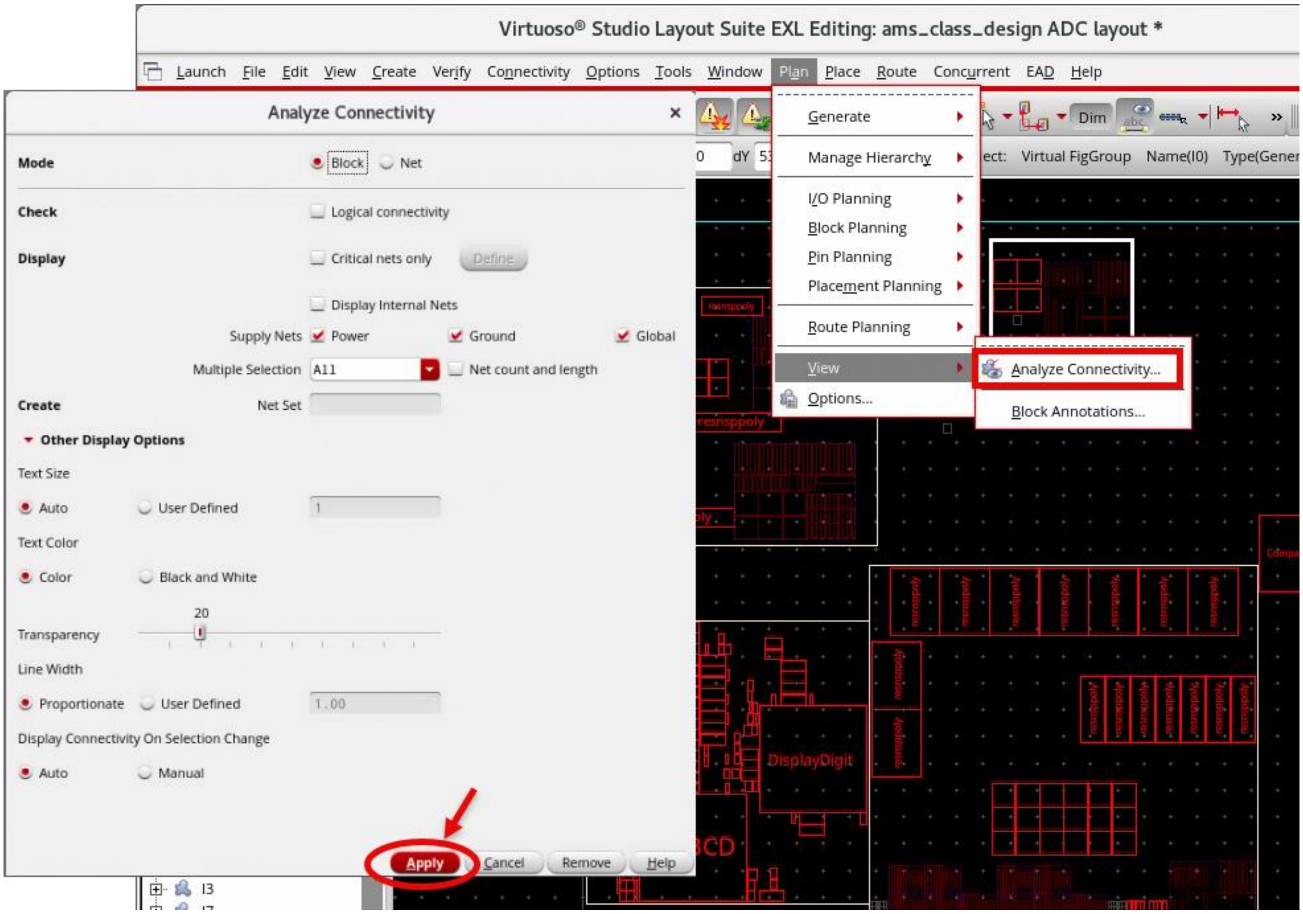
To decide upon an appropriate position for each block when planning a design, the connectivity between and within the virtual hierarchies can provide important clues. The Analyze Connectivity command is supported by the Design Planner to help with such planning decisions.

1. Select the virtual hierarchy **I0** in the *Hierarchy* data set in the Navigator assistant.

2. In the layout canvas, click the **Analyze Connectivity** icon in the Design Planning toolbar or choose the **Plan – View – Analyze Connectivity** menu command to launch the Analyze Connectivity form.

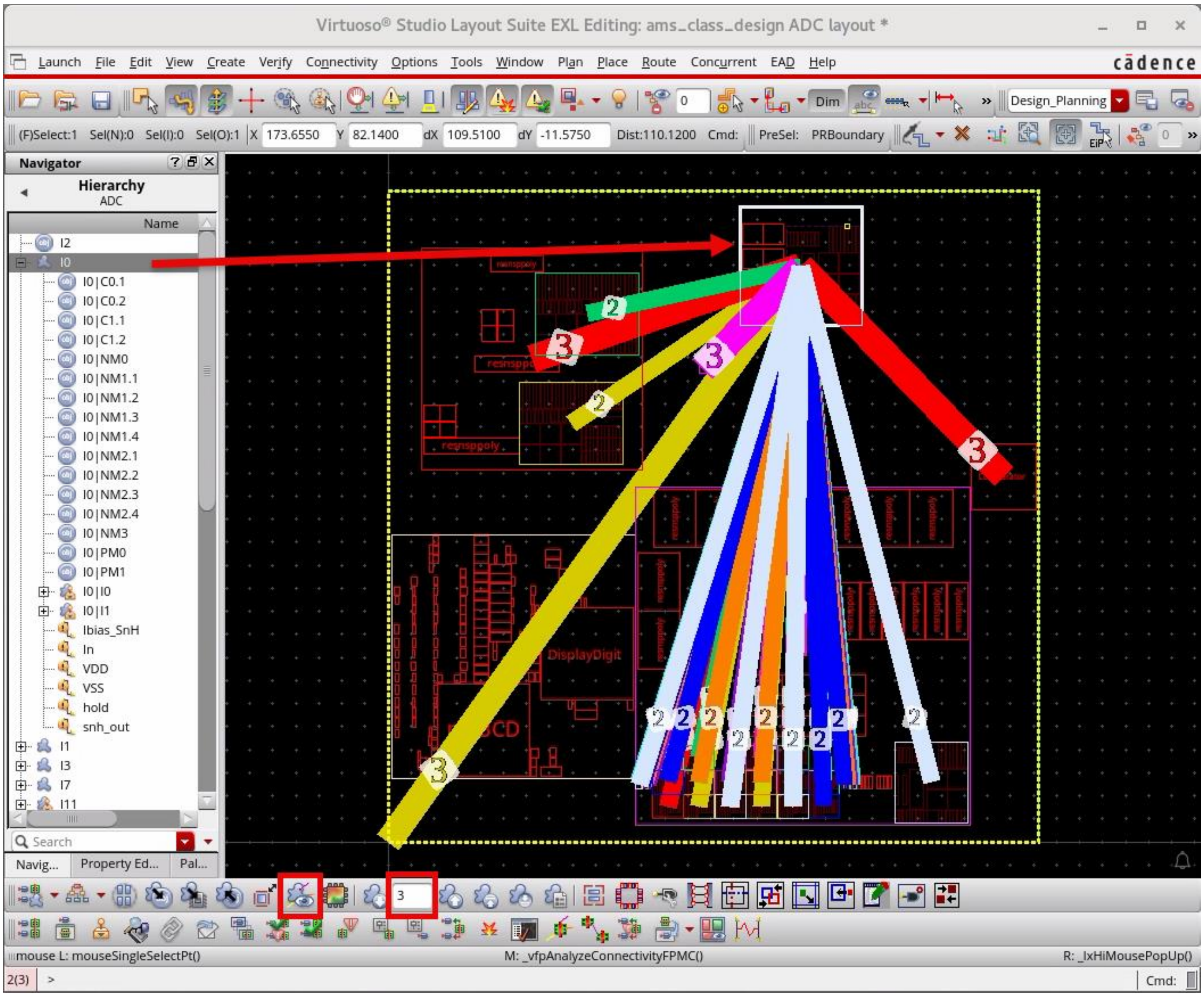
3. Keep the default values and click **Apply** to run the command.

Note: The nets can be filtered based on *Power*, *Ground*, and *Global*.



All the connections from **I0** are highlighted in the layout, as shown here. Each flightline between the connected pair is overlayed with a number that indicates the total connections along the flightline. This information can be used to position the blocks to limit the top-level routing.

The Analyze Connectivity command also displays the contents of a virtual hierarchy. Since the virtual hierarchy instance is visible, the connections to the instance are shown.

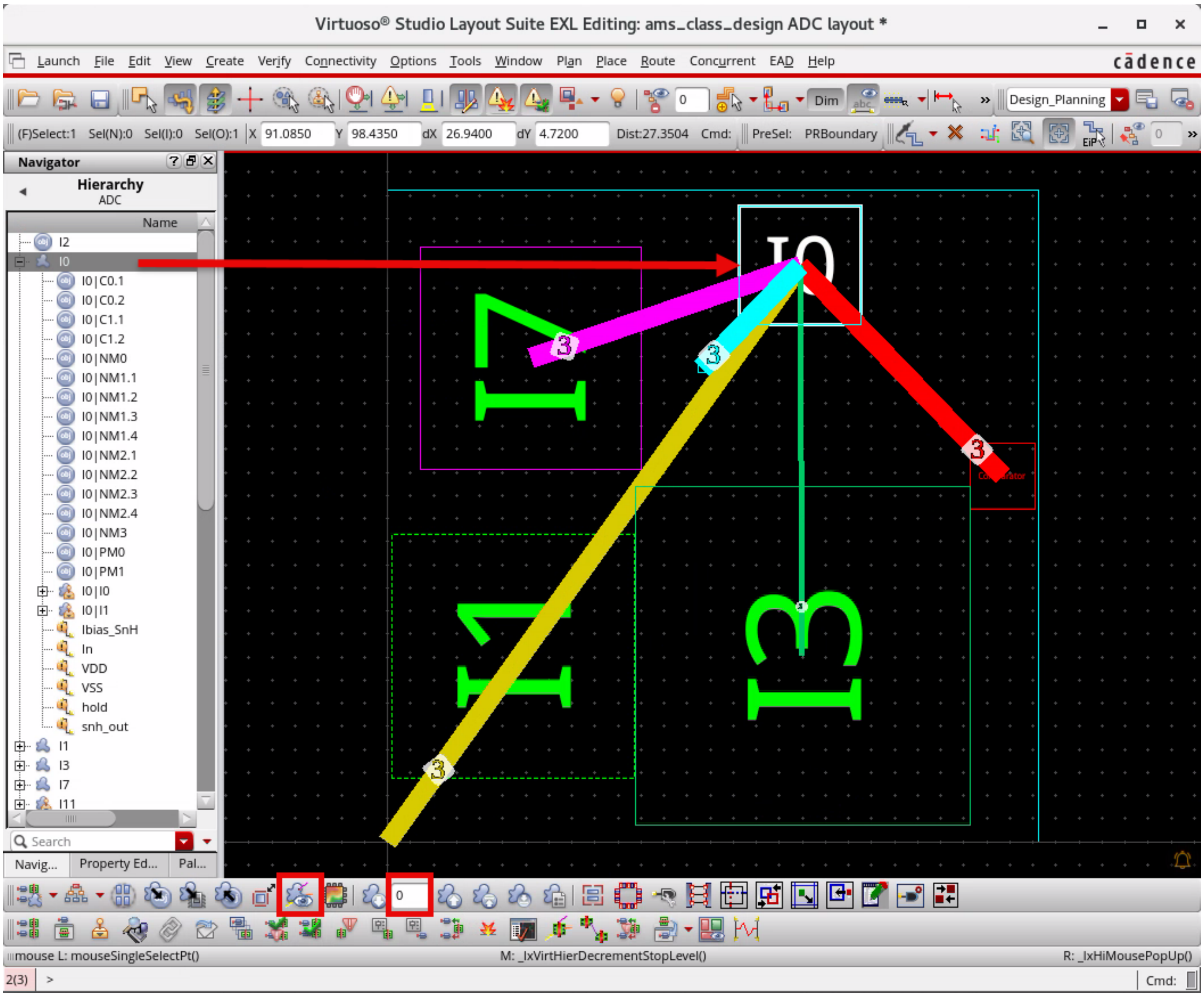


4. In the Analyze Connectivity form, click **Remove** to remove the highlighted connections.

5. In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to return to level **0,** making all the virtual hierarchies opaque again.

6. Select **I0** and in the Analyze Connectivity form, keep the default values and click **Apply** to run the command again.

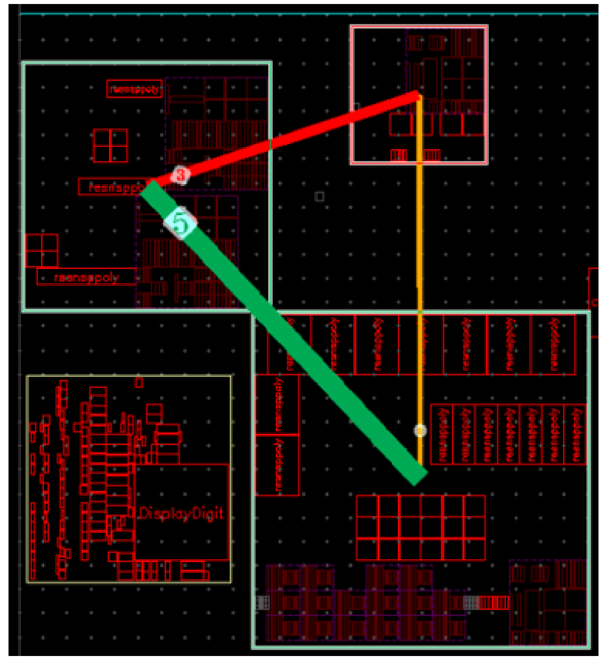
Since the virtual hierarchy is opaque, the instance connectivity is bundled into a single line to the center, as shown here.



7. In the Analyze Connectivity form, click **Remove** to remove the highlighted connections.

8. Deselect all the selected objects, if any.

9. (Optional Exercise) If two or more virtual hierarchies are selected, only the connectivity to the selected virtual hierarchies is displayed. A few examples are shown here. Explore the options as time permits.



10. **Remove** the highlighted connections and **Cancel** the Analyze Connectivity form.

11. Deselect all the selected objects, if any.

12. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned how to:

- Analyze the net connectivity in the design.

End of Lab

Module 4: Early Design Planning

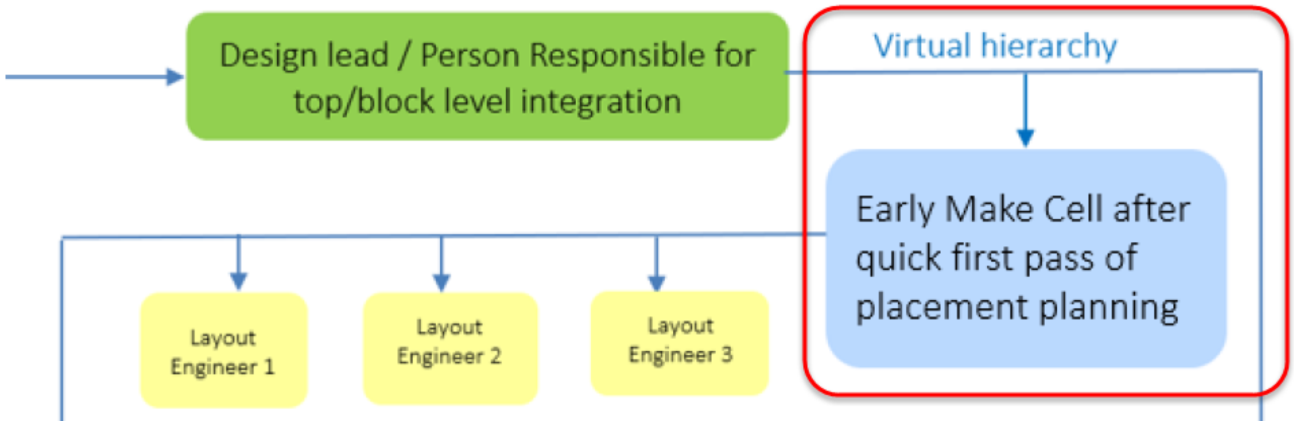
Lab 4-1 Manipulating the Virtual Hierarchy

Objective: To manipulate the virtual hierarchy in the design.

In this lab, you explore how the virtual hierarchy can be manipulated into a suitable design plan in the early design planning phase.

Using the Early Design Planning Phase

The early design planning phase is likely to be carried out by the designer responsible for the top- or block-level integration prior to assigning each block for planning to the assigned Layout Engineer. In this lab, you will see some of the tasks carried out.



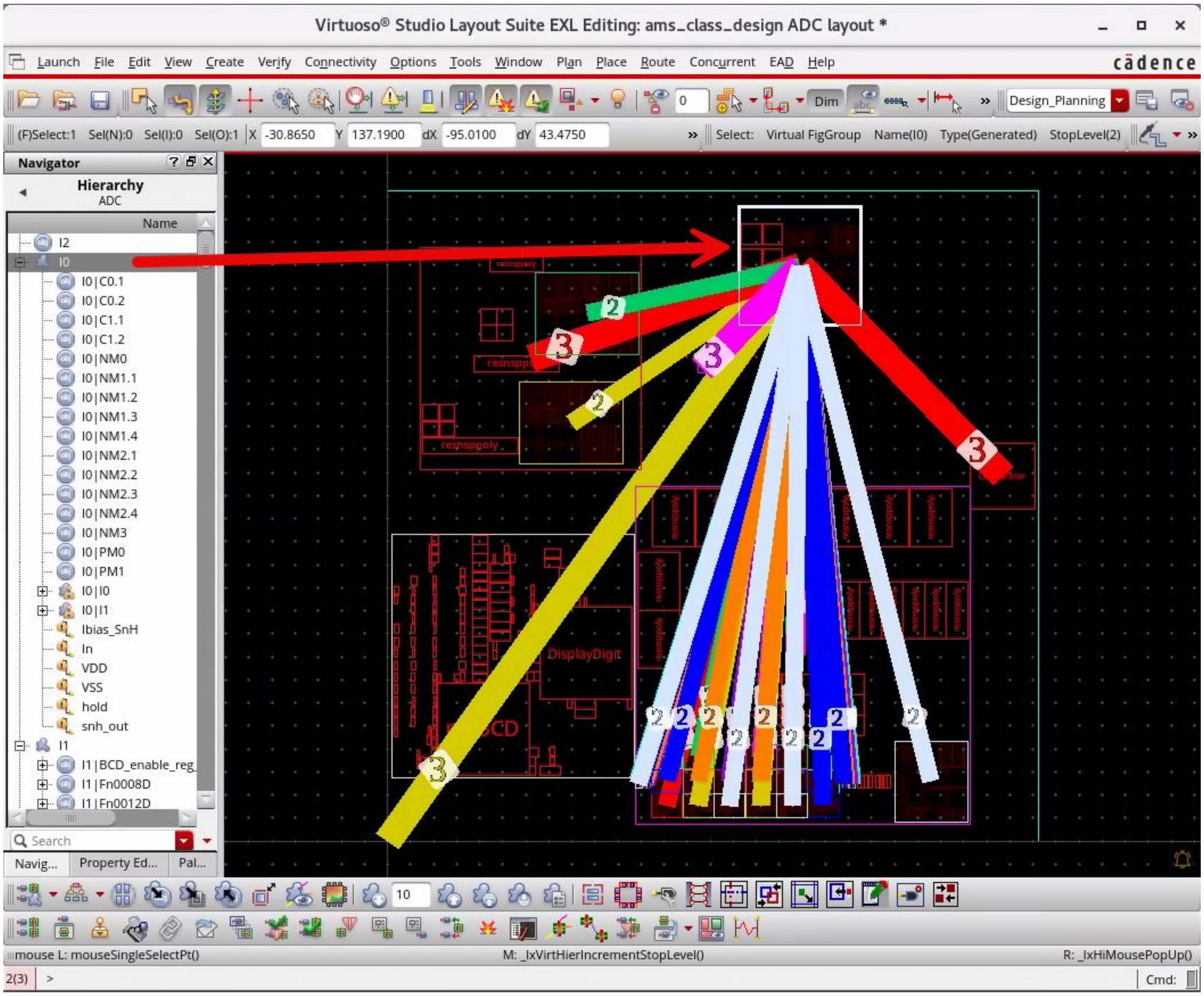
1. The schematic and layout views should be open from the previous lab. If not, re-open them now.

2. Use the previous **Lab 3-4** steps and run the **Analyze Connectivity** command on the virtual hierarchy instance **I0** in the *Hierarchy* data set in the Navigator assistant by increasing the display levels of the virtual hierarchies in the design to **3** using the **Increase Display Depth** icon in the Design Planning toolbar in the layout canvas.

   All the connections from **I0** are highlighted in the layout canvas.

   Since the virtual hierarchy instance is visible, the connections to the instance are shown.

You will use this information to determine the relative position of the virtual hierarchies.

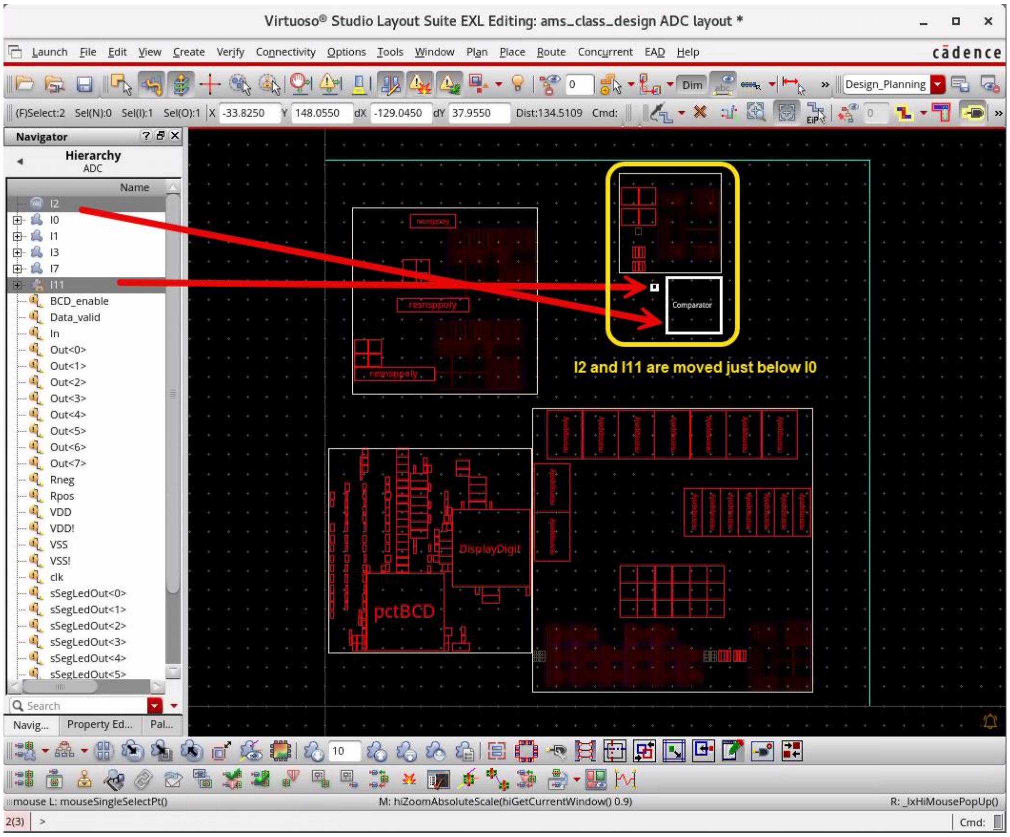


3. **Remove** the highlighted connections and **Cancel** the Analyze Connectivity form.

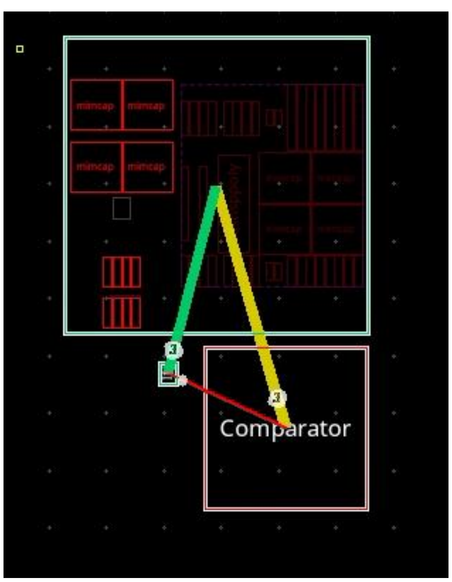
4. Deselect the virtual hierarchy instance **I0**.

5. In the layout canvas, choose the **Edit – Move** menu command or use the **Move** icon in the Edit toolbar or use the bindkey **M** to move the instances **I2** and **I11** to appear just below **I0**.

Note: In the layout canvas, choose the **Window – Toolbars** menu to activate the Edit toolbar if it is not visible.

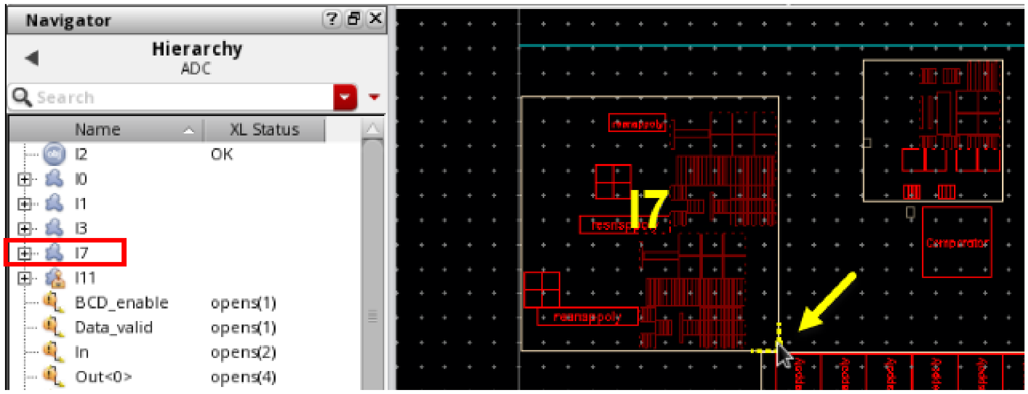


6. Looking at the relative connectivity with **Analyze Connectivity** command, you can see that the top-level connections are now shorter, as shown in the graphic here.



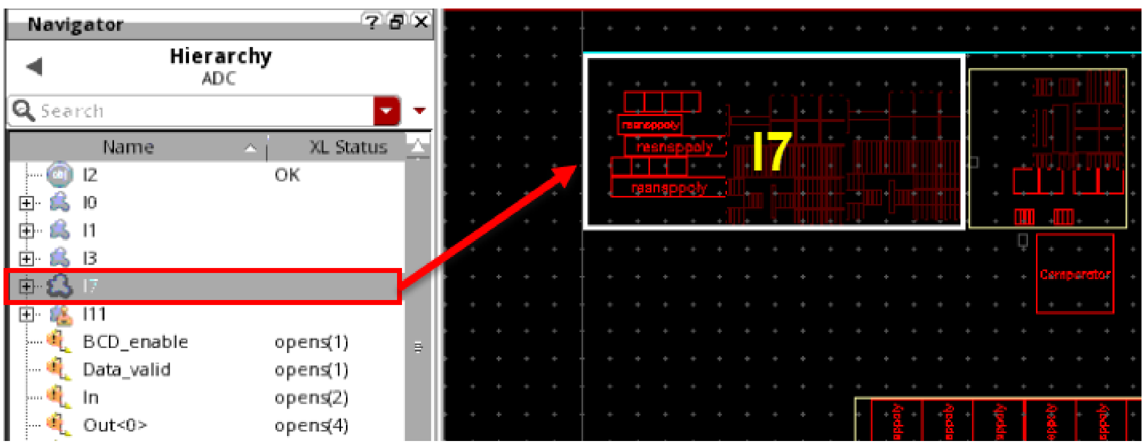
You can stretch virtual hierarchies to change their aspect ratio and achieve a tighter plan.

7. In the layout canvas, <u>with nothing selected</u>, start the Stretch command by choosing either the **Edit – Stretch** menu command or by clicking the **Stretch** icon in the Edit toolbar or by pressing the bindkey **S**, hovering the mouse over **I7,** and clicking at the bottom corner.



8. Move the cursor up and to the right to fill the space to the left of **I0**. Use the Stretch command to move **I7** to reposition and get some placement, as shown here.

   The contents are automatically repositioned inside the new boundary.

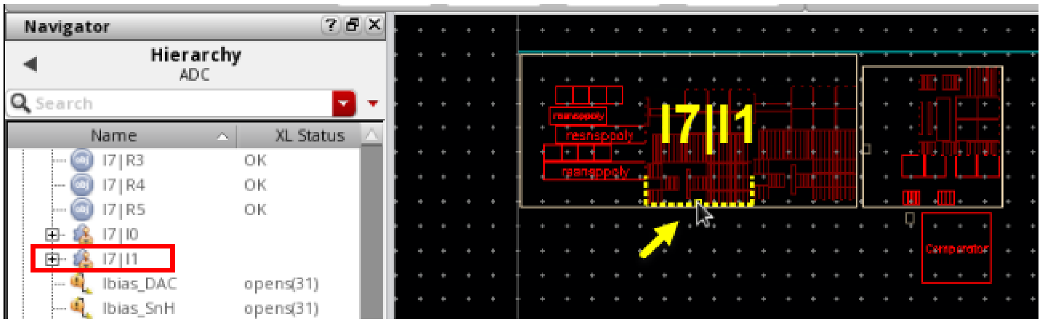


   This feature is very useful in early design planning. The contents of the virtual hierarchy can help you estimate whether the boundary is the correct size. If you want to maintain the area constantly, the *Constant area stretch* option is available by clicking **F3** when running the **Stretch** command.

   Although the contents of the virtual hierarchy clone **cannot** be selected without first doing **Edit In Place**, you can stretch the boundary.

9. Click the (**+**) button adjacent to the virtual hierarchy **I7**.

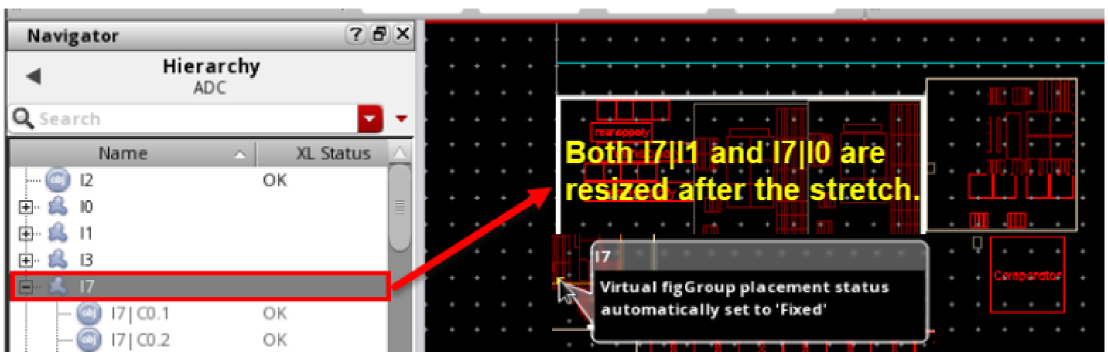
10. In the layout canvas, <u>with nothing selected</u>, start the **Stretch** command, hover the mouse over **I7|I1** and click at the bottom corner.



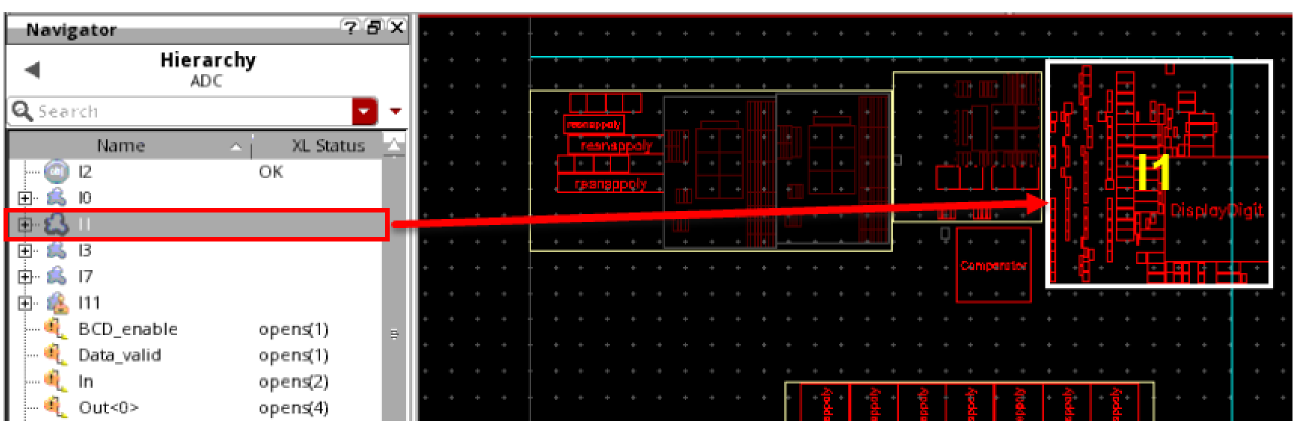
11. Stretch **I7|I1** downwards and click.

    You will see that both **I7|I1** and **I7|I0** are resized and replaced. This is because they both belong to the same clone family. If the *Auto adjust area boundary* option is set to **true** in the *Design Planning and Analysis Options* form, which is the default, the area boundary of **I7** is resized to reflect the stretch.

    Note: The info balloon informs you that the placement status is set to "Fixed". The placement status will be covered later in this lab.



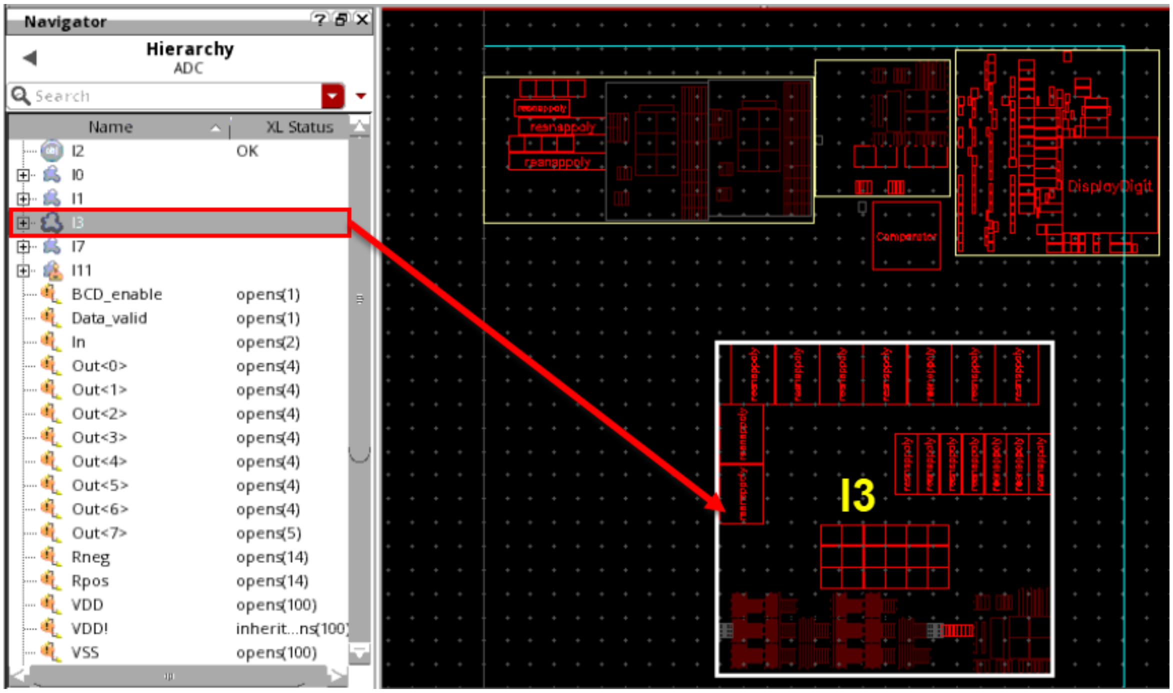
12. Move **I1** to the top-right corner, to the right of **I0**, as displayed here.



In **Step 8**, you used the **Stretch** command to change the shape of a virtual hierarchy. You can also do this using the **Chop** command, which can create a rectilinear virtual hierarchy.

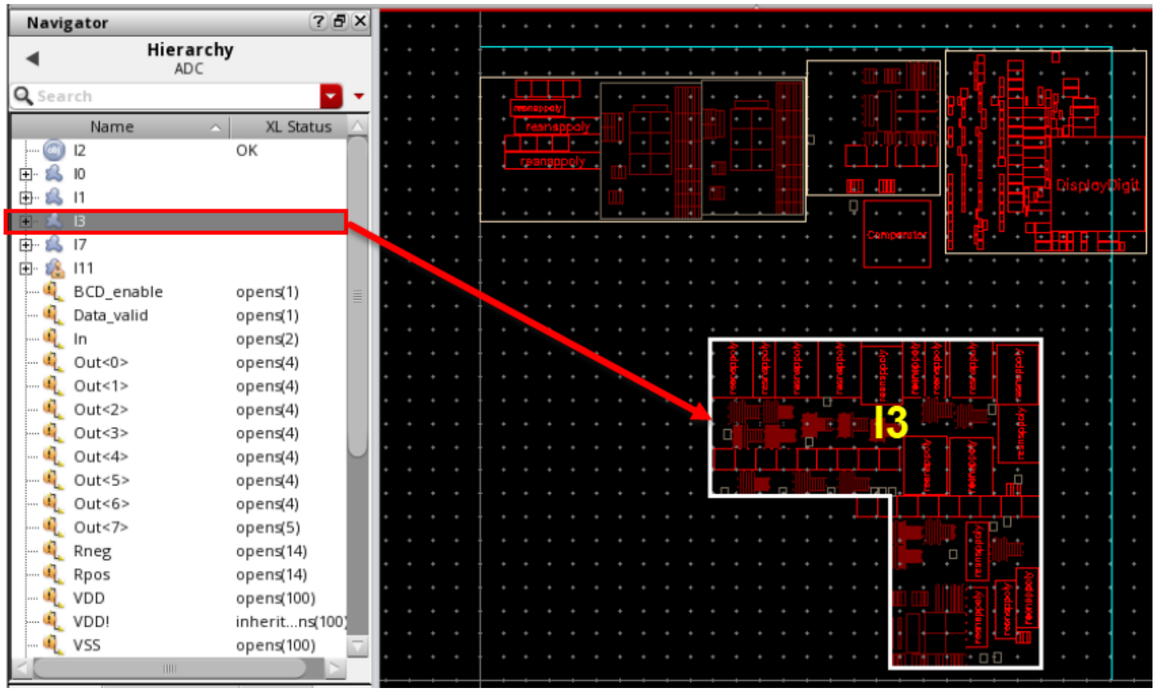
13. Select the virtual hierarchy instance **I3**.

14. In the layout canvas, choose the **Edit – Basic – Chop** menu command or press the bindkey **Shift+C** to invoke the Chop command.



15. Draw a rectangle over **I3**.

The rectangle is chopped, and the boundary is redrawn. The contents are automatically repositioned inside the new boundary.

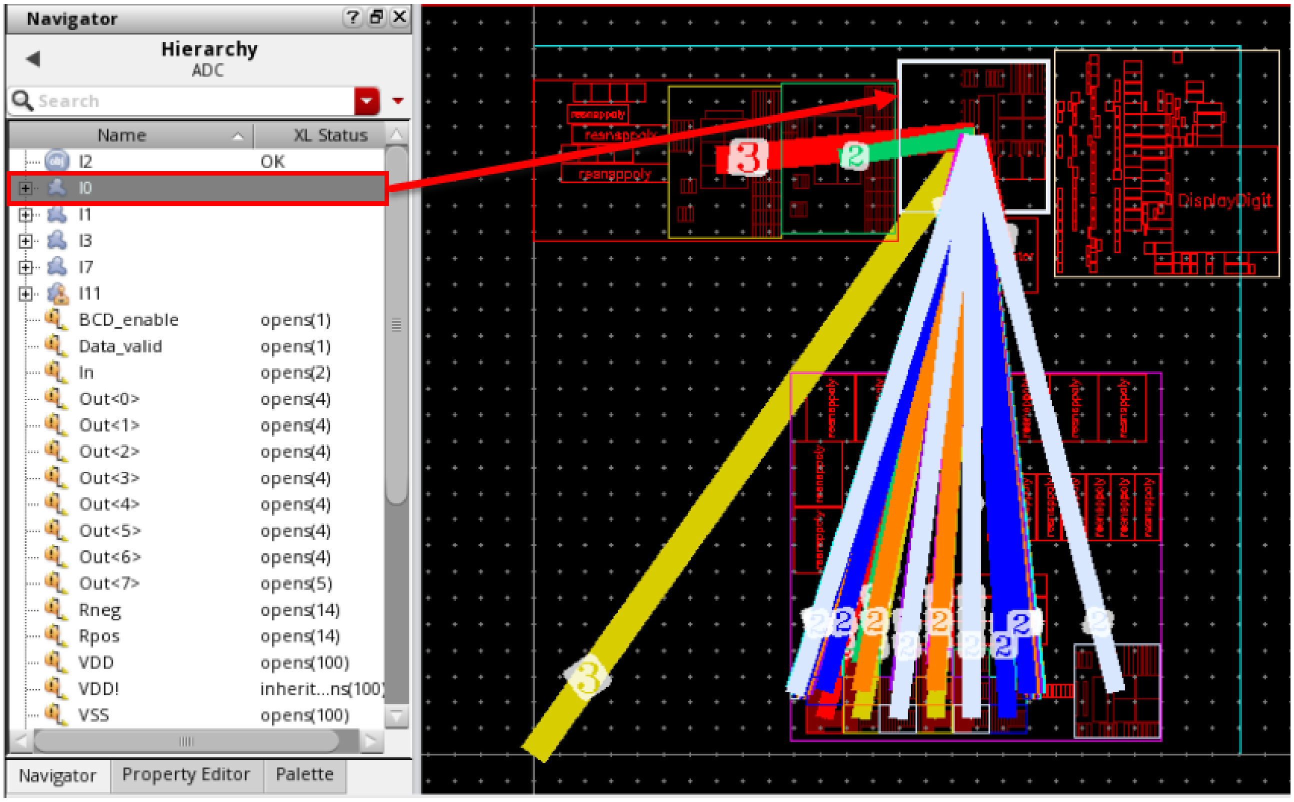


16. In the layout canvas, undo the command by choosing either the **Edit – Undo** menu command or by clicking the **Undo** icon in the Edit toolbar or by pressing the bindkey **U** to return to the rectangular boundary again for **I3**.

    You can directly edit the contents of a virtual hierarchy if the virtual hierarchy is not opaque. This avoids having to **Edit In Place** the virtual hierarchy unless it is a virtual hierarchy clone that you need to edit. Being able to directly edit the virtual hierarchy makes the editing very convenient if you work across multiple hierarchy levels.

17. Select the virtual hierarchy instance **I0** and use the **Analyze Connectivity** command to show its connections.

As displayed in the graphic here, there are many connections going inside **I3** and many unconnected **resnsppoly** instances between the connected instances.

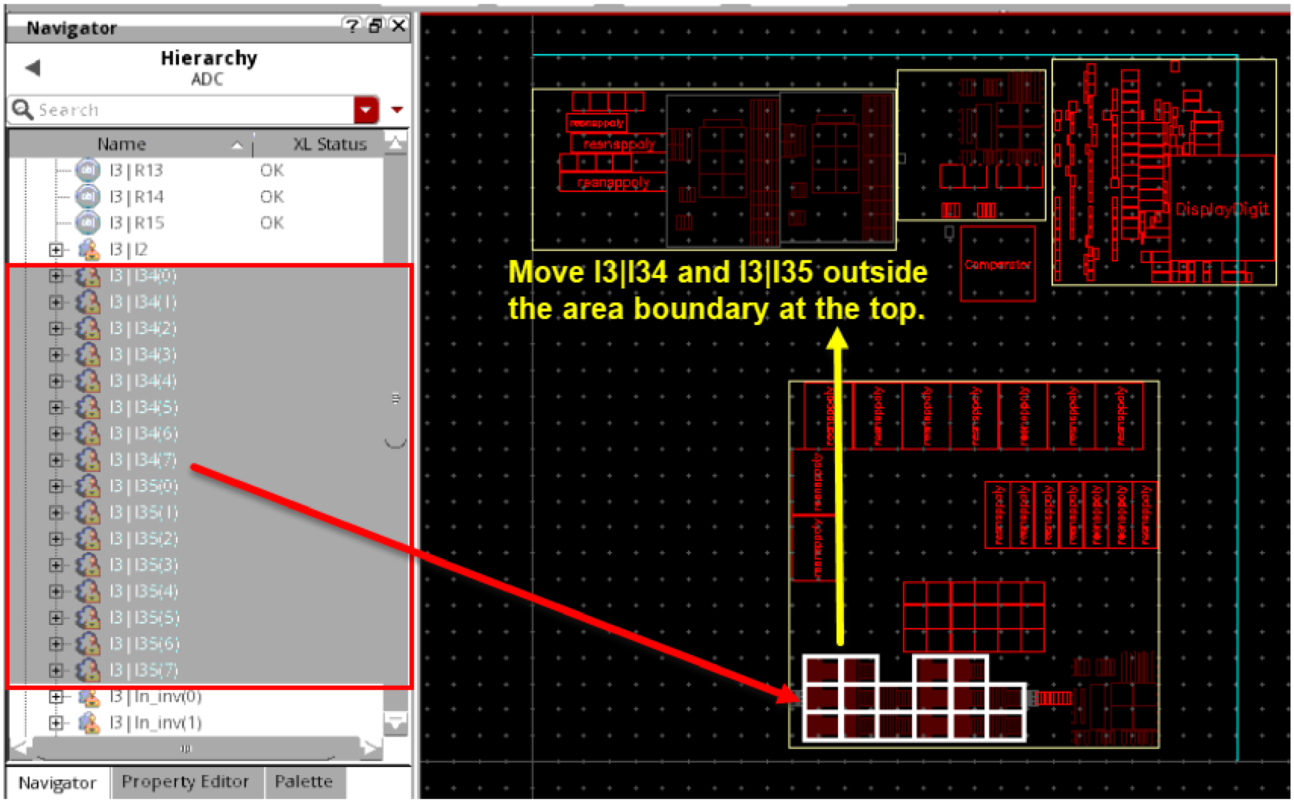


18. **Remove** the highlighted connections and **Cancel** the Analyze Connectivity form.

19. Deselect the virtual hierarchy instance **I0**.

20. Click the (**+**) button next to **I3** to expand the virtual hierarchy in the *Hierarchy* data set.

21. Select all the **I3|I34** and **I3|I35** virtual hierarchy clones and move them outside the area boundary at the top.

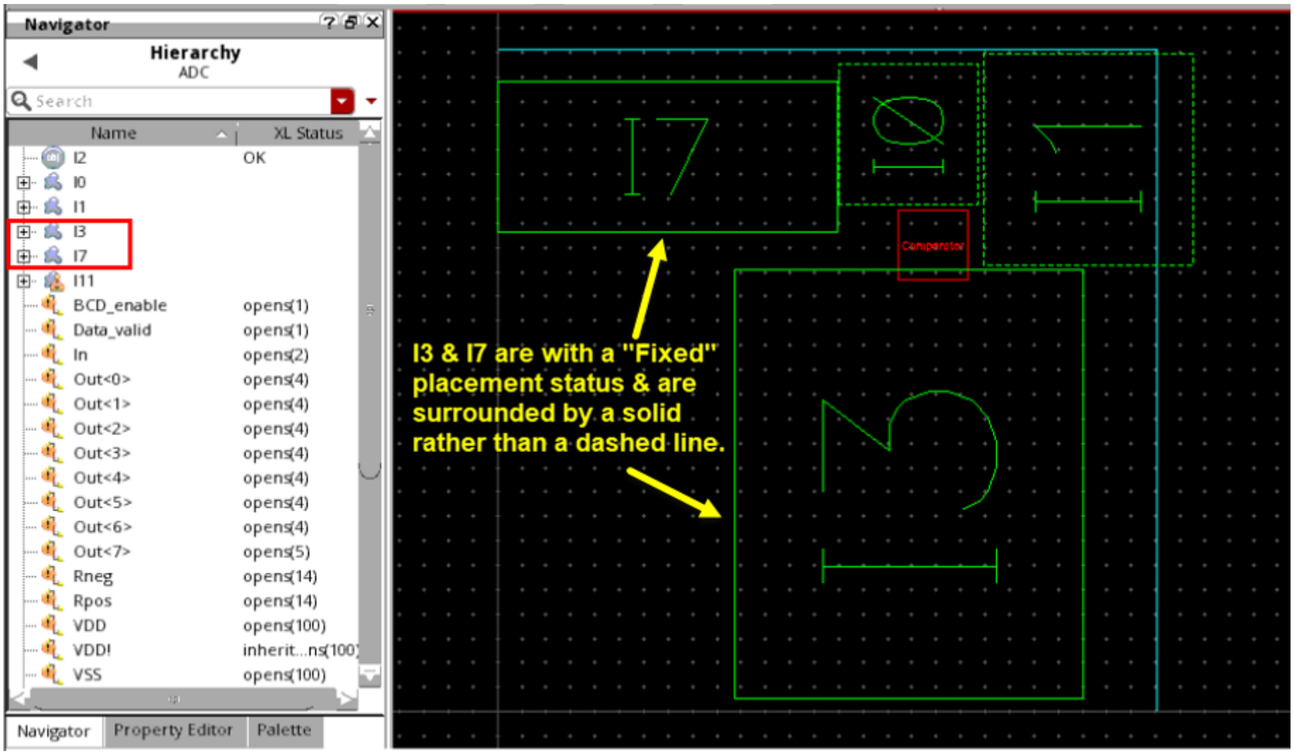


The instances are moved outside the boundary, as shown here, but they remain inside the virtual hierarchy. You have only changed the position of the instances but not moved them to the top level.

After the instances are placed, the area boundary is resized to accommodate the new placement. The info balloon informs you that the placement status has been set to “Fixed”. This ensures any more manual placement cannot cause the virtual hierarchy instances to lose their automatic placement. Further resizing of the area boundary will not trigger replacement unless the placement status is reset.

22. In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to return to level **0,** making all the virtual hierarchies opaque.

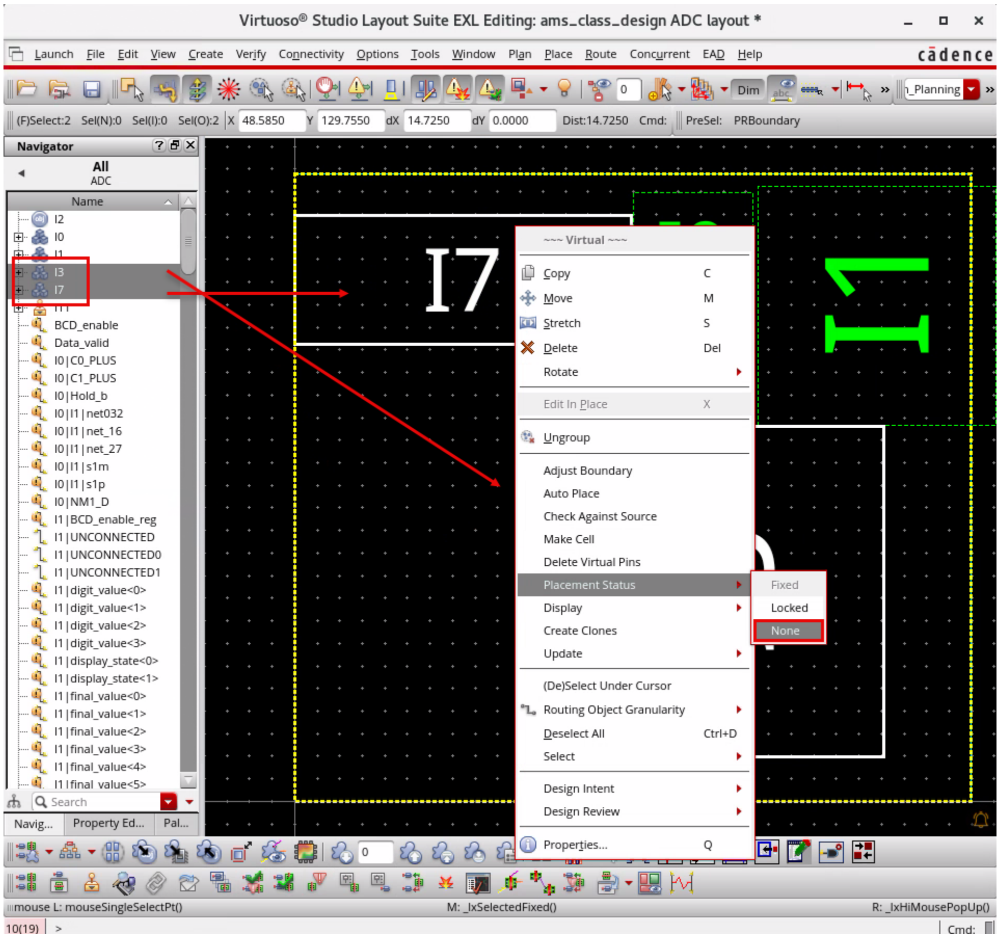
You observe that the virtual hierarchies **I3** & **I7** with a “Fixed” placement status are surrounded by a solid line rather than a dashed line, as shown here.



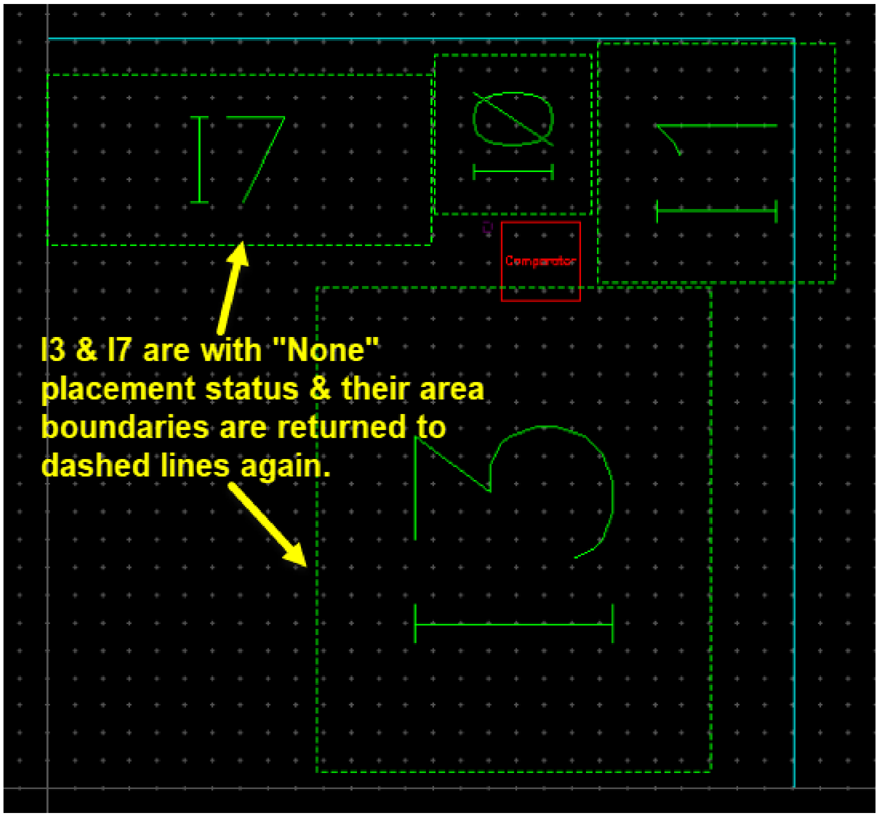
The placement status can be reset, if needed.

23. Select the virtual hierarchies **I3** & **I7**.

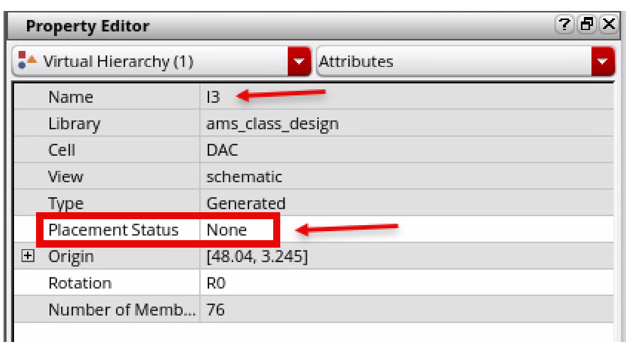
24. **Right**-click and from the *Placement Status* option, select **None** from the **Virtual** context-sensitive menu, as shown here.

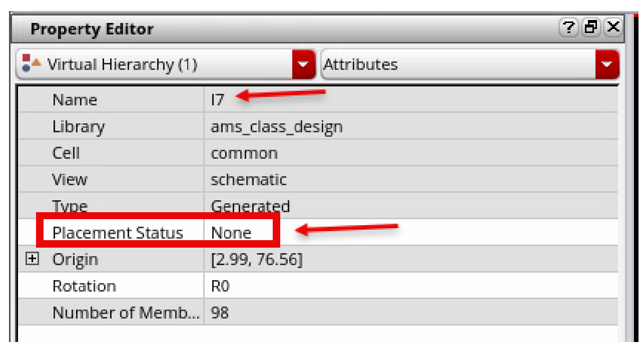


You can see that after the placement is reset to **None**, the area boundaries of **I3** & **I7** are returned to dashed lines again.



The Property Editor shows that the *Placement Status* is **None**, which means it is no longer “Fixed”.



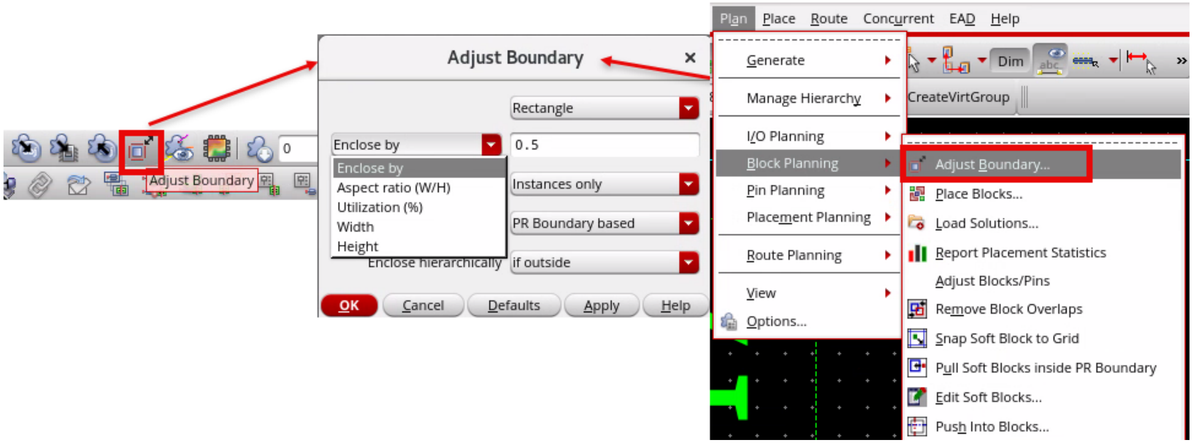


Now that you have completed some editing tasks within the virtual hierarchy, you can adjust the boundary to fit around its contents.

In the previous steps, you used the virtual hierarchy boundary to determine the position of the contents. You can also use the contents to determine the boundary size. You can adjust the boundary to fit around the virtual hierarchy contents.

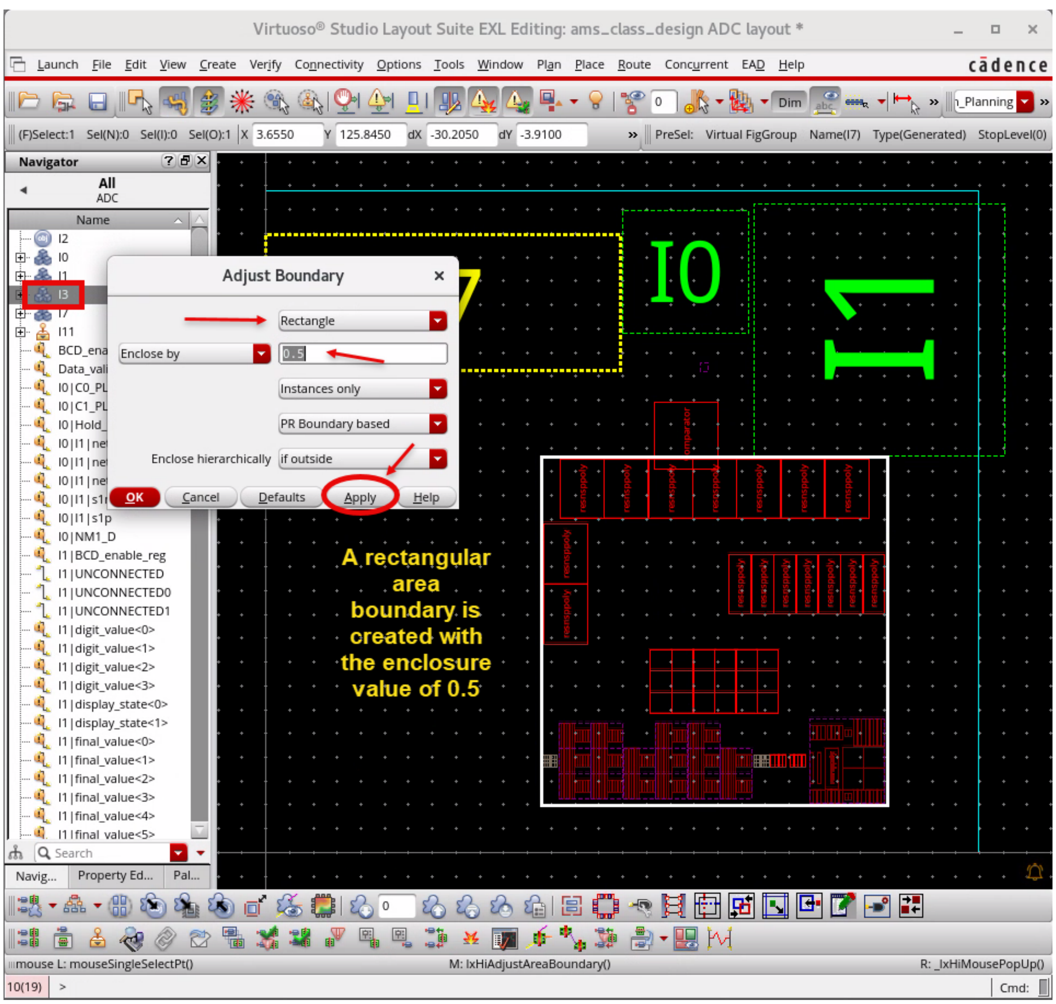
25. Select the virtual hierarchy instance **I3** and increase the Display Depth to **3**.

26. In the layout canvas, click the **Adjust Boundary** icon in the Design Planning toolbar or choose the **Plan – Block Planning – Adjust Boundary** menu command to invoke the Adjust Boundary form.



The default option is to create a rectangular boundary spaced away from the contents by the enclosure value of **0.5**.

27. Keep the default values and click **Apply** to adjust the area boundary of **I3**.



When adjusting an area boundary to be rectangular, there are additional options available to control the **Aspect ratio(W/H)**, **Utilization (%)**, and the **Width** and **Height** of the area boundary.

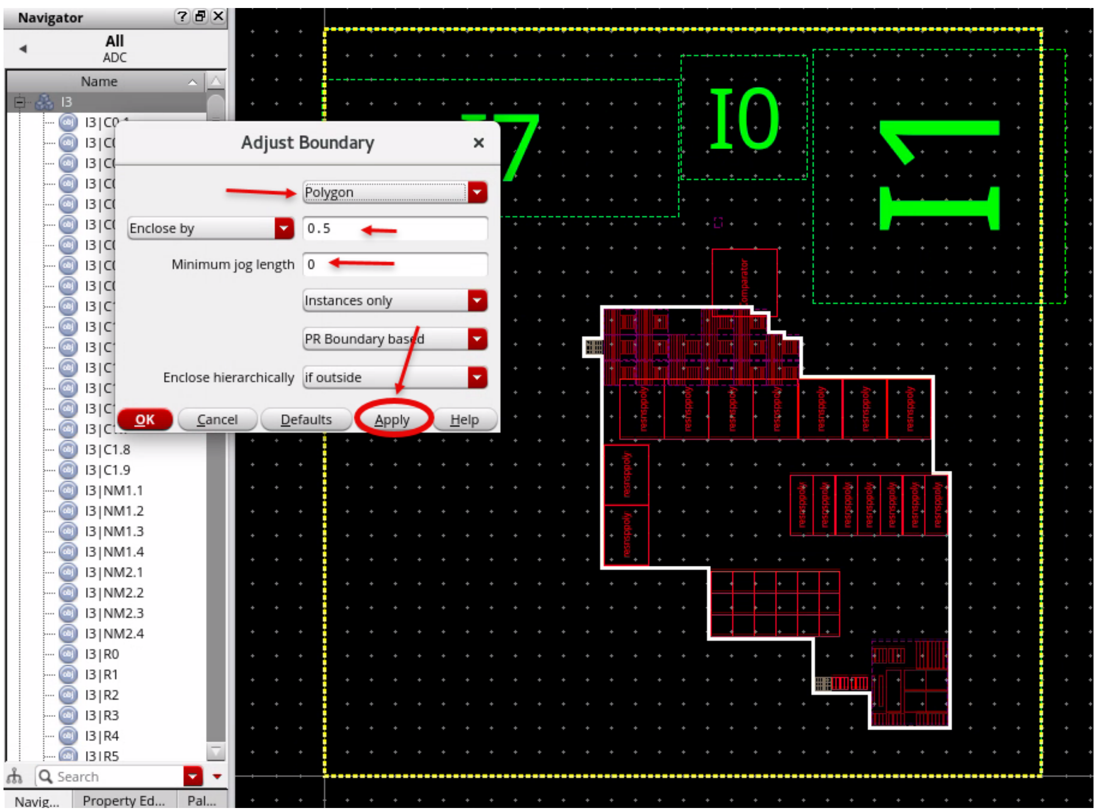
28. (Optional Exercise) Try adjusting the area boundary with these options after undoing the previous adjust area boundary edits as time permits.

29. **Undo** the previous adjust area boundary edits.

    You can also adjust a boundary to be a polygon, if required.

30. In the Adjust Boundary form, change the top field to *Polygon* and click **Apply**.

    The area boundary is adjusted to a polygon surrounding the contents of the virtual hierarchy, as displayed here.

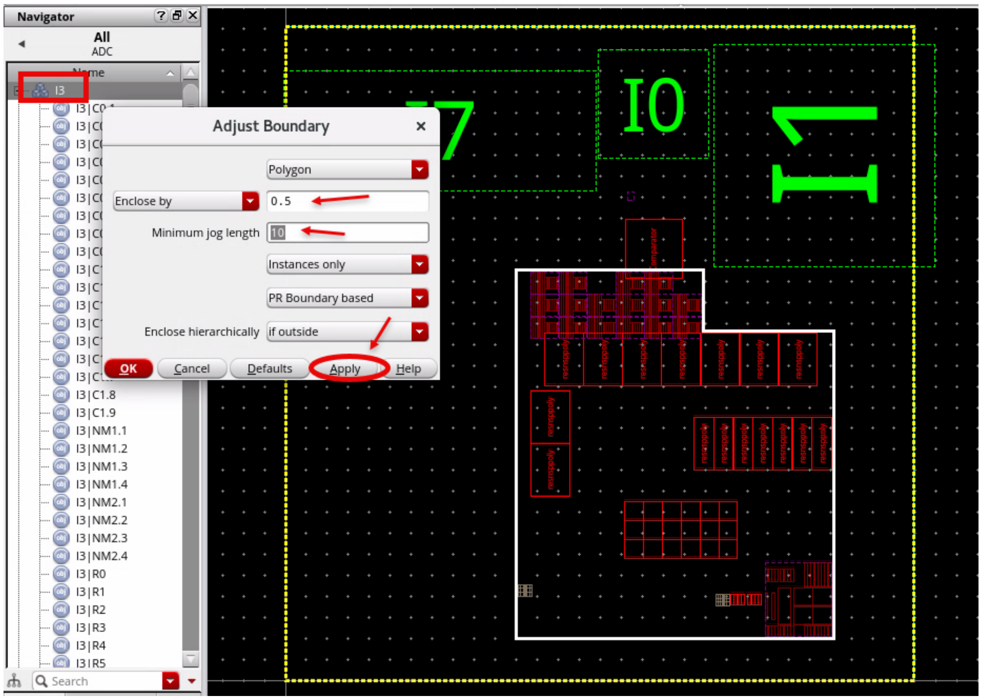


A rectilinear area boundary is created, but it has several notches, which may be undesirable.

When controlling the polygonal area boundary, you can control the minimum jog length to smoothen out unwanted notches in the boundary.

31. In the Adjust Boundary form, set the *Minimum jog length* value to **10** and click **Apply** again.

A rectilinear area boundary is created without notches, as shown here.



32. **Cancel** the Adjust Boundary form.

    The steps outlined in this lab were intended to introduce the manual editing capabilities available. You have only completed some limited moves and boundary resizing, but other commands are available to help you with design planning, such as aligning or abutting devices and creating Modgens or guard-rings. Take some time to explore the interaction further as time permits.

    In this lab, you completed the tasks that the engineer responsible for the block implementation would follow during early design planning.

33. **Close** the layout window without saving.

34. Keep the schematic window open and Virtuoso® session running for the next lab.

Summary

In this lab, you learned:

- How the virtual hierarchy can be manipulated into a suitable design plan in the early design planning phase.

End of Lab

Module 5: Using the Design Planner in the Team Environment

Lab 5-1 Creating Rows and Running Automatic Placement

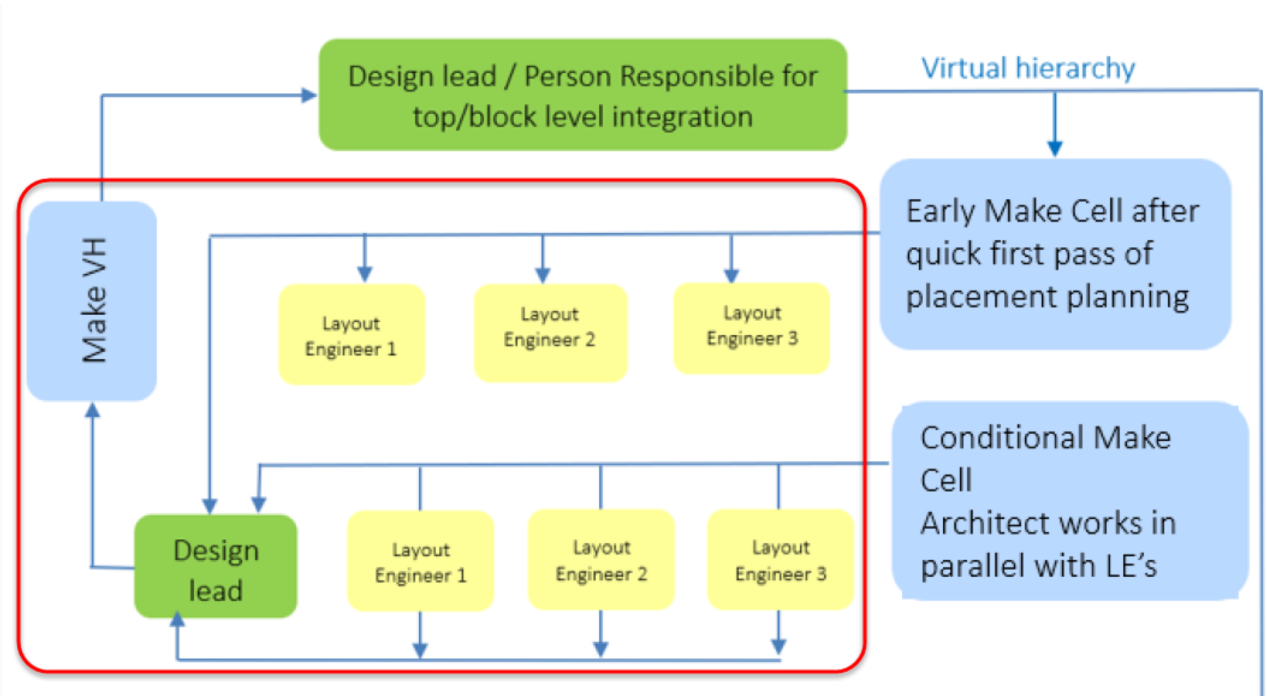
Objective: To create rows and running automatic placement in the design.

In this lab, you explore how to create rows and run automatic placement in the design.

Using the Design Planner in the Team Environment Phase

In the previous module, *Using the Early Design Planning Phase*, you completed the tasks that the engineer responsible for the block implementation would follow early in design planning.

In this module, *Using the Design Planner in the Team Environment Phase*, you will perform the steps that should be used to hand over a virtual hierarchy to a Layout Engineer for implementation.



1. The schematic window should be open from the previous lab. If not, re-open it now.

2. From the Library Manager or the Open File form in the CIW window, open the **ams_class_design/ADC/layout_place** view.

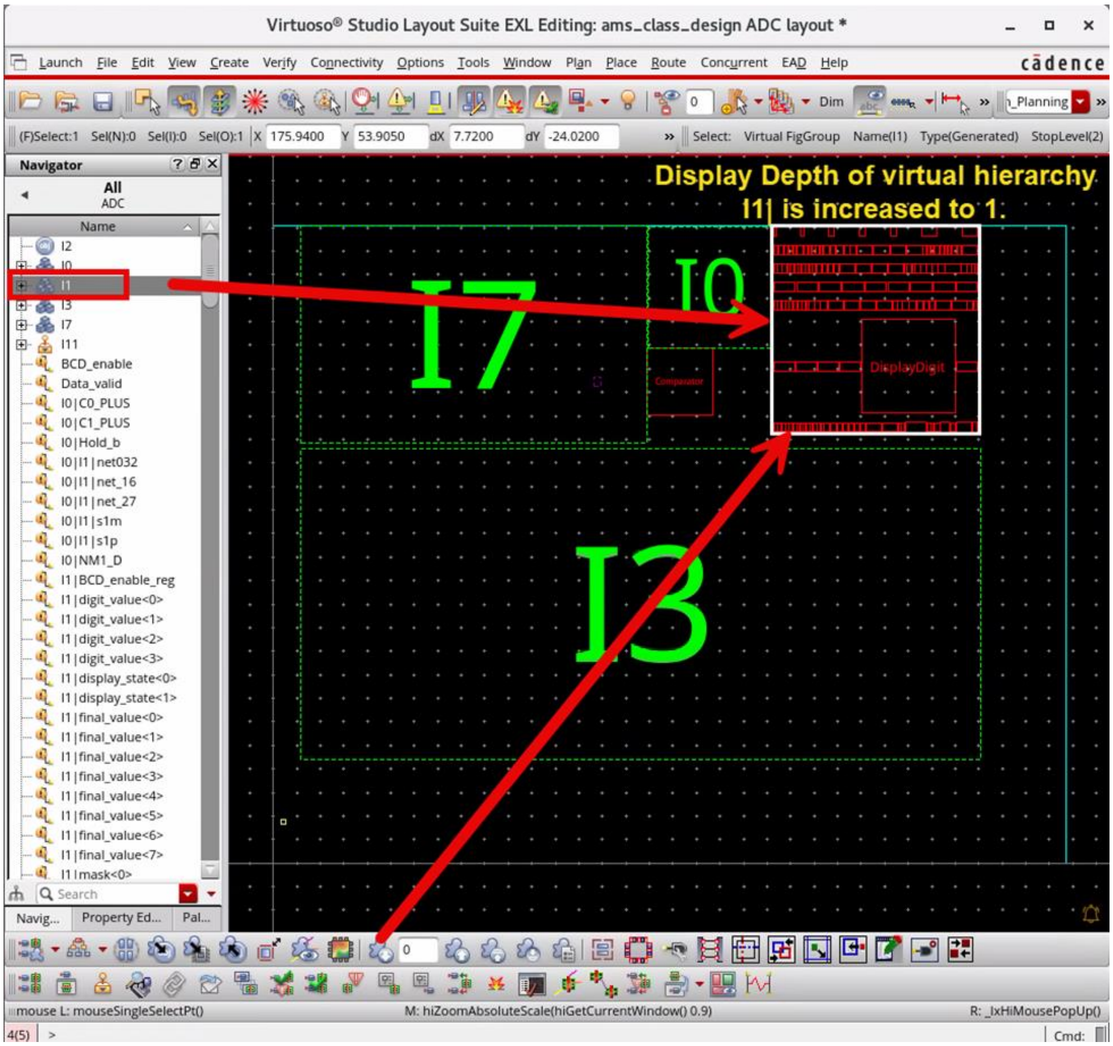
   Note: To open this design, you must be in the Layout EXL as the design contains virtual hierarchies, which are supported only in the Layout EXL from the ICADVM18.1 version onwards.

   By default, the layout view opens in the *Classic* workspace.

3. From the Workspace Configuration pull-down, select the *Design_Planning* workspace.

   Note: In the layout canvas, choose the **Window – Toolbars** menu to display the **Placement** toolbar if it is not already displayed.

4. Select the virtual hierarchy **I1** and increase the Display Depth to **1** to see its contents, as shown here.



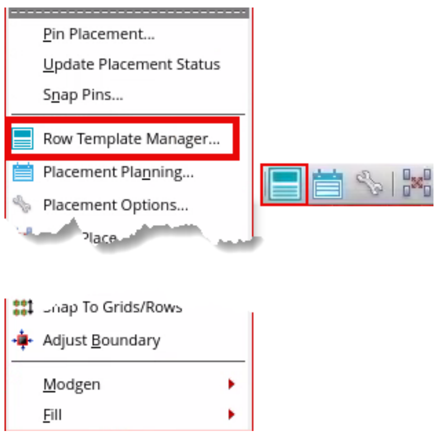
Using the Row Template Manager to View the Row Templates

I1 consists of standard cells that can be automatically placed. Automatic Placement using the *Virtuoso® Custom Digital Placer* (*VCP*) has been enhanced to be virtual hierarchy aware. Before running the *Virtuoso Placer*, you must have generated the rows in the layout canvas. Row generation involves the following steps, namely:

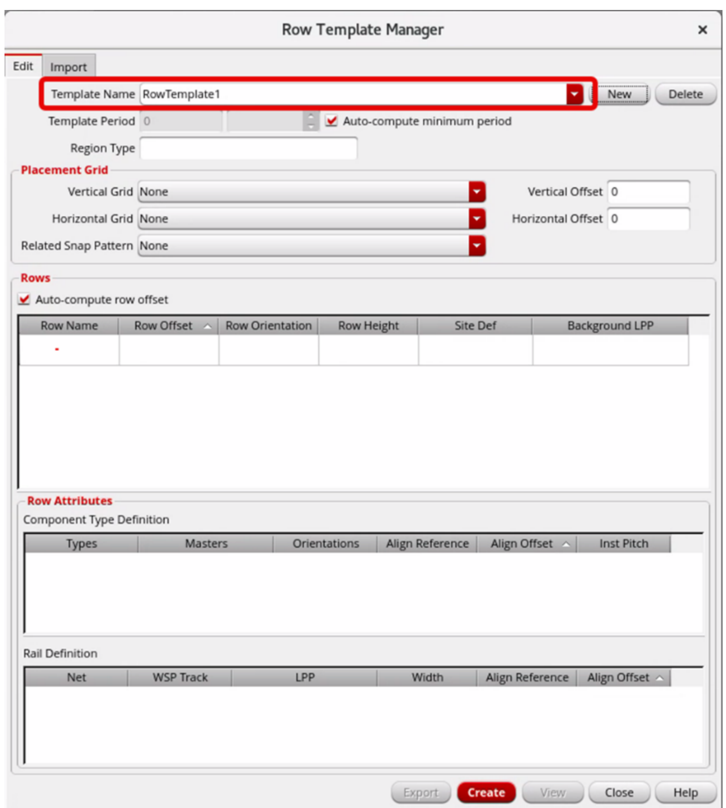
- Defining one or more row templates: Use the Row Template Manager to define the row templates.

- Defining the area where rows should be generated: Use the **Region** tab of the *Placement Planning* form to specify the area (*Boundary*) the rows must be generated.

1. In the layout canvas, choose the **Place – Row Template Manager** menu command or click the **Row Template Manager** icon in the Placement toolbar to open the Row Template Manager form.

   The Row Template Manager form appears, as shown here.

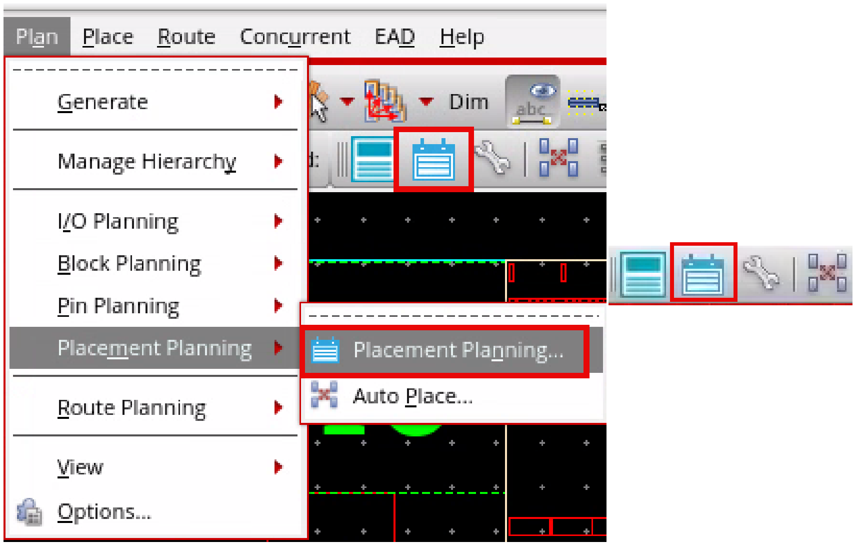
   You can see that **RowTemplate1** is already set up.

2. After reviewing the template, **Close** the Row Template Manager form.

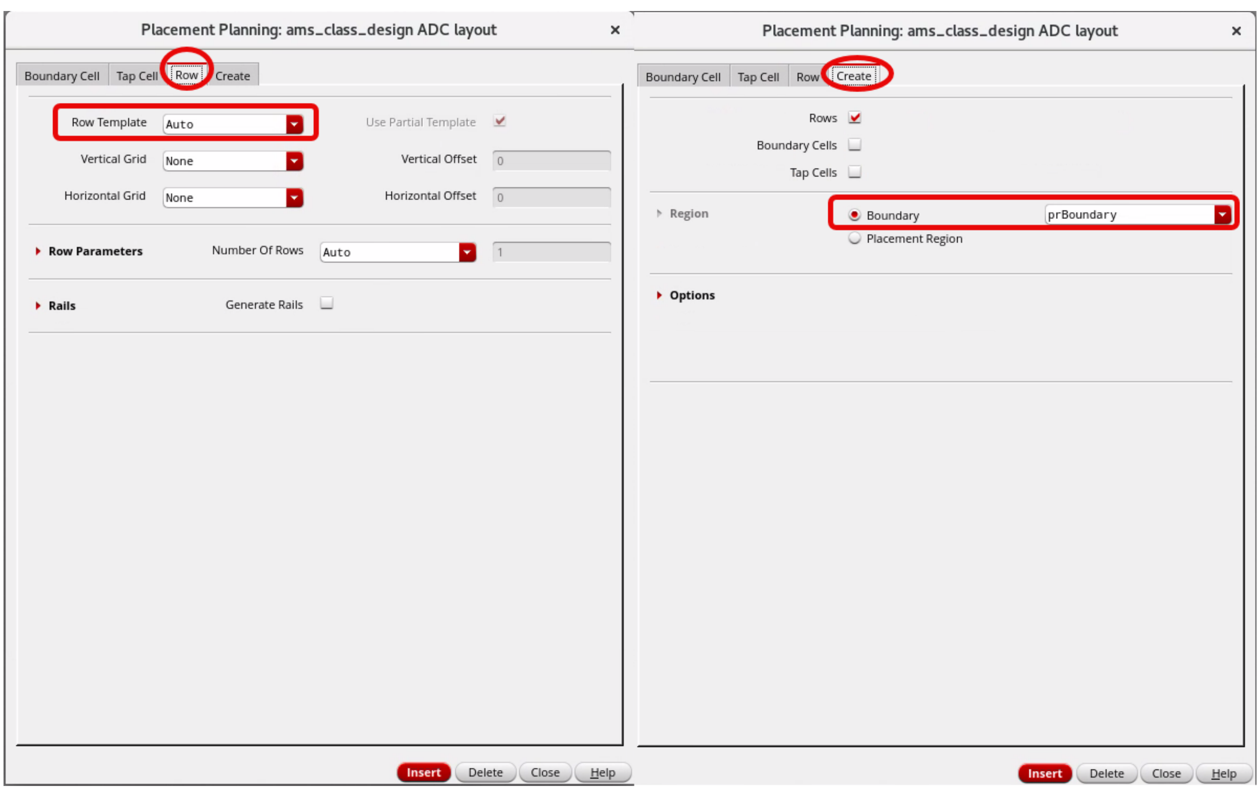
Using the Placement Planning Form to Create the Rows

After creating the row templates, you can use the *Placement Planning* form to generate the rows in the layout canvas.

1. In the layout canvas, choose **Place – Placement Planning** or **Plan – Placement Planning – Placement Planning** or click the **Placement Planning** icon in the Placement toolbar to open the Placement Planning form.



The Placement Planning form appears, as shown here.



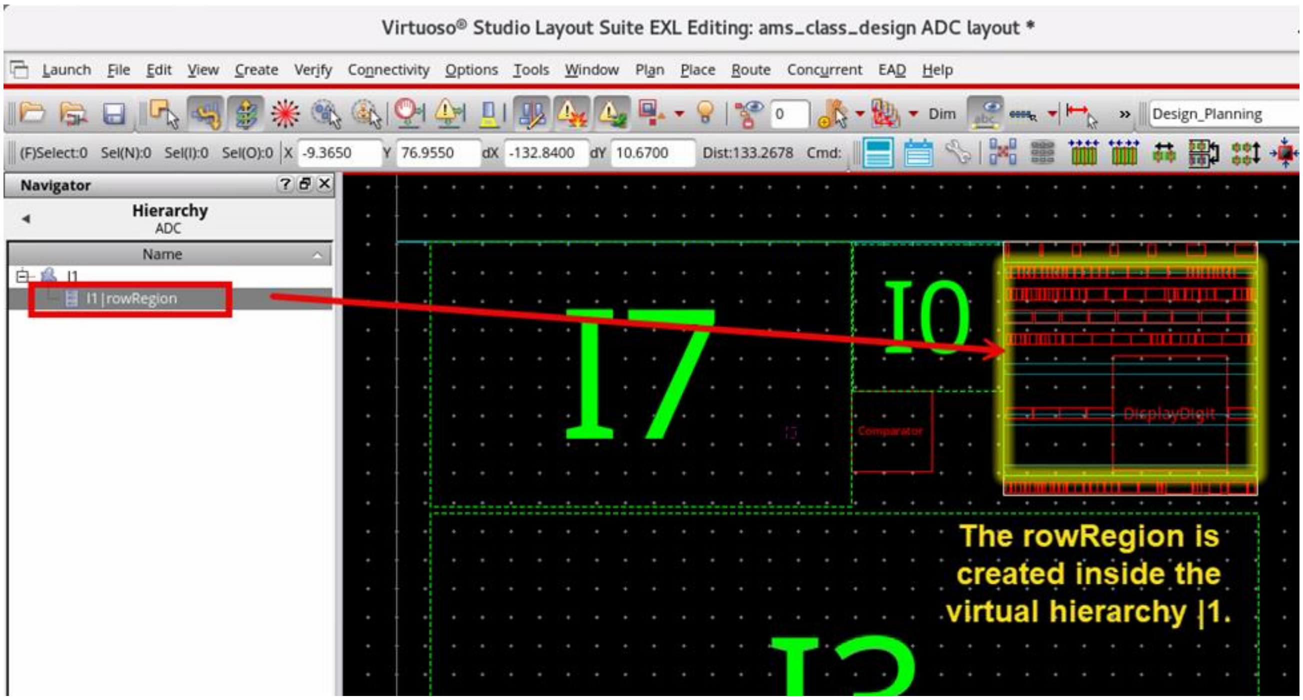
2. In the Placement Planning form, select the options, as shown here.

   a. In the **Row** tab, select **Row Template** as *RowTemplate1*.

   b. In the **Create** tab, select **Boundary** as **I1**.

   c. Keep the other tab values as default settings and click **Insert**.

   RowTemplate1 is used to create the row, and the **rowRegion** is created to correspond to the **I1** boundary, as shown here.



   The **rowRegion** is created inside the virtual hierarchy **I1**.

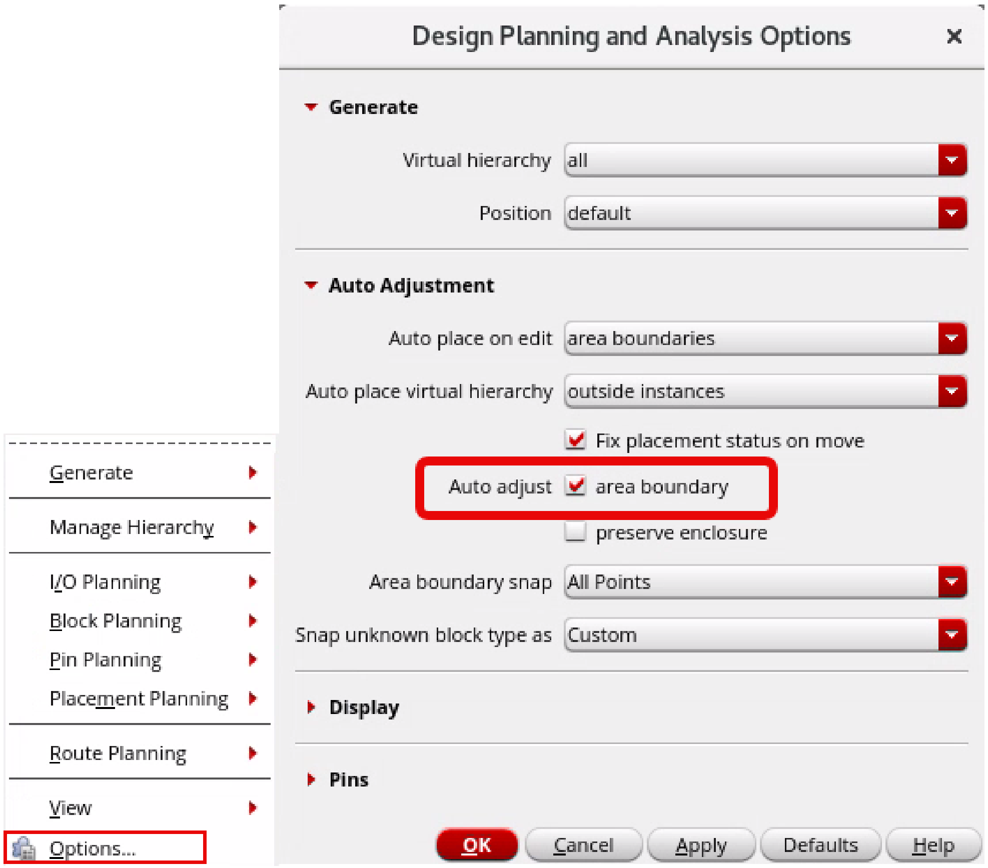
   You can see the *Number of rows created* in the CIW window.

3. **Close** the Placement Planning form.

Using the Design Planning and Analysis Options Form to Control the Area Boundary

Now you will place the standard cells in the virtual hierarchy **I1** using the automatic placement, which is performed by the *Virtuoso Placer*. **I1** has a polygonal boundary. You must ensure a rectangular boundary is not created for a polygonal virtual hierarchy.

1. In the Layout EXL canvas, choose the **Plan – Options** menu command to open the Design Planning and Analysis Options form.



The Design Planning and Analysis Options form opens, as shown here.

2. In the Design Planning and Analysis Options form, do as follows:

   a. Deselect the **Auto adjust area boundary** option.

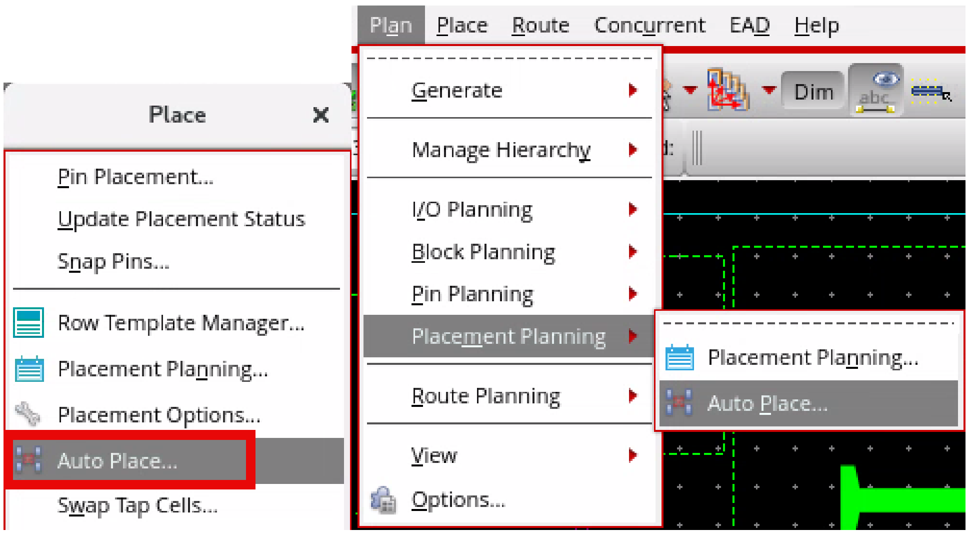
   b. Click **OK** in the form.

   This will ensure a rectangular boundary is not created for a polygonal virtual hierarchy.

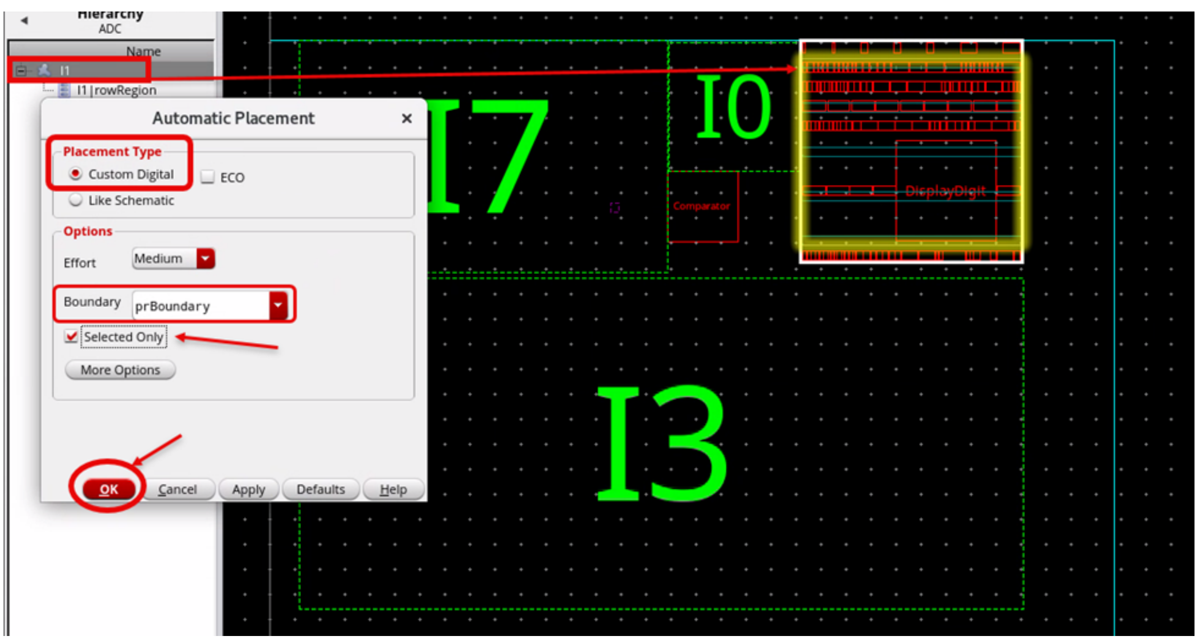
Running the Automatic Placer to Place the Standard Cells

Now you will place the standard cells in the virtual hierarchy **I1** using the Automatic Placer. Automatic placement is performed by the *Virtuoso Placer*.

1. In the Layout EXL canvas, choose the **Place – Auto Place** menu command or the **Plan – Placement Planning – Auto Place** menu command or click the **Auto Place** icon in the Placement toolbar to open the Automatic Placement form.



The Automatic Placement form appears, as shown here.



2. In the Automatic Placement form, do as follows:

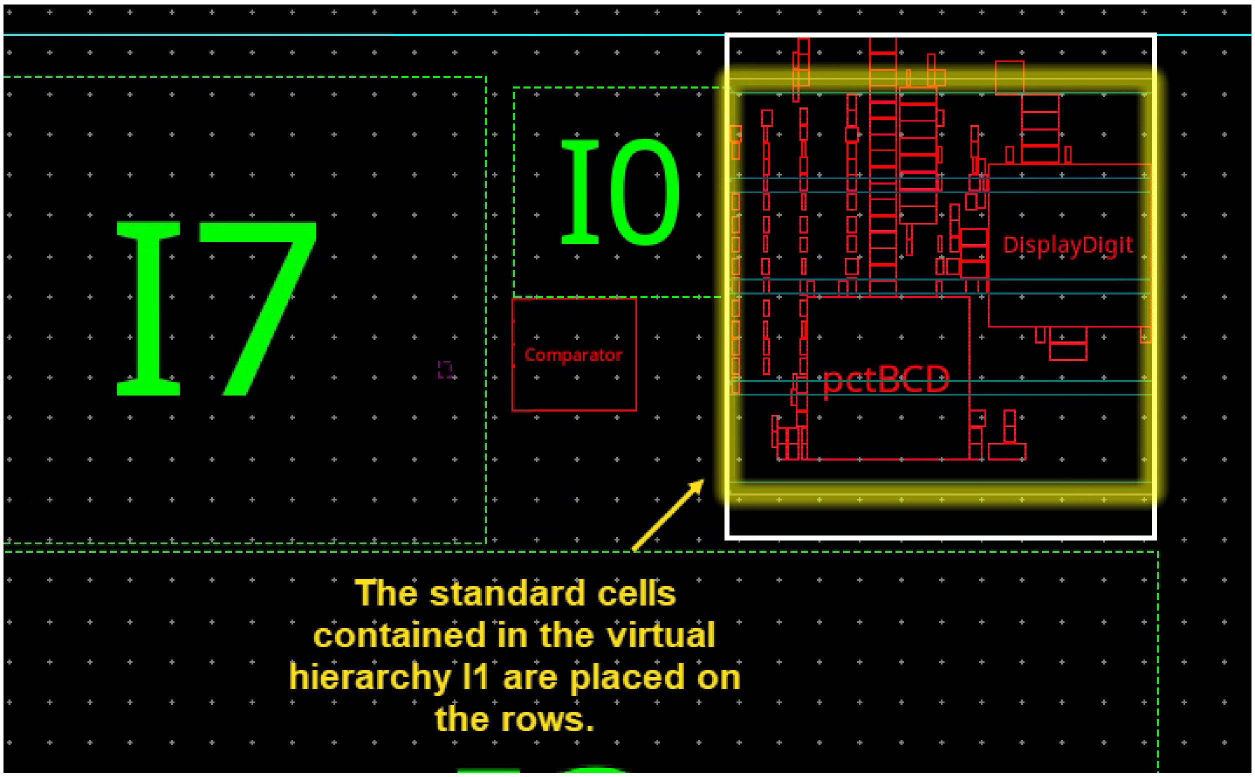
a. Choose the Placement Type as **Custom Digital**.

b. Set the Boundary to **I1|rowRegion**.

c. Enable the **Selected Only** option.

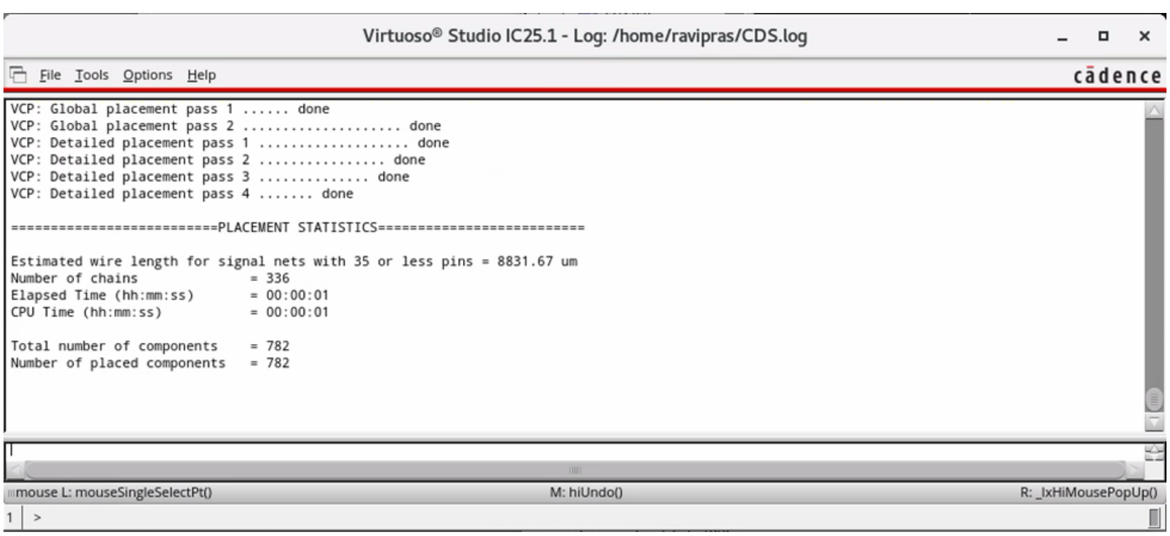
d. Click **OK** to run the placement.

The *Virtuoso Custom Digital Placer* (*VCP*) runs and places the standard cells contained in the virtual hierarchy **I1** on the rows, as shown here.



The custom digital placer runs, and the standard cells contained in the virtual hierarchy **I1** are placed on the rows.

3. Observe the Placement Statistics report in the CIW window.



The Placement Statistics report is generated dynamically in the CIW window as you run the automatic placer.

The Placement Report is available in the *CDS.log* file as well. Ignore any warning messages.

4. In the layout canvas, choose the **File – Save** menu command or click the **Save** icon in the File toolbar to save the cellview.

5. In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to make all the virtual hierarchies opaque.

6. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned how to:

- Create rows and run automatic placement in the design.

End of Lab

Lab 5-2 Running Congestion Analysis

Objective: To use the Congestion Analysis assistant to check for the availability of routing resources.

In the Early Design Planning module, you manipulated the virtual hierarchy and completed some design planning tasks aided by the virtual hierarchy enhancements. You can use the contents of the virtual hierarchy to create a footprint that is more realistic than one based only on estimates. In this lab, you explore how to use the Congestion Analysis assistant to check whether there are enough resources available for routing.

Setting Environment Variables to Run Congestion Analysis in the Design Planning Environment

To run Congestion Analysis in the design planning environment, you must update the following two environment variables to change their default settings:

```
ia cmapEnablePinOptimizeStep boolean t
rte genRoutingStyle cyclic “Chip Assembly”
```

Note: These environment variables are already set up in the *.cdsinit* file in your working directory. You can open and review them.

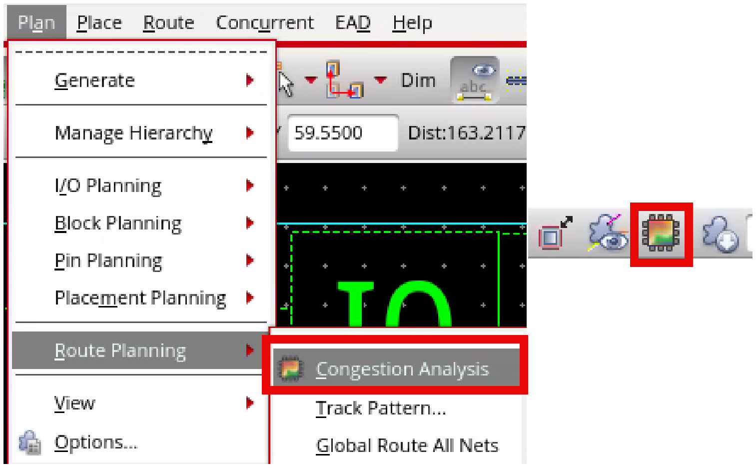
Note: The Pin Optimize option can be enabled from **Route – Design Setup – Congestion Analysis** and selecting the **Enable Pin Optimization** option in the Congestion Analysis form.

Note: Routing Style can be selected from **Route – Design Setup** and selecting the option from the *Design Style* field or from **Window – Assistants – Wire Assistant**, Automatic section, select the option from the *Design Style* field.

Once these variables are set up, they can be verified in the CIW, as shown here.

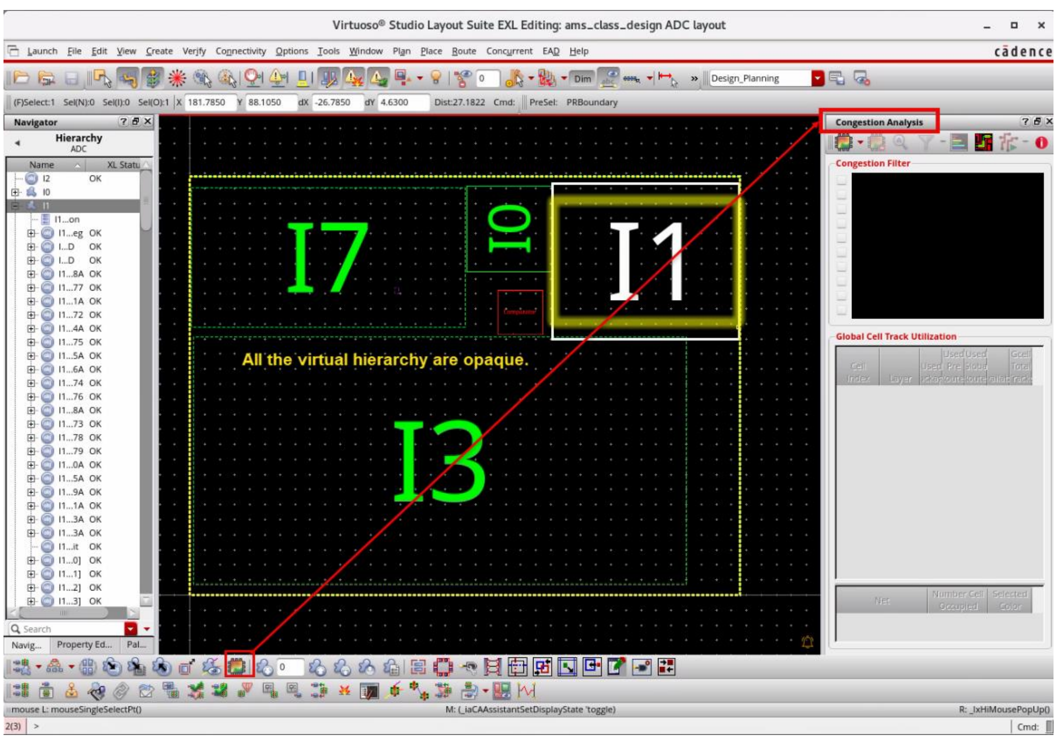
Activating the Congestion Analysis Assistant in Layout Canvas

1. In the layout canvas, choose the **Plan – Route Planning – Congestion Analysis** menu command or click the **Congestion Analysis** icon in the Design Planning toolbar to open the Congestion Analysis assistant in the layout window.



Note: You can invoke the Congestion Analysis assistant from **Window – Assistants – Congestion Analysis** as well.

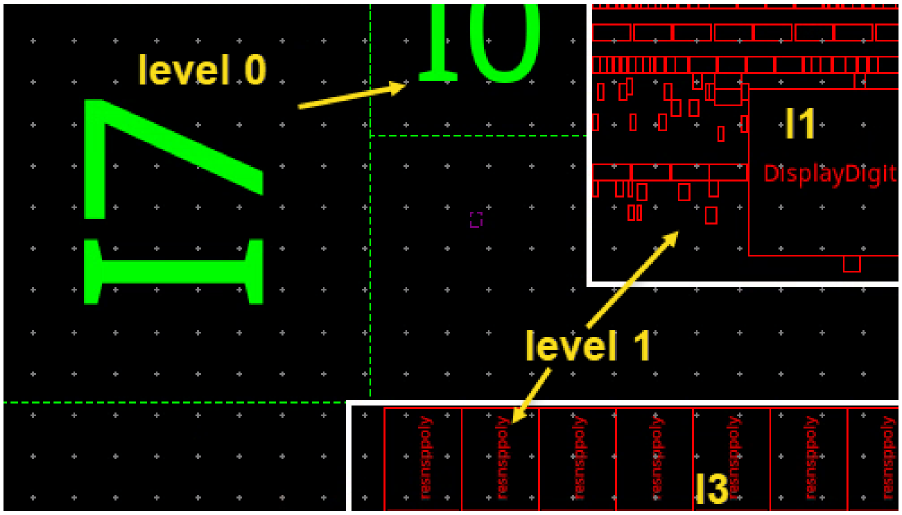
You see the layout window with Congestion Analysis assistant enabled here.



2. Set the Display Depth for the virtual hierarchies as follows:

   a. **Level 1** for the virtual hierarchies **I1** and **I3**

b. **Level 0** for the other virtual hierarchies

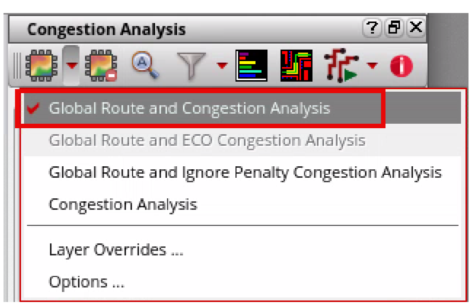


Important: The virtual hierarchy display depth level is important when running the Congestion Analysis.

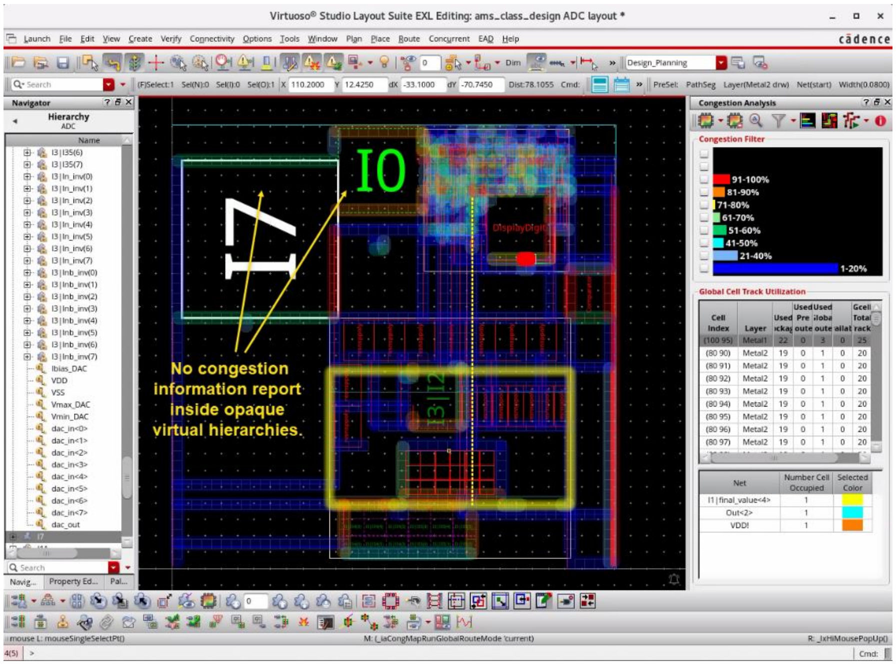
- If the contents of a virtual hierarchy have not been placed, it is recommended to set the level to **0**, i.e., opaque. In this case, the virtual hierarchy is considered as a blockage and congestion information is reported only for the interface nets of the virtual hierarchy.
- If the virtual hierarchy is visible, the congestion information for its contents is reported. This is recommended if you have completed some placement within the virtual hierarchy and would like to check the congestion.

Running Global Route and Congestion Analysis

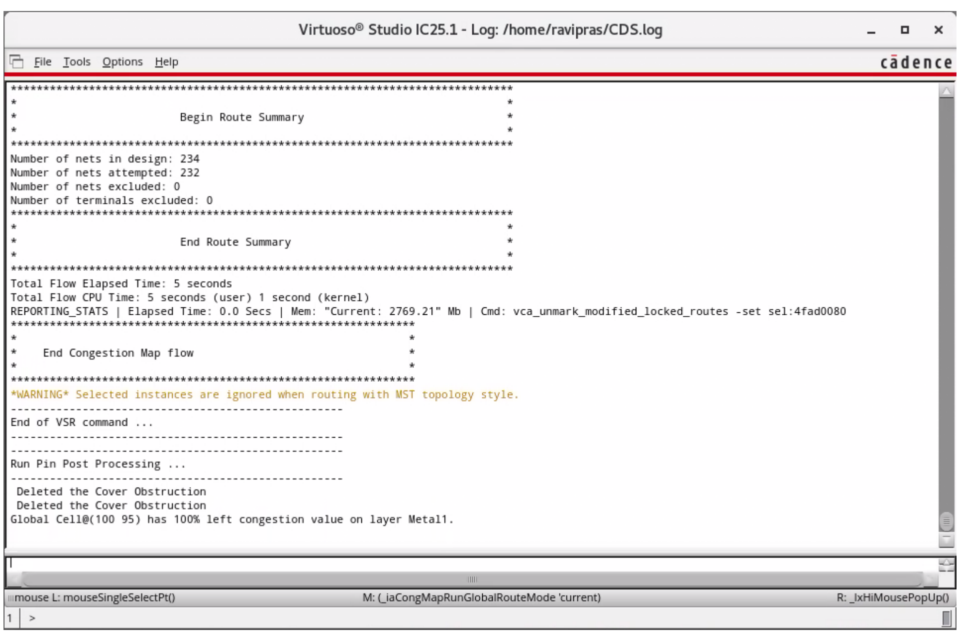
1. In the layout window, in the Congestion Analysis assistant, click the top-left icon and select **Global Route and Congestion Analysis**, as shown here.



The Global Route command runs, and the results are displayed in the layout canvas as well as in the *Congestion Analysis* assistant. Color coding is used to show the level of congestion in each global cell known as GCell. A GCell is a global cell used for routing and congestion analysis.



2. Check the CIW window for the *Route Summary* information.



The *Route Summary* is available in the *CDS.log* file as well.

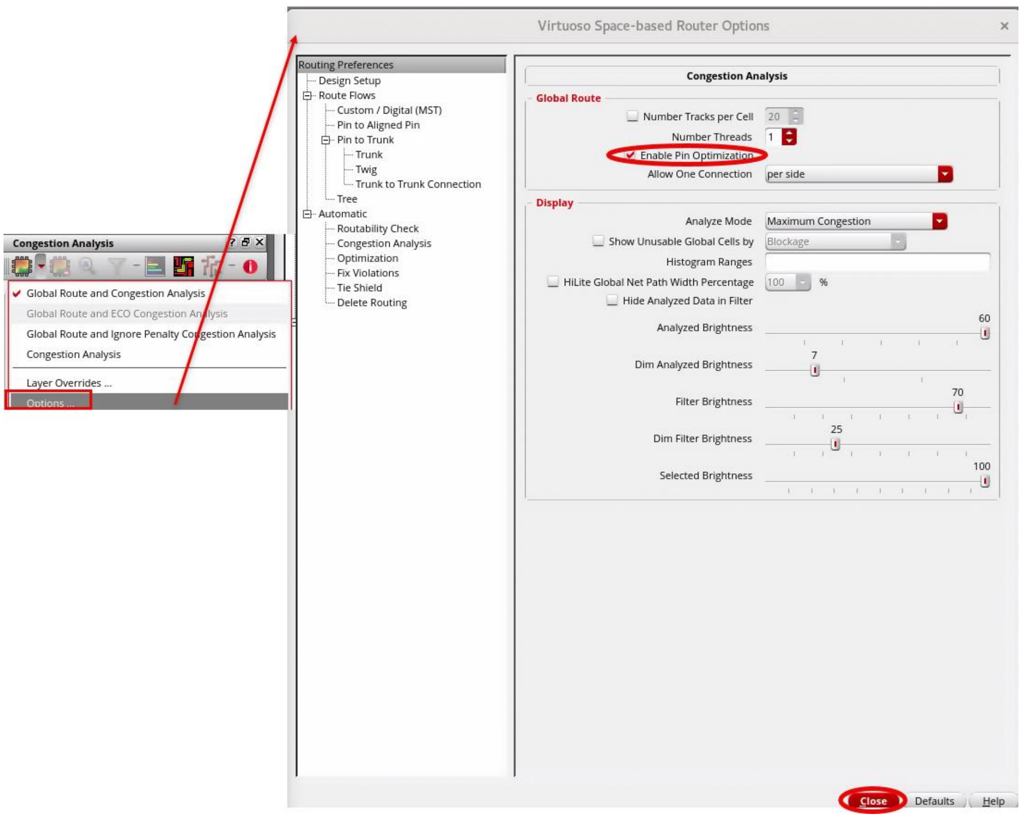
When a virtual hierarchy is opaque, only it's external or interface nets are globally routed. It is unnecessary to route the contents as they have not been placed yet, so the congestion is of limited value.

For opaque virtual hierarchies, the pins are temporarily elongated around the area boundary so that the router can make connections for the interface nets at the optimum position in the context of the top level.

3. Pin Optimization can be enabled from **Congestion Analysis – Options** to open the Virtuoso Space-based Router Options form.

   a. Select the **Enable Pin Optimization** option.

   b. **Close** the form.

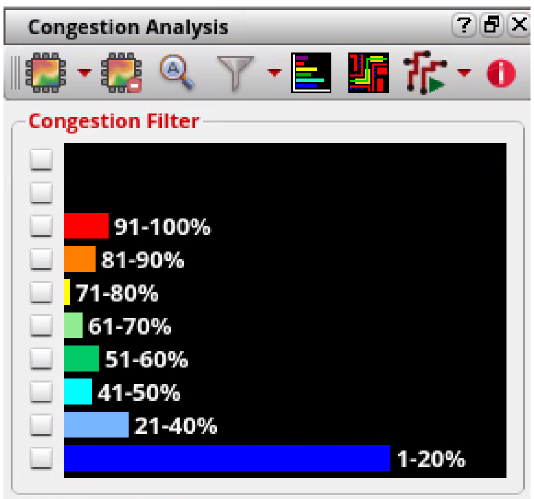


Note: Pin optimization can be enabled in the *.cdsinit* file as well.

Analyzing the Congestion

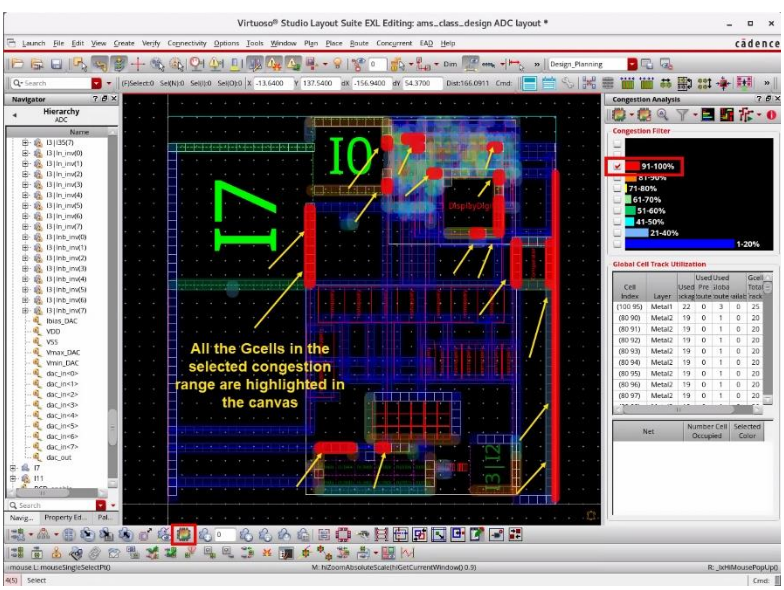
The Congestion Analysis histogram shows the breakdown of the congestion across the GCells. The results are displayed in ranges that can be filtered, allowing you to review only the areas of interest. The design plan in the example here is not heavily congested, and the majority of GCells reflect congestion in the lower ranges. This shows that the footprint may be too large and that the design area could be reduced.

You can use this information for two purposes: To ensure that there are enough routing resources between the virtual hierarchies for the top-level routing and that the virtual hierarchies themselves are not under or over congested.

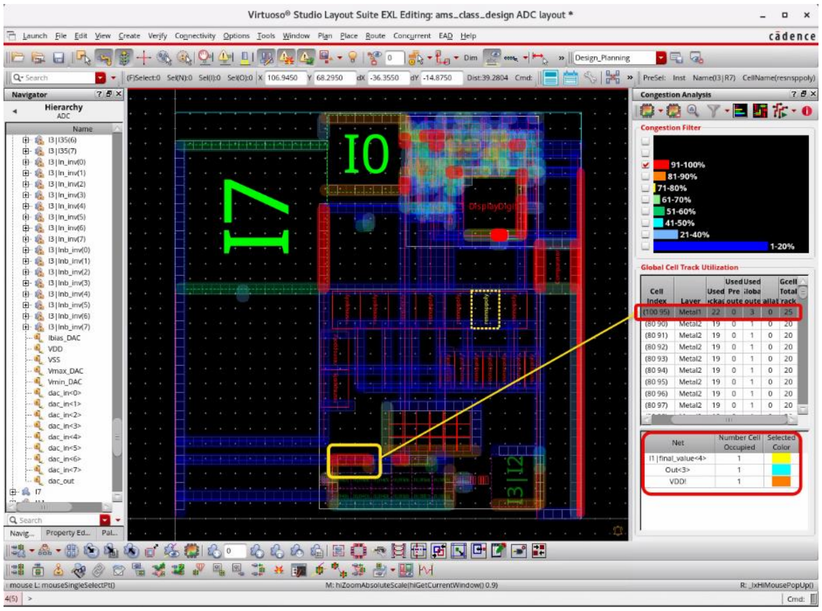


1. In the *Congestion Analysis* assistant, click the **91-100%** checkbox in the *Filter by Histogram* pane.

   All the GCells in the selected congestion range are highlighted in the canvas.



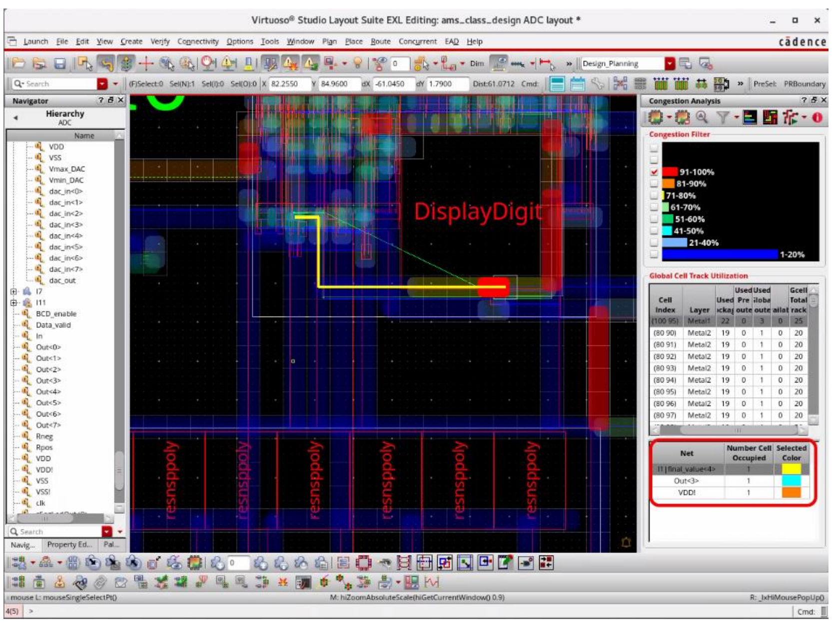
2. Select the top entry in the *Global Cell Track Utilization* table as highlighted in the graphic here.



The GCell is highlighted in the layout canvas, and the breakdown of the congestion is shown in the utilization table. The bottom pane shows the nets that route through the selected GCell.

3. Select the nets shown in the bottom pane of the *Congestion Analysis* assistant.

The selected nets are displayed in the layout canvas and the Navigator assistant, as shown here.



This congestion information can be used to refine the design plan in several ways. You can:

- Resize the virtual hierarchies to decrease (or increase) the congestion, either between or inside the virtual hierarchies.
- Move instances or blocks at the top level or inside the virtual hierarchy to get a more balanced congestion score.
- Set global bias constraints to push the nets away or to pull them towards an area.

Note: More information on Congestion Analysis and using global bias constraints can be found in the Congestion Assistant Workshop in the Cadence® Learning and Support portal.

4. **Close** the *Congestion Analysis* assistant by clicking the **Hide** (**X**) button in the top-right corner of the layout canvas.

5. In the layout canvas, choose the **File – Save** menu command or click the **Save** icon in the File toolbar to save the cellview.

6. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned how to:

- Use the Congestion Analysis assistant to check whether there are enough resources available for routing.

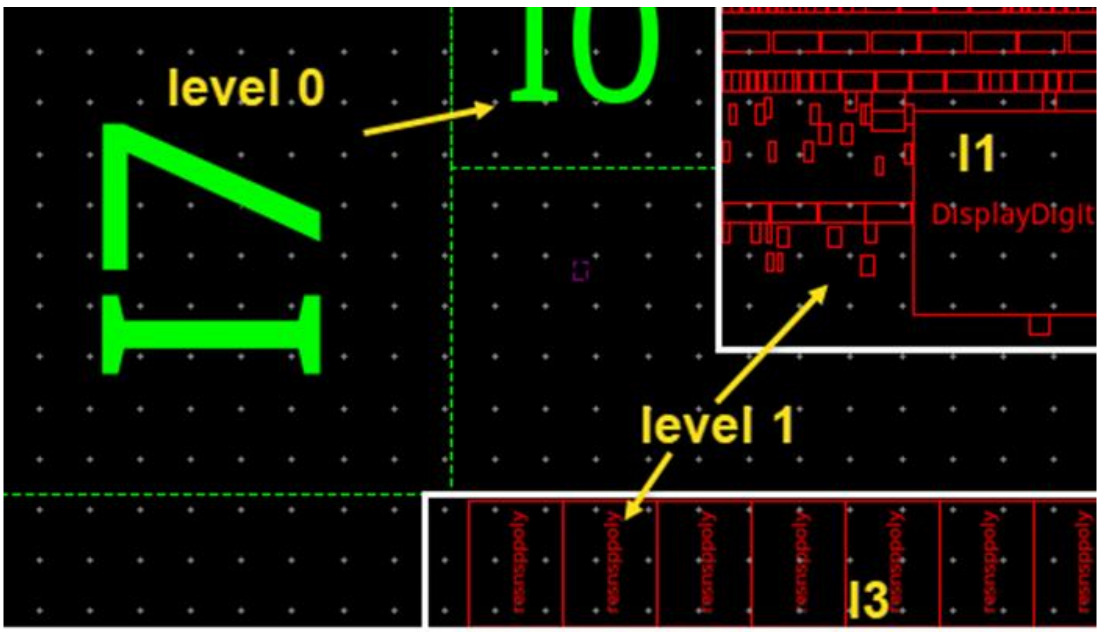
End of Lab

Lab 5-3 Creating Real Layout Views for the Virtual Hierarchies

Objective: To create real layout views for the virtual hierarchies.

In the previous lab, you checked the congestion inside and around the virtual hierarchy, which an engineer responsible for the block implementation is likely to do. The next step is to create layout views for the virtual hierarchies so that other engineers in the design team can work on them. These layouts need to have a PR Boundary and pins for the interface nets created in suitable positions. In this lab, you explore how to create real layout views using the Make Cell and Remaster commands.

1. Make sure the virtual hierarchy display depth is set to *Level 1* for **I1** and **I3** and *Level 0* for the other virtual hierarchies from the previous lab. If not, set it now.



Important: Some virtual hierarchies have a solid line rather than a dashed line. This is to indicate that they have a fixed placement status. You can check in the Property Editor or **right**-click the virtual hierarchy context menu and observe that virtual hierarchies **I0**, **I2** & **I3** have "Fixed" placement status.

Running *Make Cell* with the *Create pins: Congestion aware* Option

At this stage of the design planning process, it is unlikely that the contents of the virtual hierarchy have been placed in their final positions. The contents are only contained within the boundary using a suitable utilization. This means that when creating pin positions, it does not make sense to consider their absolute positions. Reducing the virtual hierarchy display depth to **0** makes the virtual hierarchy opaque, allowing the *Create pins: Congestion aware* option in the Make Cell form to create pins anywhere on the boundary. This is recommended for the virtual hierarchies that are not yet placed.

1. Follow the steps given here to create a layout view using the Make Cell command.

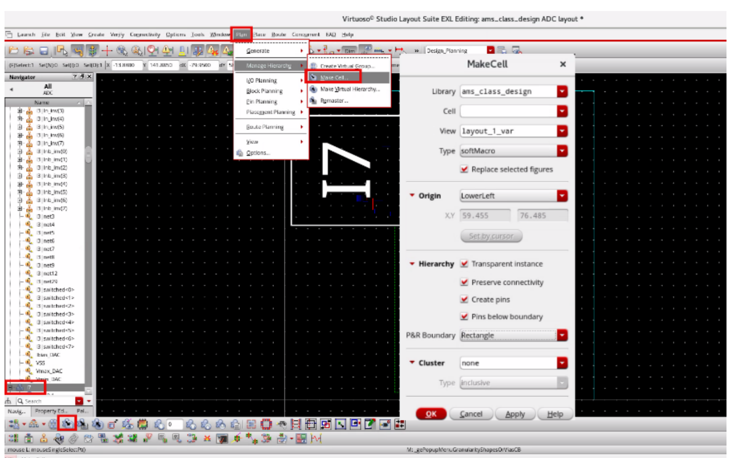
   a. In the layout canvas, select the virtual hierarchy **I7**.

   Note that the virtual hierarchy is represented using an amoeba-shaped icon.

   b. Click the **Make Cell** icon in the Design Planning toolbar or choose the **Plan – Manage Hierarchy – Make Cell** menu command to open the Make Cell form.

   c. Set the options, as shown here in the Make Cell form:

   d. Click **OK** to run the Make Cell command.

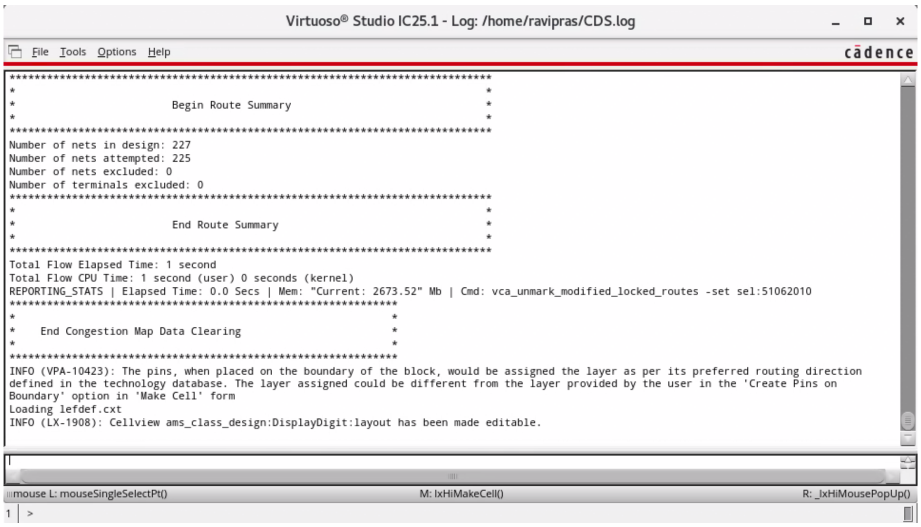
   Running the Make Cell command on the selected virtual hierarchy **I7** creates a layout view for **I7** using the selected options. The area boundary is updated to a PR Boundary.

   When running **Make Cell** on a virtual hierarchy clone, it is important to have the *All clones* option selected so that the operation applies to all the clones in the family. If the option is not selected, different layouts are created for the same schematic master. It is recommended to always leave this option set to **true**.

Viewing the Results of the Make Cell Command

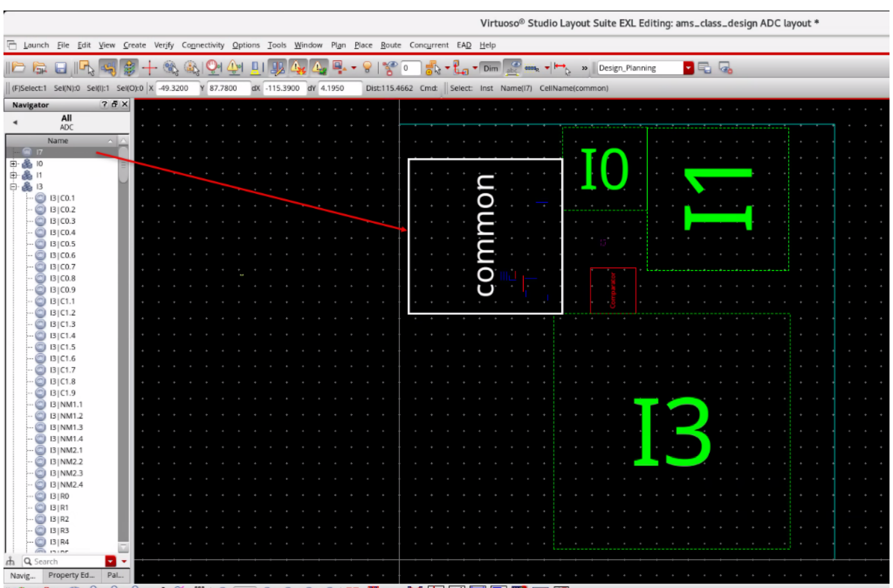
You will now examine the results of the Make Cell command.

2. Check the CIW window for the *Route Summary* information.



This information is available in the *CDS.log* file as well.

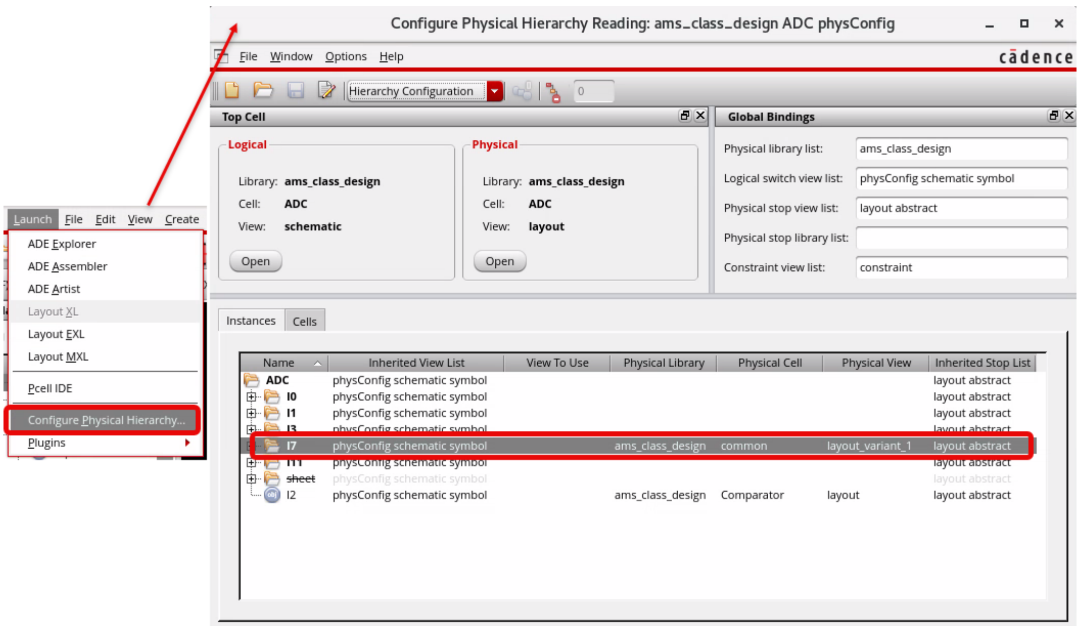
3. In the Navigator assistant, in the *Hierarchy* view, select **I7**.



You can notice that **I7** is no longer a virtual hierarchy. The icon in the Navigator assistant has changed to reflect that the virtual hierarchy is now an instance of a layout. The XL Status is **OK**, showing that the created instance is fully XL compliant. In the layout canvas, the instance is now colored red.

Note: The layout instance **I7** is now represented using the icon shown here in the Navigator assistant.

4. In the layout canvas, choose **Launch – Configure Physical Hierarchy** to open the Configure Physical Hierarchy (CPH) form.



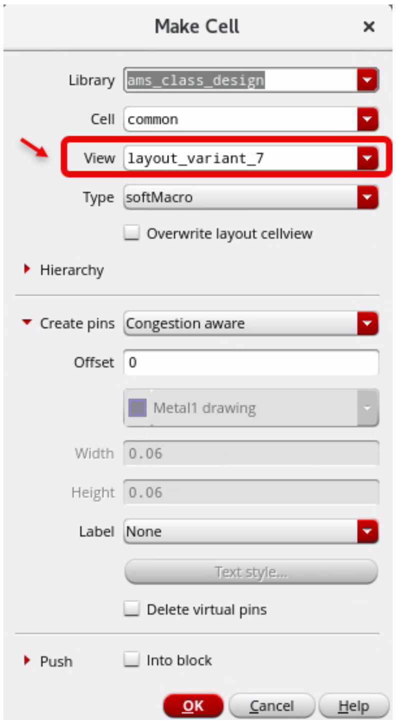
a. In the Configure Physical Hierarchy form, in the *Hierarchy Configuration* cyclic section, review the *Instances* tab.

   The *physConfig* view has been updated so that the newly created layout view automatically replaces the virtual hierarchy. A physical binding is created to match the options set in the Make Cell form.

b. Leave the CPH form open.

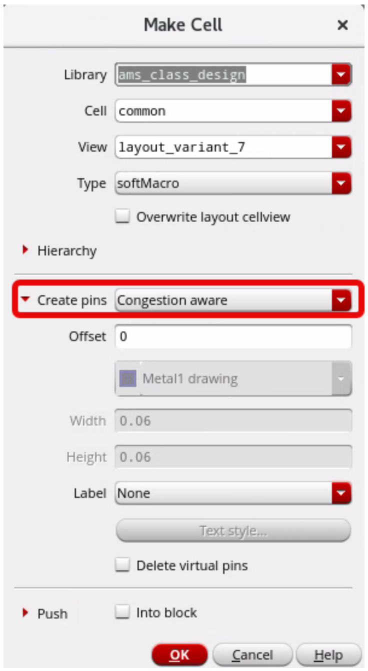
Layout Variant View

By default, Make Cell creates a layout variant view. You can edit the fields if you want the new layout variant to be created in a different name.



Make Cell with Create pins: C*ongestion aware* Option

When you select the **Create pins: Congestion aware** option in the Make Cell form, the global router is run, and pins are created on the boundary for the interface nets of the selected virtual hierarchy. Interface nets are the ones that cause the route to cross the virtual hierarchy boundary.



As the option name implies, the pin creation is congestion aware, and pins are created on the boundary with the layer determined by the router.

When a virtual hierarchy is opaque, the position of instances inside a virtual hierarchy is not considered. Instead, the pin targets are elongated around the boundary for the router to make connections at the closest point on the boundary. This is where the pins are eventually created.

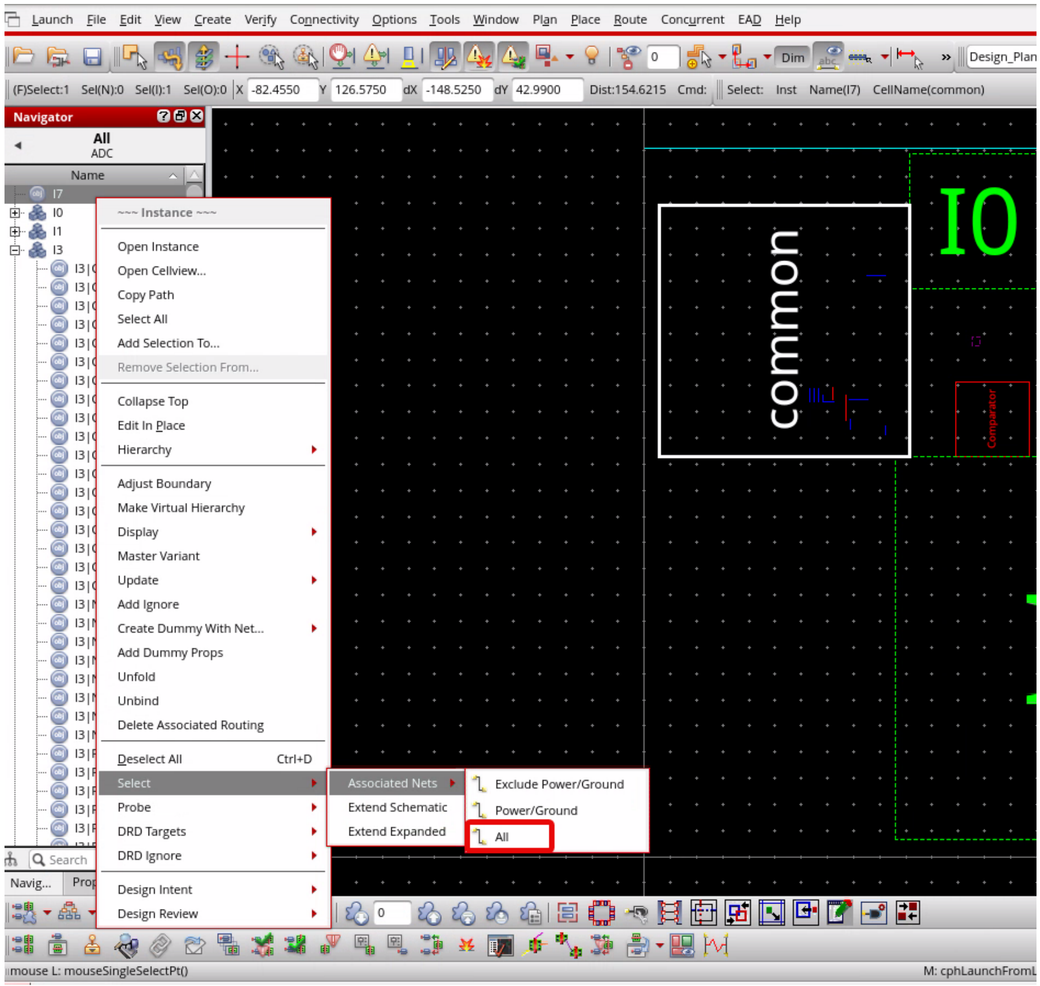
Viewing the Pins Created from the *Make Cell* with the *Create pins: Congestion aware* Option

You will view the pins created from the global routing when the **Make Cell** command with the *Create pins: Congestion aware* option is run.

1. Follow the steps given here to view the pins created.

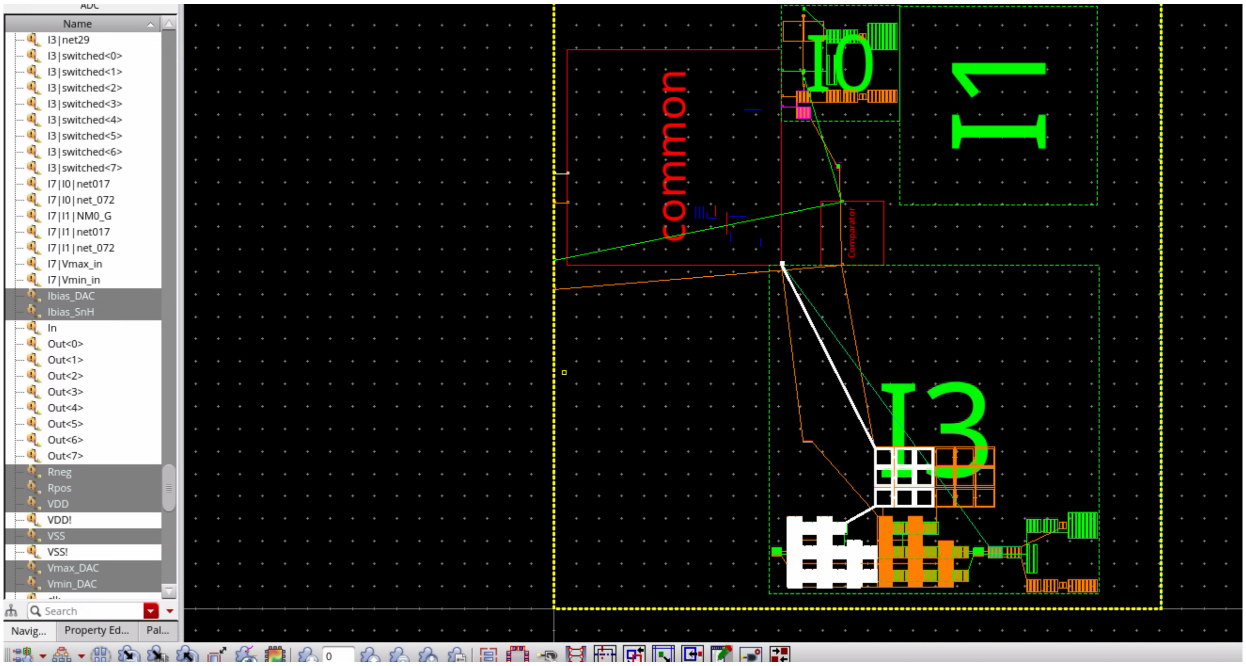
    a. In the layout canvas, select the newly created layout instance **I7**.

b. **Right**-click the selected instance **I7** and choose **Select – Associated Nets – All** from the context-sensitive menu, as shown here.



You can observe that pins are created on the boundary, attempting to minimize the top-level net length.

c. Choose the **View – Zoom To Area** menu command or press the bindkey **Z** to zoom near the layout instance **I7** to see the pins created.



d. Observe the **Nets** information in the Navigator assistant **OBJECTS** category as shown here.

This newly created layout can now be passed to the Layout Engineer who is assigned to implement the block, providing them with the boundary and pin positions defined in the context of the top level.

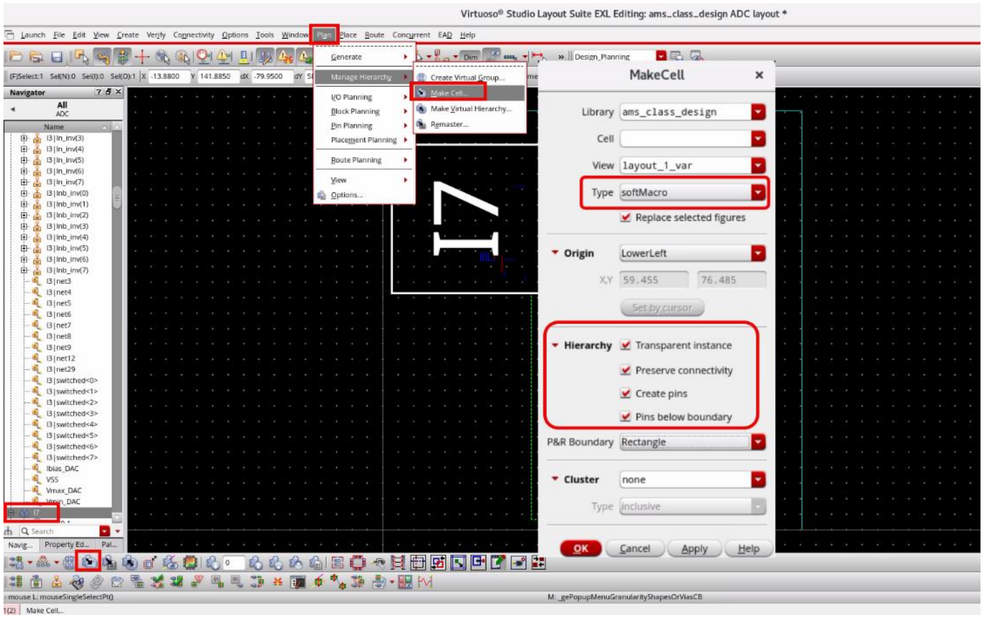
Running *Make Cell* with the *Create pins: Below boundary* Option

There are two other pin creation modes: *Below boundary* and *On boundary*. These modes do not run global routing and allow you to specify the pin layer and size. You will examine the *Create pins: Below boundary* option now.

1. **Undo** the Make Cell in the layout canvas that was completed in the previous section.

   I7 reverts from layout instance to virtual hierarchy again.

2. Select **I7**, open the Make Cell form to set the options shown below and click **OK**.



Note that the *Create pins* option is set to **Below boundary**. This enables the pin layer, width, height and label options, making them user-configurable. In this mode, the pins are created at the bottom corner so that further pin optimization may be required. The *Type* field is set to **softMacro** to enable pin optimization.

The *View* field in the Make Cell form is pre-populated with *layout_variant_2*. Each time the Make Cell command is run, the view name for the new variant is incremented to prevent the cellview from being overwritten. You can update the value in the field if needed.

I7 changes to the layout instance again after running the Make Cell command.

Viewing the Pins Created from the *Make Cell* with the *Create pins: Below boundary* Option

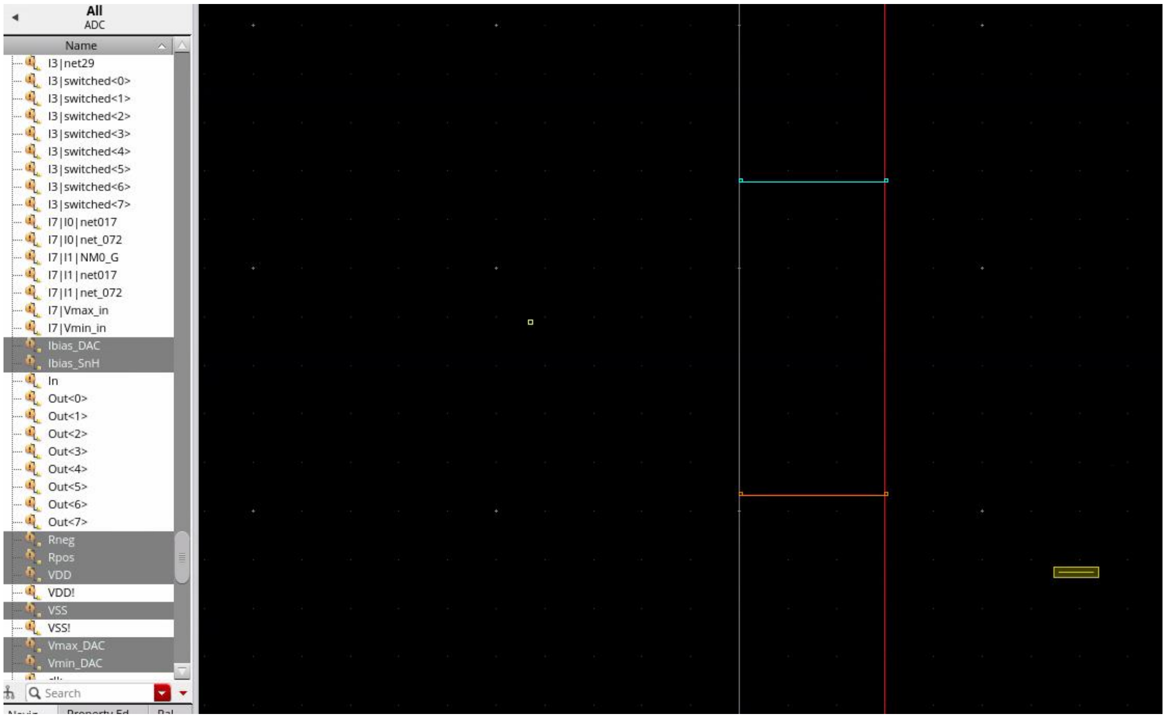
1. Follow the steps given here to view the pins created.

   a. In the layout canvas, select the newly created layout instance **I7**.

   b. **Right**-click the selected instance **I7** and choose **Select – Associated Nets – All** from the context-sensitive menu, as shown here.

   c. **Zoom** in to the bottom left corner of the layout instance **I7**.

You see the pins created.



Pin Optimization Using the Pin Placement Form

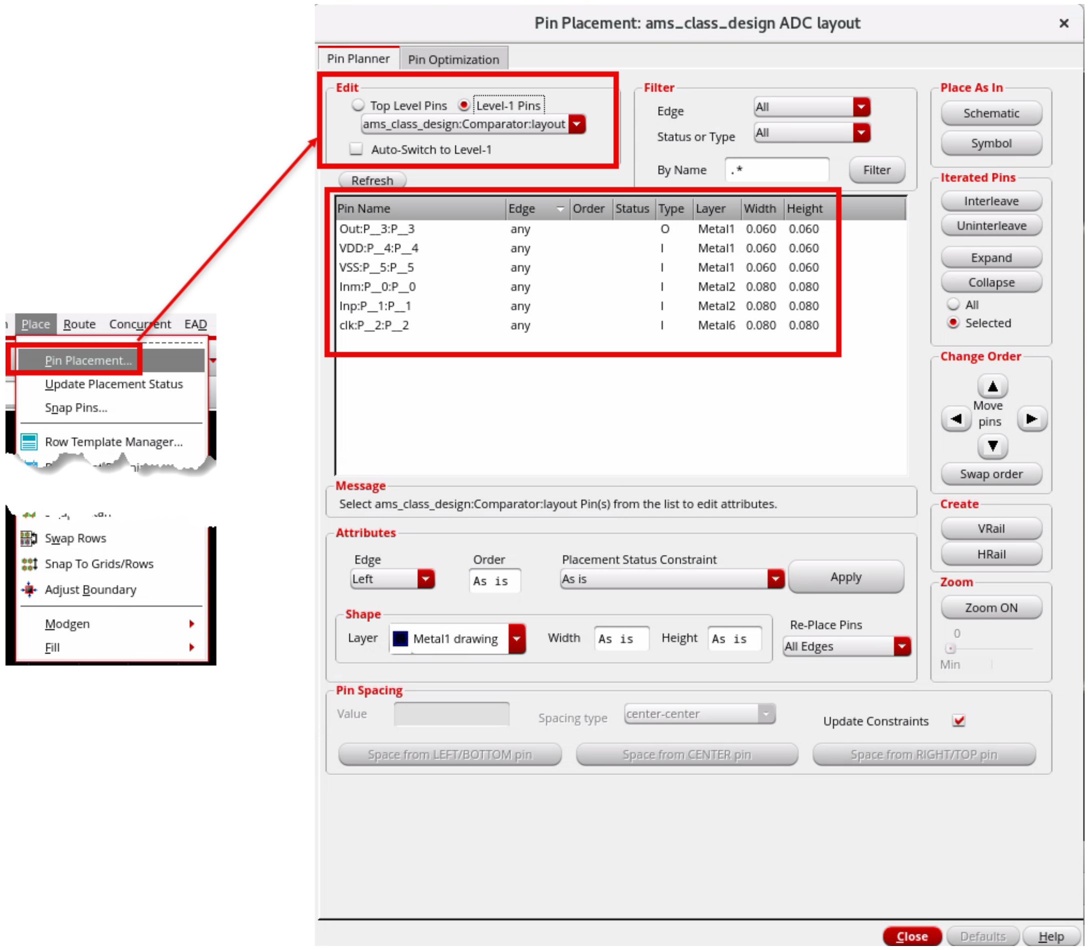
The pins created in this *Create pins: Below boundary* option mode will require further optimization before the layout is handed over for implementation.

1. Follow the steps given here to do Pin Optimization.

   a. In the layout window, select the layout instance **I7**.

   b. Choose **Place – Pin Placement** to open the Pin Placement form.

Tip: When running the Make Cell command, you set *Type* to **softMacro** to make the pins editable from the top level.



c. (Optional Exercise) The pin positions can be further edited after creation with the *Pin Planner*/*Pin Optimization* commands. Explore the options as time permits.

d. Close the Pin Placement form after you have explored these options.

Running *Make Cell* with the *Create pins: On boundary* Option

As you have seen earlier, there are two other pin creation modes: *Below boundary* and *On boundary*. These modes do not run global routing and allow you to specify the pin layer and size. You will examine the *Create pins: On boundary* option now.

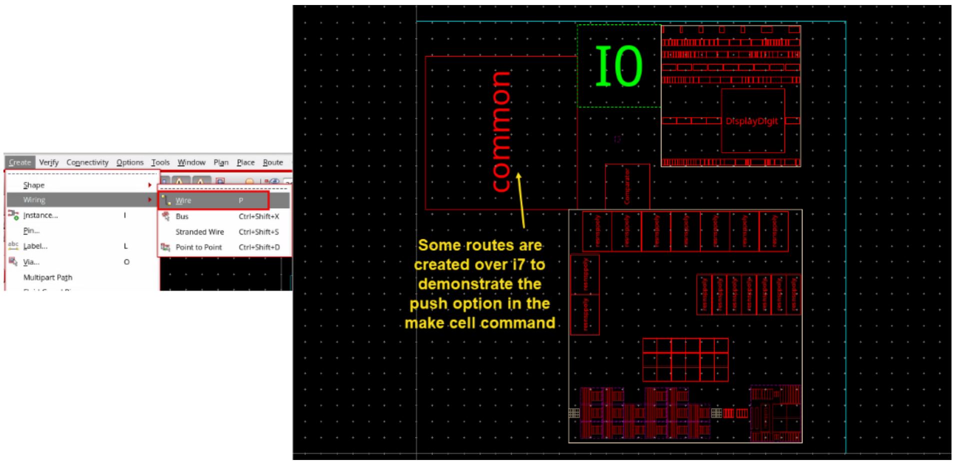
1. **Undo** the Make Cell in the layout canvas that was completed in the previous section.

   I7 reverts from layout instance to virtual hierarchy again.

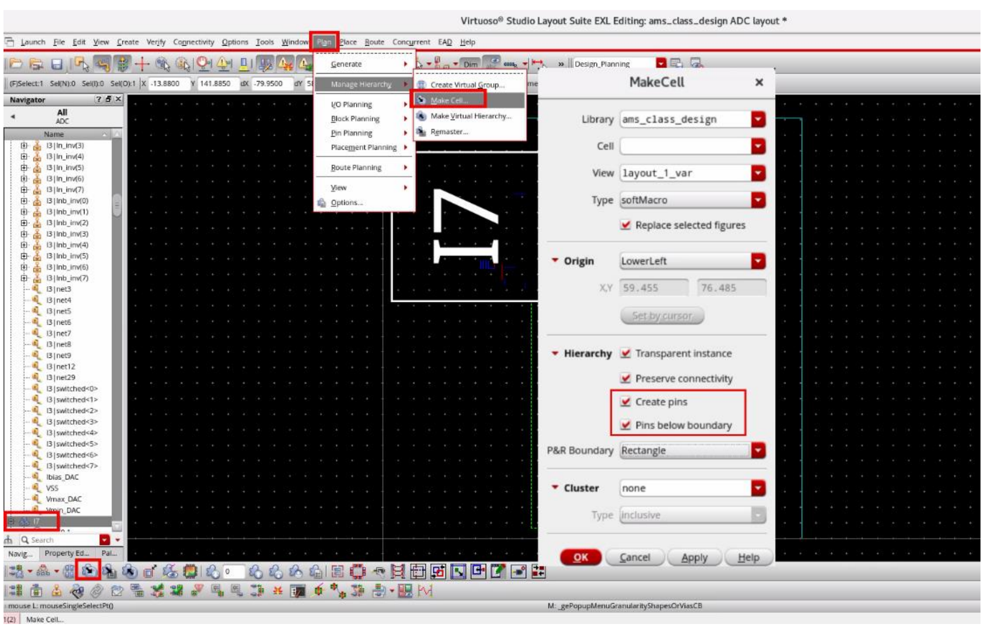
Creating Routes to Demonstrate the Push Option in Make Cell

Now, you will use the Create Wire command to create some routes over **I7** to demonstrate the Push option in the Make Cell command.

1. In the layout canvas, choose the **Create – Wiring – Wire** menu commands or use the **Create Wire** icon in the Create toolbar or use the bindkey **P** to create some routes over **I7**.



2. Select **I7**, open the Make Cell form to set the options, as shown here and click **OK**.



I7 changes to the layout instance again after running the Make Cell command.

Viewing the Routes Created Which Are Pushed as Blockages

Running the Make Cell command with the options selected here creates pins on the boundary based on the flightlines of the interface nets and pushes the routes over the top of the virtual hierarchy inside the layout cell as blockages.

1. Follow the steps given here to view the routes which are pushed as blockages.

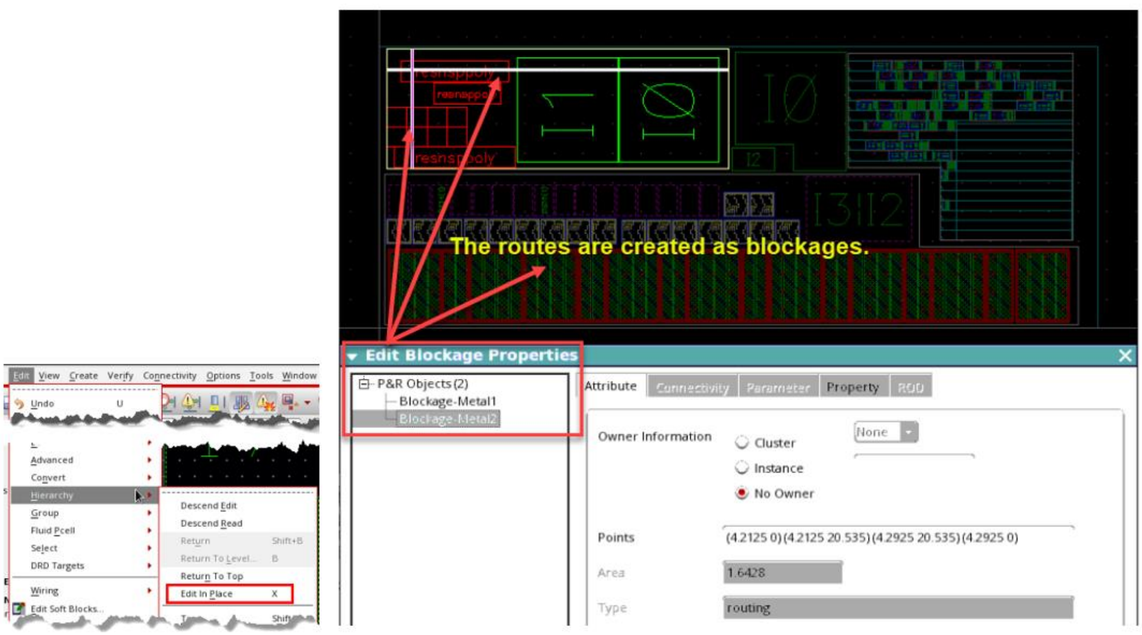
   a. In the layout canvas, select the layout instance **I7**.

   b. Choose the **Edit – Hierarchy – Edit In Place** menu command or press the bindkey **X** to edit in place of the instance **I7**.

   c. Select the routes created in the layout cell to examine their properties.

   d. Choose the **Edit – Basic – Properties** menu command or click the **Edit Properties** icon in the Edit toolbar or press the bindkey **Q** to query the routes in the layout canvas.

   You can see these are blockages. This is needed because when the layout is being implemented out of context, and the layout designer needs to know the position of nets at higher levels.



Important: If the *Routes as blockages* option had not been selected during the Make Cell run as shown here, top-level nets would have been pushed down as routes instead of blockages.

e. Choose **Edit – Hierarchy – Return To Top** to exit the *Edit In Place* mode in the layout canvas and return to the top-level after viewing the blockages.

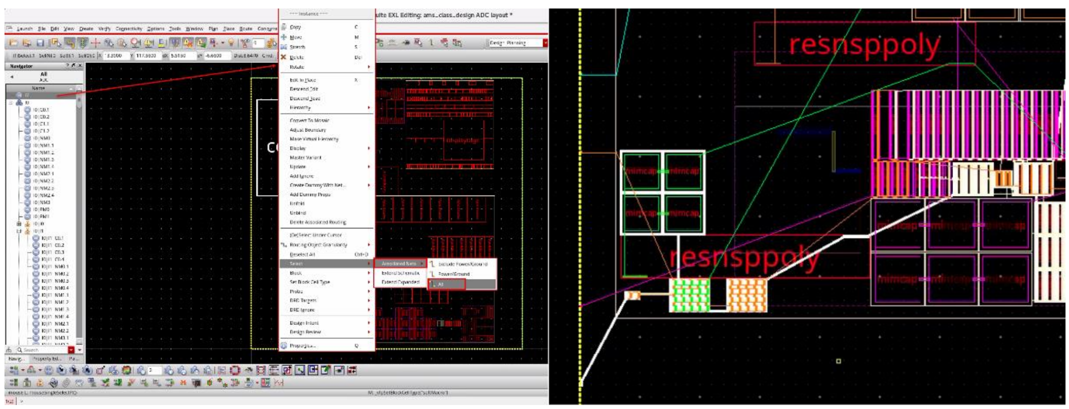
Viewing the Pins Created from the *Make Cell* with the *Create pins: On boundary* Option

1. Follow the steps given here to review the pin positions created during the Make Cell run.

   a. In the layout canvas, select the newly created layout instance **I7**.

   b. **Right**-click the selected instance **I7** and choose **Select – Associated Nets – All** from the context-sensitive menu, as shown here.

   c. **Zoom** in near the layout instance **I7** to review the pin positions created during the Make Cell run.



When the *On boundary* option is used, the pins are created on the boundary as expected. This mode differs from the *Congestion aware* mode as the global router is not run, so the engineer determines the layer and size of the pins. Additionally, the *On boundary* option is not affected by opaque virtual hierarchy. The flightlines are drawn between the current positions of the connected virtual hierarchy contents. The pins are created where the flightlines cross the virtual hierarchy boundary.

Note: In the layout canvas, choose the **Options – Display** menu commands or press the bindkey **E** to open the Display Options form and select the **Instance Pins** option in the Display Controls section to view the instance pins.

In the *Congestion aware* mode, the position of instances inside opaque virtual hierarchies are not used, and the global router is free to create the pins anywhere on the boundary.

Pin positions can also be created from detailed routing if required. This can help later in the design flow for cases that require a complex placement or where the placement is more complete or in a smaller design where one Layout Engineer is responsible for planning multiple blocks.

If you are re-running Make Cell after a Make Virtual Hierarchy run, preserved virtual pins will be used to create pins, taking advantage of the previously created pins.

Running *Remaster* to Handle Layouts Created Outside of the Virtual Hierarchy

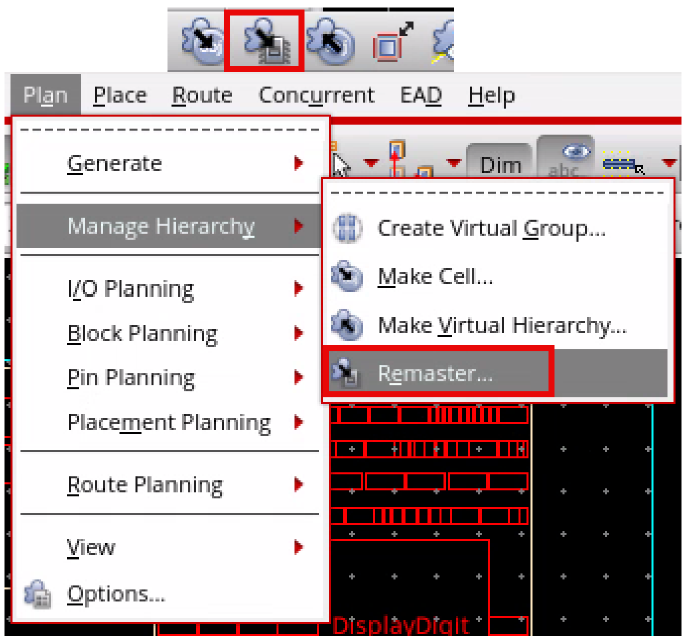
There may be cases where you create a virtual hierarchy for a schematic, and another engineer has implemented the layout outside of the Design Planning environment. You will see how to replace a virtual hierarchy with an existing layout.

1. Follow the steps given here to handle the layouts created outside of the virtual hierarchy.

   a. In the layout canvas, click the **Set Default Display Depth 0** button in the Design Planning toolbar to return to level **0,** making all the virtual hierarchies opaque.

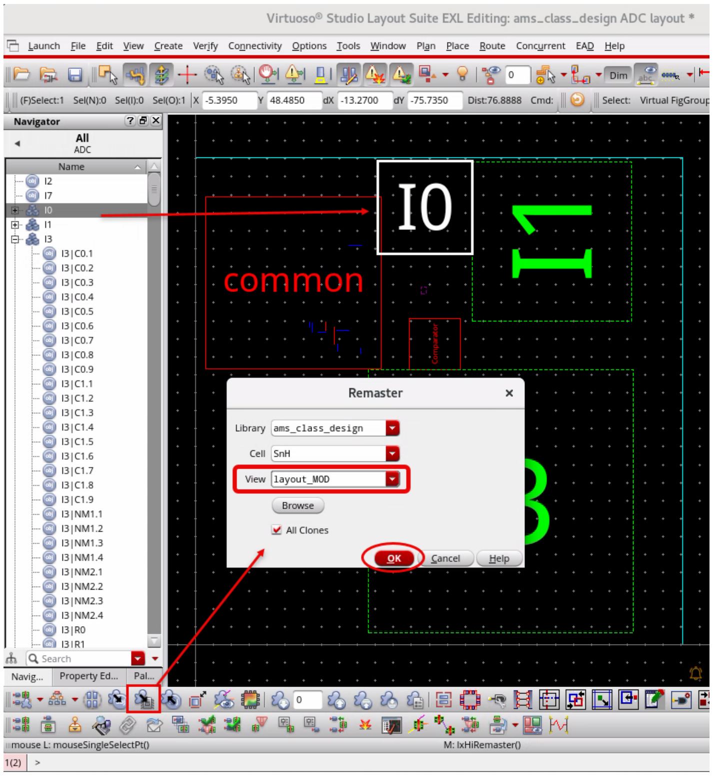
   b. Select the virtual hierarchy **I0**.

   c. Click the **Remaster** button in the Design Planning toolbar or choose the **Plan – Manage Hierarchy – Remaster** menu command to open the Remaster form.

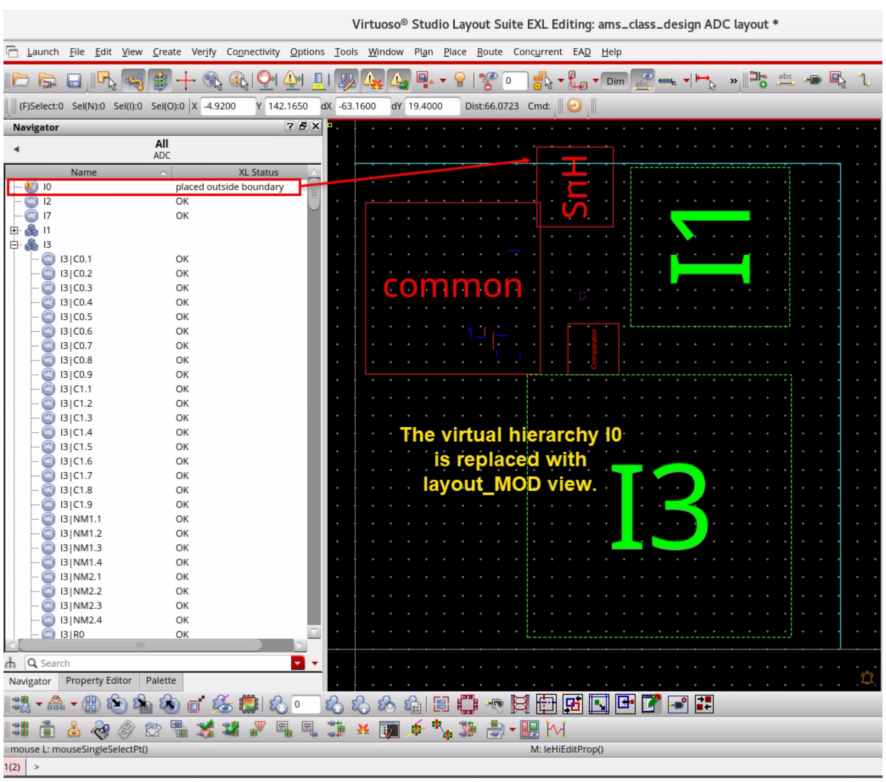


   d. In the Remaster form, the *Library* and *Cell* fields are pre-populated.

e. Using the *View* pull-down, select **layout_MOD** and click **OK**.



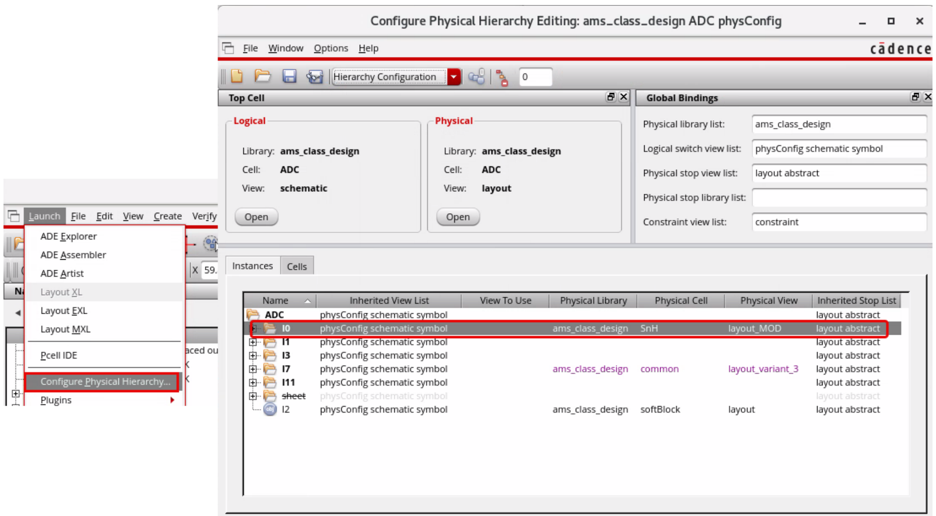
You see that the virtual hierarchy **I0** is replaced with the **layout_MOD** view.



Note: In the Navigator assistant, the XL Status shows as *placed outside boundary*.

2. If the Configure Physical Hierarchy form is not open, choose **Launch – Configure Physical Hierarchy** to re-open the CPH form again.

   a. Review the *Instances* tab.



You can observe that after remastering is done, and as for the Make Cell command, the physical bindings in Configure Physical hierarchy are handled automatically.

3. Keep the CPH form open.

4. In the layout canvas, choose the **File – Save** menu command or click the **Save** icon in the File toolbar to save the cellview.

5. Keep the schematic, layout and Virtuoso session running for the next lab.

Summary

In this lab, you learned how to:

- Create real layout views using the Make Cell and Remaster commands.

End of Lab

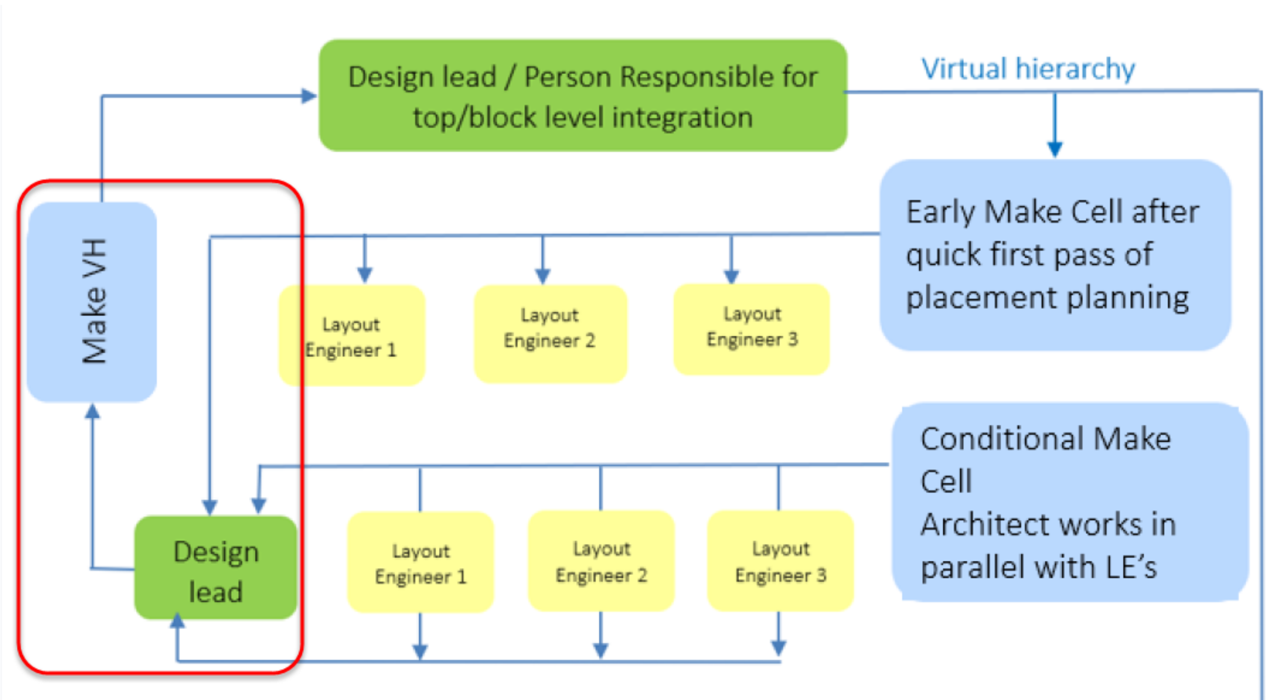
Lab 5-4 Creating Virtual Hierarchies for the Layout Views

Objective: To create virtual hierarchies for the layout views.

In this lab, you explore how to create a virtual hierarchy using the Make Virtual Hierarchy command.

Creating a Virtual Hierarchy Using the Make Virtual Hierarchy Command

You will bring a layout view back into the virtual hierarchy using the Make Virtual Hierarchy command. In a real design scenario, this task is likely to be performed by the Layout Engineer responsible for implementing the top level. To do so, the Layout Engineer will bring a layout implemented by another engineer back into the virtual hierarchy to make some changes to the embedded cellview in the context of the top level.



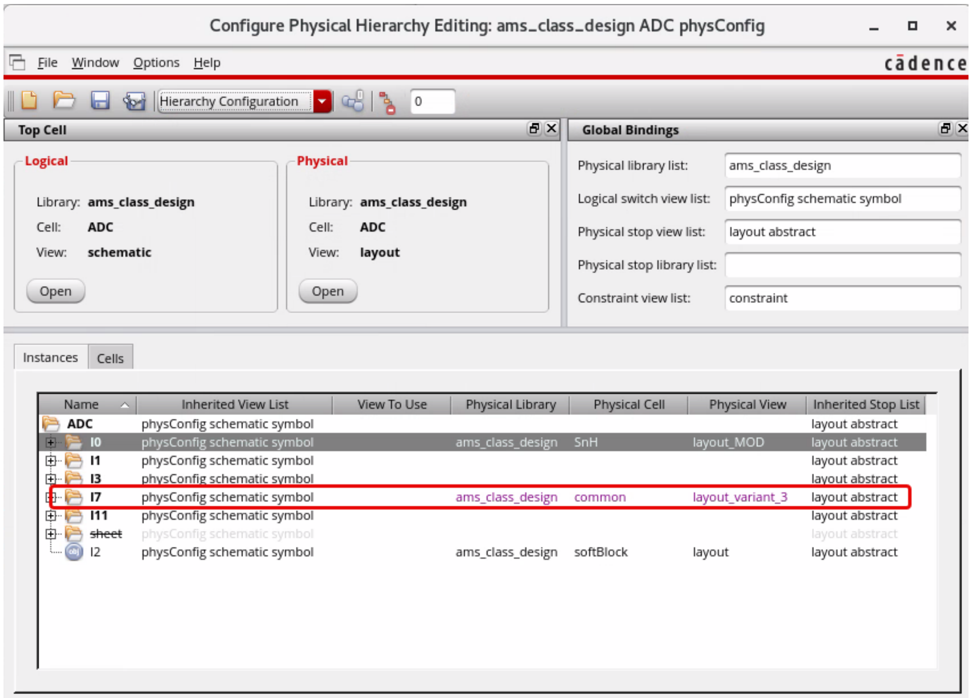
Running the Make Virtual Hierarchy Command

1. Follow the steps given here to create a virtual hierarchy using the Make Virtual Hierarchy command.

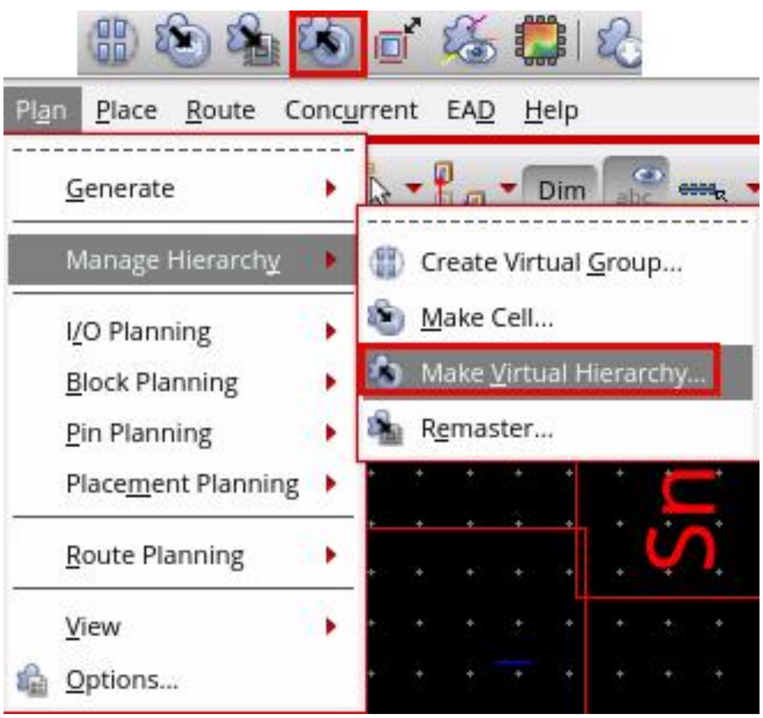
   a. In the layout canvas, in the Design Planning toolbar, *Increase Display Depth* to **1** with no virtual hierarchies selected.

   b. Select the layout instance **I7**.

This is the cell that Make Cell was run in the previous **Lab 5-3,** and it has *layout_variant_3* view as seen in the CPH form shown here.



c. Click the **Make Virtual Hierarchy** button in the Design Planning toolbar or choose the **Plan – Manage Hierarchy – Make Virtual Hierarchy** menu command to open the Make Virtual Hierarchy form.

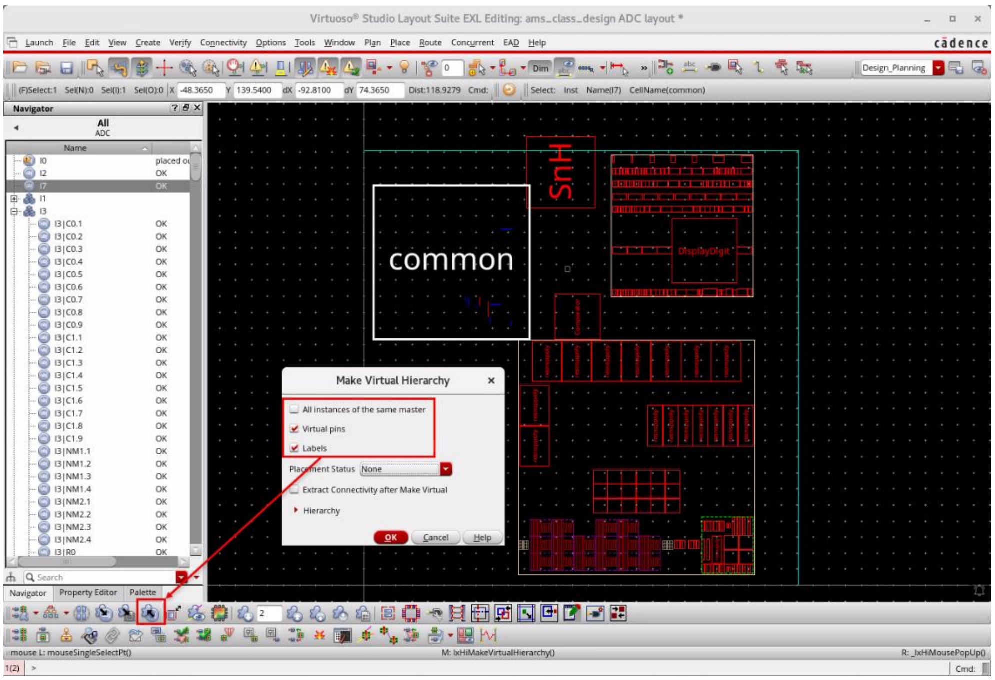


The Make Virtual Hierarchy form appears.

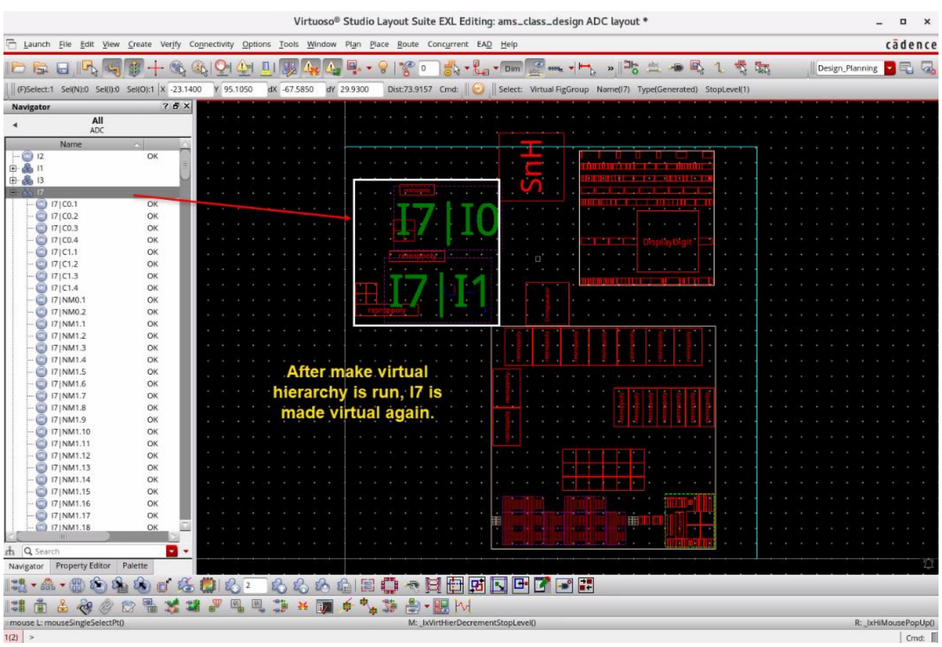
Note: The Make Virtual Hierarchy command has three options:

- *All instances of the same master* – If a master has multiple instances, adds all instances of the master as virtual hierarchy clones.
- *Virtual pins* – Preserves all pins as shapes with connectivity to be used to create pins in the same location on a future Make Cell.
- *Labels* – Creates virtual pins with labels so that the pin connectivity can be visualized in the layout canvas.

4. Set the options as shown here and click **OK**.



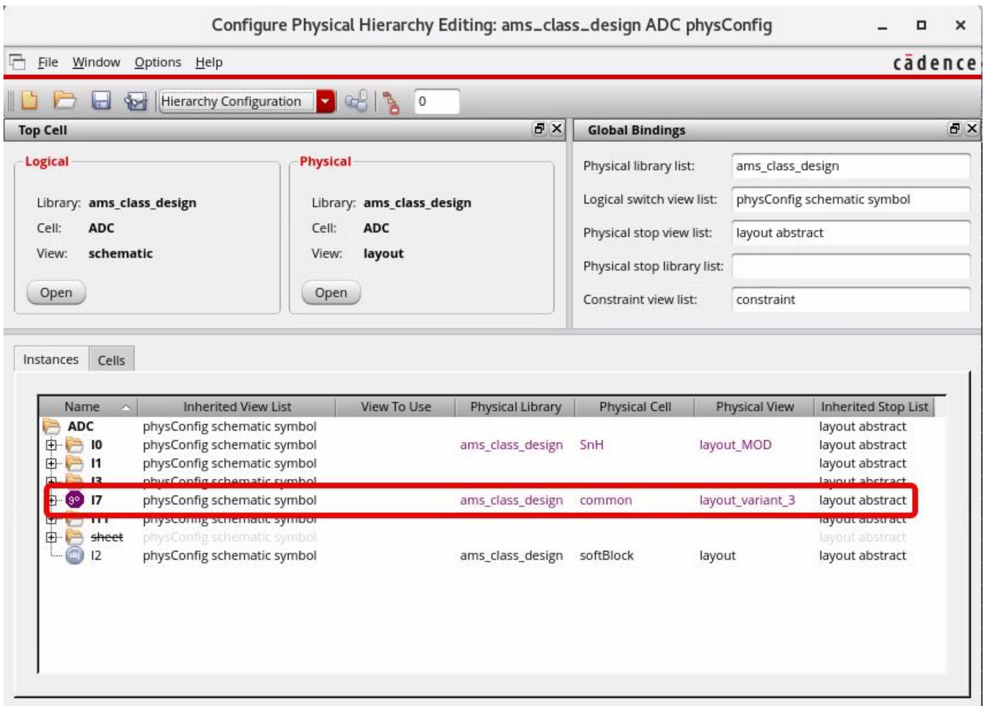
After Make Virtual Hierarchy is run, **I7** is made virtual again, as shown here.



The virtual hierarchy **I7** is represented using an amoeba-shaped icon as seen in the Navigator assistant.

2. If the Configure Physical Hierarchy form is not open, choose **Launch – Configure Physical Hierarchy** to re-open the CPH form again.

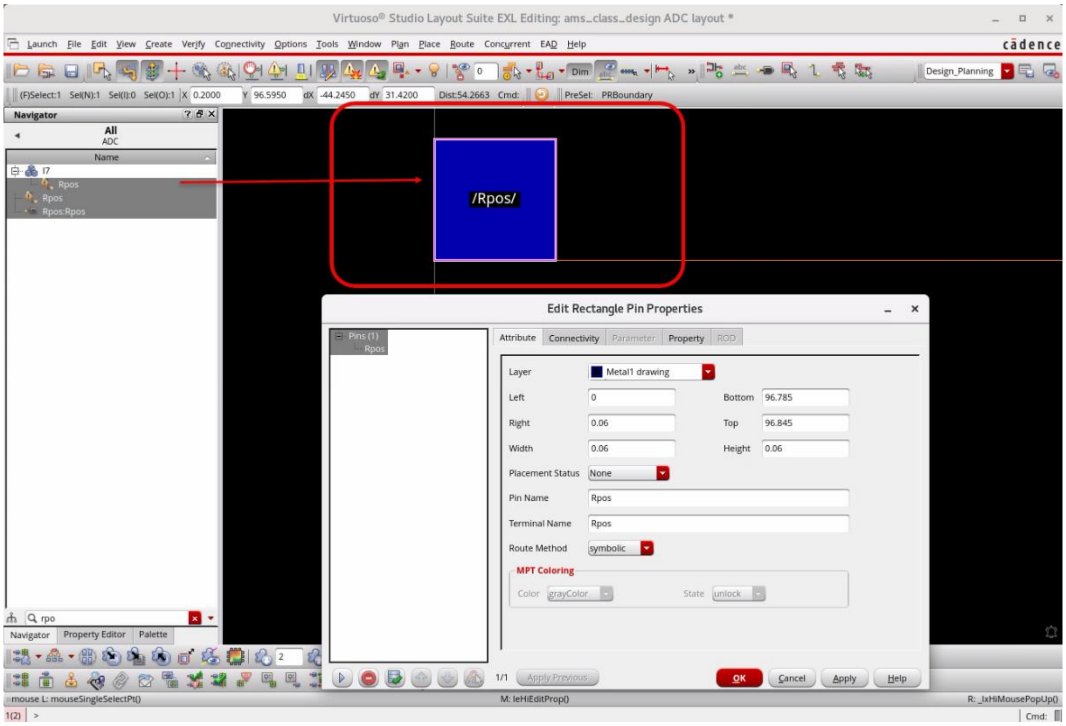
   a. Review the **Instances** tab, as shown here.



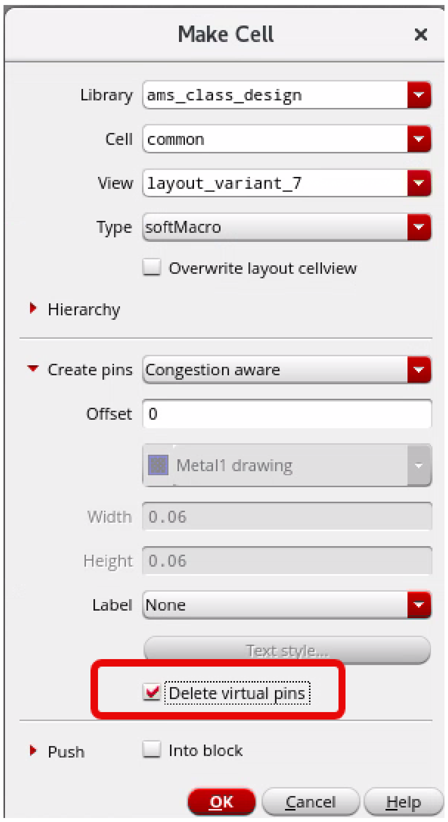
The command makes a virtual hierarchy from the layout view. The command also automatically updates the *physConfig* view so that the virtual hierarchy is used instead of the layout. A force descend has been added for **I7** so that *layout_variant_3* is no longer used in the layout.

Note: The layout view on disk is not altered in any way by Make Virtual Hierarchy.

3. Zoom in to the edge of the **I7** to locate any of the virtual pins and review the properties.



Note: The virtual pin is not a real pin but a shape with connectivity. Pins are not needed to carry the connectivity with the virtual hierarchy, as the layout is flat. When the *Virtual pins* option is selected during the Make Virtual Hierarchy run, virtual pins are created. This preserves the pin locations that will be used for a subsequent Make Cell run to avoid having to recreate the pins. If you prefer recreating the virtual pins, you must run Make Cell with the *Delete virtual pins* option selected.



4. In the layout canvas, choose the **File – Save** menu command or click the **Save** icon in the File toolbar to save the cellview.

5. **Close** all the design windows and forms that are opened.

6. In the CIW window, choose **File – Exit** to close the Virtuoso session.

Summary

In this lab, you learned how to:

- Create a virtual hierarchy using the *Make Virtual Hierarchy* command.

End of Lab